Replication Code Appendix

ADH China Shock Political Event Study Extension

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This appendix prints the replication source files used to construct the final results. Files are included directly from `../src/` at render time.

1 00_config.R

```r
# extension-adh2013/src/00_config.R

if (!requireNamespace("here", quietly = TRUE)) {
  stop("Package 'here' is required. Install it before sourcing the pipeline config.")
}

REPLICATION_DIR <- here::here()
SRC_DIR <- here::here("src")
OUTPUT_DIR <- here::here("output")
INTERMEDIATE_DIR <- here::here("output", "intermediate")
TABLE_DIR <- here::here("output", "tables")
FIGURE_DIR <- here::here("output", "figures")
DIAGNOSTIC_DIR <- here::here("output", "diagnostics")

CONFIG <- list(
  replication_dir = REPLICATION_DIR,
  src_dir = SRC_DIR,
  output_dir = OUTPUT_DIR,
  intermediate_dir = INTERMEDIATE_DIR,
  table_dir = TABLE_DIR,
  figure_dir = FIGURE_DIR,
  diagnostic_dir = DIAGNOSTIC_DIR,

  first_election_year = 1972L,
  diagnostic_first_election_year = 1952L,
  last_election_year = 2020L,
  reference_year = 1988L,

  outcome_var = "rep_margin",
  outcome_label = "Republican presidential vote margin",
  exposure_var = "exposure",
  exposure_label = "ADH China import exposure, 1990-2007",
  exposure_units = "thousand dollars per worker",
  weight_var = "adh_weight",

  crosswalk_weight = "m5_weight",
  crosswalk_missing_weight_policy = "fallback_m2",
  renormalize_crosswalk_weights = FALSE,
  weight_sum_tolerance = 1e-6,
  min_retained_vote_share = 0.98,
  max_abs_vote_identity_error = 1e-6,
  max_missing_cz_year_share = 0,

  require_balanced_panel = TRUE,
  require_no_duplicate_county_years = TRUE,
  require_no_vcov_repair_for_main = TRUE,
  allow_vcov_repair = FALSE,
  allow_failed_diagnostics = FALSE,

  conley_cutoff_km = 500,
  conley_cutoffs_km = c(250, 500, 750, 1000),
  conley_distance = "spherical",
  nw_lag = 1L,
  dk_lag = 1L,
  main_se_type = "cluster_cz",

  standardize_interacted_controls = TRUE,
  export_bandwidth_diagnostic = TRUE,
  export_crosswalk_maps = TRUE,
  run_crosswalk_sensitivity = FALSE,
  run_sixth_extensions = TRUE,
  extension_crosswalk_weight = "m5_weight",
  extension_crosswalk_missing_weight_policy = "fallback_m2",
  save_single_rds = FALSE,
  rds_chunk_size_mib = 45,
  county1990_shapefile_path = Sys.getenv("COUNTY1990_SHAPEFILE", unset = file.path(REPLICATION_DIR, "spatial-data", "counties-1990")),
  nhgis_extract_dir = Sys.getenv("NHGIS_1990_COUNTY_EXTRACT_DIR", unset = file.path(REPLICATION_DIR, "spatial-data", "nhgis_1990_county"))
)

CONFIG$baseline_controls <- c(
  "l_shind_manuf_cbp",
  "l_sh_popedu_c",
  "l_sh_popfborn",
  "l_sh_empl_f",
  "l_sh_routine33",
  "l_task_outsource"
)

CONFIG$interacted_controls <- CONFIG$baseline_controls

CONFIG$core_interacted_controls <- c(
  "l_shind_manuf_cbp",
  "l_sh_popedu_c",
  "l_sh_popfborn"
)
```

2 01_helpers.R

```r
# extension-adh2013/src/01_helpers.R

required_packages <- function(config = NULL) {
  pkgs <- c(
    "dplyr", "fixest", "ggplot2", "haven", "jsonlite", "kableExtra", "knitr",
    "purrr", "readr", "readxl", "rlang", "stringr", "tibble", "tidyr"
  )
  if (!is.null(config) && isTRUE(config$export_crosswalk_maps)) pkgs <- c(pkgs, "sf")
  unique(pkgs)
}

optional_packages <- function() {
  c("ipumsr")
}

configured_shapefile_defaults <- function(config) {
  c(
    file.path(config$replication_dir, "spatial-data", "counties-1990"),
    file.path(config$replication_dir, "spatial-data", "nhgis_1990_county")
  )
}

load_required_packages <- function(config = NULL) {
  pkgs <- required_packages(config)
  missing <- pkgs[!vapply(pkgs, requireNamespace, FUN.VALUE = logical(1), quietly = TRUE)]
  if (length(missing) > 0) {
    stop(
      "Missing required R packages: ", paste(missing, collapse = ", "),
      ". Install them before running the pipeline, or restore the project dependency manifest.",
      call. = FALSE
    )
  }
  invisible(lapply(pkgs, library, character.only = TRUE))
  options(dplyr.summarise.inform = FALSE)
}

ensure_output_dirs <- function(config) {
  dirs <- c(config$output_dir, config$intermediate_dir, config$table_dir,
            config$figure_dir, config$diagnostic_dir)
  invisible(lapply(dirs, dir.create, recursive = TRUE, showWarnings = FALSE))
}

weight_slug <- function(weight_col) {
  stringr::str_replace(weight_col, "_weight$", "")
}

finalize_config <- function(config) {
  allowed_weights <- c("m2_weight", "m4_weight", "m5_weight", "m6_weight")
  if (!config$crosswalk_weight %in% allowed_weights) {
    stop("CONFIG$crosswalk_weight must be one of: ", paste(allowed_weights, collapse = ", "))
  }

  allowed_policies <- c("fail", "fallback_m4", "fallback_m2", "identity_if_same_fips")
  if (!config$crosswalk_missing_weight_policy %in% allowed_policies) {
    stop(
      "CONFIG$crosswalk_missing_weight_policy must be one of: ",
      paste(allowed_policies, collapse = ", ")
    )
  }

  if (is.null(config$interacted_controls)) {
    config$interacted_controls <- config$baseline_controls
  }
  missing_interacted <- setdiff(config$interacted_controls, config$baseline_controls)
  if (length(missing_interacted) > 0) {
    stop("CONFIG$interacted_controls must be drawn from CONFIG$baseline_controls. Invalid: ",
         paste(missing_interacted, collapse = ", "))
  }

  slug <- weight_slug(config$crosswalk_weight)
  config$crosswalk_weight_slug <- slug
  config$event_years <- seq(config$first_election_year, config$last_election_year, by = 4)
  config$diagnostic_event_years <- seq(config$diagnostic_first_election_year, config$last_election_year, by = 4)
  config$source_decades <- sort(unique(year_to_source_decade(config$diagnostic_event_years)))
  if (is.null(config$main_se_type) || !nzchar(config$main_se_type)) {
    config$main_se_type <- paste0("conley_", config$conley_cutoff_km, "km")
  }

  # Treat common placeholder strings as unset. This avoids silently skipping maps
  # when COUNTY1990_SHAPEFILE was previously set to something like
  # "path/to/nhgis_1990_county.shp".
  if (is.null(config$county1990_shapefile_path) ||
      !nzchar(config$county1990_shapefile_path) ||
      grepl("^path/to", config$county1990_shapefile_path, ignore.case = TRUE)) {
    config$county1990_shapefile_path <- file.path(config$replication_dir, "spatial-data", "counties-1990")
  }
  if (is.null(config$nhgis_extract_dir) || !nzchar(config$nhgis_extract_dir)) {
```

```
  config$nhgis_extract_dir <- file.path(config$replication_dir, "spatial-data", "nhgis_1990_county")
}

config$analysis_panel_rds <- file.path(
  config$intermediate_dir, paste0("analysis_panel_aa2021_rep_margin_", slug, ".rds")
)
config$analysis_panel_full_rds <- file.path(
  config$intermediate_dir, paste0("analysis_panel_full_aa2021_rep_margin_", slug, ".rds")
)
config$diagnostic_panel_rds <- file.path(
  config$intermediate_dir, paste0("analysis_panel_aa2021_rep_margin_1952start_", slug, ".rds")
)
config$diagnostic_panel_full_rds <- file.path(
  config$intermediate_dir, paste0("analysis_panel_full_aa2021_rep_margin_1952start_", slug, ".rds")
)
config$cz_centroids_csv <- file.path(
  config$intermediate_dir, "cz_centroids_1990_population_weighted.csv"
)
config$event_study_rds <- file.path(
  config$intermediate_dir, paste0("event_study_results_aa2021_rep_margin_", slug, ".rds")
)
config$event_study_csv <- file.path(
  config$intermediate_dir, paste0("event_study_coefficients_aa2021_rep_margin_", slug, ".csv")
)
config$all_specs_rds <- file.path(
  config$intermediate_dir, paste0("event_study_results_all_specs_aa2021_rep_margin_", slug, ".rds")
)
config$all_specs_csv <- file.path(
  config$intermediate_dir, paste0("event_study_coefficients_all_specs_aa2021_rep_margin_", slug, ".csv")
)

config$dependency_manifest_json <- file.path(config$diagnostic_dir, "dependency_manifest.json")
config$pipeline_manifest_json <- file.path(config$diagnostic_dir, "pipeline_manifest.json")
config$output_manifest_csv <- file.path(config$diagnostic_dir, "output_manifest.csv")
config$output_manifest_json <- file.path(config$diagnostic_dir, "output_manifest.json")
config$validation_checks_csv <- file.path(config$diagnostic_dir, "pipeline_validation_checks.csv")
config$crosswalk_diagnostics_csv <- file.path(
  config$diagnostic_dir, paste0("ftz_", slug, "_crosswalk_source_diagnostics.csv")
)
config$crosswalk_zero_support_csv <- file.path(
  config$diagnostic_dir, paste0("ftz_", slug, "_undefined_or_zero_support_counties.csv")
)
config$crosswalk_row_diagnostics_csv <- file.path(
  config$diagnostic_dir, paste0("ftz_", slug, "_crosswalk_row_diagnostics.csv")
)
config$crosswalk_issue_vote_report_csv <- file.path(
  config$diagnostic_dir, paste0("ftz_", slug, "_crosswalk_issue_vote_report.csv")
)
config$crosswalk_preflight_summary_csv <- file.path(
  config$diagnostic_dir, "ftz_m5_m6_zero_support_preflight_summary.csv"
)
config$crosswalk_preflight_counties_csv <- file.path(
  config$diagnostic_dir, "ftz_m5_m6_zero_support_preflight_counties.csv"
)
config$all_weight_support_csv <- file.path(
  config$diagnostic_dir, "ftz_all_weight_support_by_source_county.csv"
)
config$target_weight_support_county_csv <- file.path(
  config$diagnostic_dir, "ftz_weight_support_by_1990_county.csv"
)
config$target_weight_support_cz_csv <- file.path(
  config$diagnostic_dir, "ftz_weight_support_by_cz.csv"
)
config$county_bridge_classification_csv <- file.path(
  config$diagnostic_dir, paste0("unmatched_source_county_classification_", slug, ".csv")
)
config$mainland_retention_csv <- file.path(
  config$diagnostic_dir, paste0("county_bridge_retention_by_scope_", slug, ".csv")
)
config$baseline_control_summary_csv <- file.path(
  config$diagnostic_dir, "baseline_control_summary.csv"
)
config$baseline_control_correlation_csv <- file.path(
  config$diagnostic_dir, "baseline_control_correlation_matrix.csv"
)
config$interacted_control_variance_csv <- file.path(
  config$diagnostic_dir, "interacted_control_variance_diagnostics.csv"
)
config$conley_neighbor_counts_csv <- file.path(
  config$diagnostic_dir, "conley_neighbor_count_by_cutoff.csv"
)
config$vcov_rank_csv <- file.path(
  config$diagnostic_dir, "event_study_vcov_rank_by_se_type.csv"
)
config$se_warning_diagnostics_csv <- file.path(
  config$diagnostic_dir, "event_study_se_warning_diagnostics.csv"
)
config$crosswalk_comparison_csv <- file.path(
  config$diagnostic_dir, "crosswalk_specification_comparison_coefficients.csv"
)
```

```
config$crosswalk_comparison_png <- file.path(
  config$figure_dir, "fig_crosswalk_specification_comparison.png"
)
config$crosswalk_comparison_pdf <- file.path(
  config$figure_dir, "fig_crosswalk_specification_comparison.pdf"
)
config$crosswalk_comparison_delta_csv <- file.path(
  config$diagnostic_dir, "crosswalk_specification_comparison_deltas_vs_pure_m2.csv"
)
config$crosswalk_comparison_delta_png <- file.path(
  config$figure_dir, "fig_crosswalk_specification_comparison_deltas_vs_pure_m2.png"
)
config$crosswalk_comparison_delta_pdf <- file.path(
  config$figure_dir, "fig_crosswalk_specification_comparison_deltas_vs_pure_m2.pdf"
)
config$crosswalk_support_county_map_png <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_counties.png"
)
config$crosswalk_support_county_map_pdf <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_counties.pdf"
)
config$crosswalk_support_cz_map_png <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_czs.png"
)
config$crosswalk_support_cz_map_pdf <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_czs.pdf"
)
config$crosswalk_support_county_map_simplified_png <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_counties_simplified.png"
)
config$crosswalk_support_county_map_small_png <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_counties_small.png"
)
config$crosswalk_support_county_map_simplified_pdf <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_counties_simplified.pdf"
)
config$crosswalk_support_cz_map_simplified_png <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_czs_simplified.png"
)
config$crosswalk_support_cz_map_small_png <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_czs_small.png"
)
config$crosswalk_support_cz_map_simplified_pdf <- file.path(
  config$figure_dir, "fig_crosswalk_unavailable_weights_1990_czs_simplified.pdf"
)
config$outcome_duplicate_report_csv <- file.path(
  config$diagnostic_dir, "aa2021_presidential_duplicate_county_years.csv"
)
config$outcome_validation_csv <- file.path(
  config$diagnostic_dir, "aa2021_presidential_county_year_validation.csv"
)
config$national_vote_totals_csv <- file.path(
  config$diagnostic_dir, "aa2021_presidential_national_vote_totals.csv"
)
config$state_vote_totals_csv <- file.path(
  config$diagnostic_dir, "aa2021_presidential_state_vote_totals.csv"
)
config$bridge_diagnostics_csv <- file.path(
  config$diagnostic_dir, paste0("county_bridge_diagnostics_by_year_", slug, ".csv")
)
config$unmatched_source_counties_csv <- file.path(
  config$diagnostic_dir, paste0("unmatched_source_counties_", slug, ".csv")
)
config$missing_cz_years_csv <- file.path(
  config$diagnostic_dir, paste0("missing_cz_year_outcomes_", slug, ".csv")
)
config$panel_coverage_csv <- file.path(
  config$diagnostic_dir, paste0("panel_coverage_by_year_", slug, ".csv")
)
config$spec_status_csv <- file.path(config$diagnostic_dir, "event_study_spec_status.csv")
config$model_diagnostics_csv <- file.path(config$diagnostic_dir, "event_study_model_diagnostics.csv")
config$controls_manifest_csv <- file.path(config$diagnostic_dir, "event_study_controls_manifest.csv")
config$vcov_diagnostics_csv <- file.path(config$diagnostic_dir, "event_study_vcov_diagnostics.csv")
config$vcov_eigenvalues_csv <- file.path(config$diagnostic_dir, "event_study_vcov_eigenvalues.csv")
config$pretrend_tests_csv <- file.path(config$diagnostic_dir, "event_study_pretrend_tests.csv")
config$pretrend_coefficients_csv <- file.path(config$diagnostic_dir, "event_study_pretrend_coefficients.csv")

config$table_tex <- file.path(config$table_dir, paste0("tab_event_study_rep_margin_main_specs_", slug, ".tex"))
config$table_csv <- file.path(config$table_dir, paste0("tab_event_study_rep_margin_main_specs_", slug, ".csv"))
config$table_pdf <- file.path(config$table_dir, paste0("tab_event_study_rep_margin_main_specs_", slug, ".pdf"))
config$table_standalone_tex <- file.path(
  config$table_dir, paste0("tab_event_study_rep_margin_main_specs_", slug, "_standalone.tex")
)
config$figure_pdf <- file.path(config$figure_dir, paste0("fig_event_study_rep_margin_main_specs_", slug, ".pdf"))
config$figure_png <- file.path(config$figure_dir, paste0("fig_event_study_rep_margin_main_specs_", slug, ".png"))

# Sixth-section extension outputs. These are intentionally keyed to the
# primary crosswalk slug when relevant, but are otherwise single-project
# diagnostics that should not be re-estimated separately for every
# crosswalk-sensitivity specification.
```

```
  config$bartik_first_stage_csv <- file.path(config$diagnostic_dir, "bartik_first_stage_diagnostics.csv")
  config$bartik_balance_csv <- file.path(config$diagnostic_dir, "bartik_balance_correlations.csv")
  config$bartik_pretrend_csv <- file.path(config$diagnostic_dir, "bartik_pretrend_placebos.csv")
  config$bartik_data_availability_csv <- file.path(config$diagnostic_dir, "bartik_rotemberg_data_availability.csv")
  config$bartik_identification_memo_md <- file.path(config$diagnostic_dir, "bartik_identification_memo.md")
  config$bartik_first_stage_plot_png <- file.path(config$figure_dir, "fig_bartik_first_stage.png")
  config$bartik_first_stage_plot_pdf <- file.path(config$figure_dir, "fig_bartik_first_stage.pdf")
  config$bartik_industry_shift_summary_csv <- file.path(config$diagnostic_dir, "bartik_industry_shift_summary.csv")
  config$bartik_preperiod_workfile_csv <- file.path(config$diagnostic_dir, "bartik_preperiod_workfile_diagnostics.csv")

  config$alternative_outcomes_coefficients_csv <- file.path(config$diagnostic_dir, "alternative_political_outcome_event_studies.csv")
  config$alternative_outcomes_summary_csv <- file.path(config$diagnostic_dir, "alternative_political_outcome_summary.csv")
  config$alternative_outcomes_plot_png <- file.path(config$figure_dir, "fig_alternative_political_outcomes.png")
  config$alternative_outcomes_plot_pdf <- file.path(config$figure_dir, "fig_alternative_political_outcomes.pdf")
  config$alternative_outcomes_status_csv <- file.path(config$diagnostic_dir, "alternative_political_outcome_status.csv")
  config$alternative_outcomes_decomposition_plot_png <- file.path(config$figure_dir,
    "fig_alternative_political_outcome_decomposition.png")
  config$alternative_outcomes_decomposition_plot_pdf <- file.path(config$figure_dir,
    "fig_alternative_political_outcome_decomposition.pdf")
  config$alternative_outcomes_vote_choice_plot_png <- file.path(config$figure_dir, "fig_alternative_political_outcomes_vote_choice.png")
  config$alternative_outcomes_vote_choice_plot_pdf <- file.path(config$figure_dir, "fig_alternative_political_outcomes_vote_choice.pdf")
  config$alternative_outcomes_vote_volume_plot_png <- file.path(config$figure_dir, "fig_alternative_political_outcomes_vote_volume.png")
  config$alternative_outcomes_vote_volume_plot_pdf <- file.path(config$figure_dir, "fig_alternative_political_outcomes_vote_volume.pdf")

  config$nanda_cz_panel_csv <- file.path(config$intermediate_dir, "nanda_2004_2022_cz_panel.csv")
  config$nanda_outcomes_coefficients_csv <- file.path(config$diagnostic_dir, "nanda_outcome_event_studies.csv")
  config$nanda_outcomes_summary_csv <- file.path(config$diagnostic_dir, "nanda_outcome_summary.csv")
  config$nanda_outcomes_plot_png <- file.path(config$figure_dir, "fig_nanda_outcome_event_studies.png")
  config$nanda_outcomes_plot_pdf <- file.path(config$figure_dir, "fig_nanda_outcome_event_studies.pdf")
  config$nanda_outcomes_status_csv <- file.path(config$diagnostic_dir, "nanda_outcome_event_study_status.csv")
  config$nanda_county_validation_csv <- file.path(config$diagnostic_dir, "nanda_county_year_validation.csv")
  config$nanda_cz_validation_csv <- file.path(config$diagnostic_dir, "nanda_cz_year_validation.csv")
  config$nanda_outlier_rates_csv <- file.path(config$diagnostic_dir, "nanda_outlier_rates.csv")
  config$nanda_outcome_availability_csv <- file.path(config$diagnostic_dir, "nanda_outcome_year_availability.csv")
  config$nanda_turnout_plot_png <- file.path(config$figure_dir, "fig_nanda_turnout_registration.png")
  config$nanda_turnout_plot_pdf <- file.path(config$figure_dir, "fig_nanda_turnout_registration.pdf")
  config$nanda_vote_plot_png <- file.path(config$figure_dir, "fig_nanda_presidential_vote_ratios.png")
  config$nanda_vote_plot_pdf <- file.path(config$figure_dir, "fig_nanda_presidential_vote_ratios.pdf")
  config$nanda_partisanship_plot_png <- file.path(config$figure_dir, "fig_nanda_partisanship_indices.png")
  config$nanda_partisanship_plot_pdf <- file.path(config$figure_dir, "fig_nanda_partisanship_indices.pdf")

  config$adhm2020_summary_csv <- file.path(config$diagnostic_dir, "adhm2020_public_data_summary.csv")
  config$adhm2020_regressions_csv <- file.path(config$diagnostic_dir, "adhm2020_mechanism_regressions.csv")
  config$adhm2020_outcome_dictionary_csv <- file.path(config$diagnostic_dir, "adhm2020_outcome_dictionary.csv")
  config$adhm2020_status_csv <- file.path(config$diagnostic_dir, "adhm2020_mechanism_regression_status.csv")
  config$adhm2020_heterogeneity_availability_csv <- file.path(config$diagnostic_dir, "adhm2020_heterogeneity_data_availability.csv")
  config$adhm2020_plot_png <- file.path(config$figure_dir, "fig_adhm2020_mechanism_regressions.png")
  config$adhm2020_plot_pdf <- file.path(config$figure_dir, "fig_adhm2020_mechanism_regressions.pdf")

  config$subperiod_exposure_csv <- file.path(config$intermediate_dir, "adh_subperiod_exposure_by_cz.csv")
  config$subperiod_exposure_summary_csv <- file.path(config$diagnostic_dir, "adh_subperiod_exposure_summary.csv")
  config$subperiod_event_study_coefficients_csv <- file.path(config$diagnostic_dir, "subperiod_exposure_event_study_coefficients.csv")
  config$subperiod_event_study_status_csv <- file.path(config$diagnostic_dir, "subperiod_exposure_event_study_status.csv")
  config$subperiod_event_study_plot_png <- file.path(config$figure_dir, "fig_subperiod_exposure_event_study.png")
  config$subperiod_event_study_plot_pdf <- file.path(config$figure_dir, "fig_subperiod_exposure_event_study.pdf")
  config$subperiod_first_stage_csv <- file.path(config$diagnostic_dir, "subperiod_exposure_first_stage.csv")
  config$subperiod_equality_tests_csv <- file.path(config$diagnostic_dir, "subperiod_exposure_equality_tests.csv")
  config$subperiod_selected_horizons_csv <- file.path(config$diagnostic_dir, "subperiod_exposure_selected_horizons.csv")

  config
}

fips_to_gisjoin <- function(fips) {
  fips <- stringr::str_pad(as.character(fips), width = 5, side = "left", pad = "0")
  paste0("G", substr(fips, 1, 2), "0", substr(fips, 3, 5), "0")
}

gisjoin_to_fips <- function(gisjoin) {
  gisjoin <- as.character(gisjoin)
  out <- ifelse(is.na(gisjoin), NA_character_, paste0(substr(gisjoin, 2, 3), substr(gisjoin, 5, 7)))
  stringr::str_pad(out, width = 5, side = "left", pad = "0")
}

year_to_source_decade <- function(year) {
  dplyr::case_when(
    year < 1980 ~ 1970L,
    year < 1990 ~ 1980L,
    year < 2000 ~ 1990L,
    year < 2010 ~ 2000L,
    year < 2020 ~ 2010L,
    TRUE ~ 2020L
  )
}

read_aa_rdata_object <- function(path, object_name = NULL) {
  e <- new.env(parent = emptyenv())
  loaded <- load(path, envir = e)
  if (!is.null(object_name)) return(e[[object_name]])
  candidates <- setdiff(loaded, ".Random.seed")
  if (length(candidates) != 1) stop("Expected exactly one non-.Random.seed object in: ", path)
```

```
  e[[candidates]]
}

sum_preserve_all_na <- function(x) {
  if (length(x) == 0 || all(is.na(x))) NA_real_ else sum(x, na.rm = TRUE)
}

read_ftz_crosswalk <- function(source_decade, zip_path, county_fips_1990_identity = NULL,
                               weight_col = "m5_weight",
                               missing_weight_policy = "fail",
                               renormalize_weights = FALSE,
                               tolerance = 1e-8) {
  if (source_decade == 1990L) {
    ids <- unique(as.character(county_fips_1990_identity))
    return(tibble::tibble(
      source_decade = 1990L,
      gisjoin_source = fips_to_gisjoin(ids),
      gisjoin_1990 = fips_to_gisjoin(ids),
      county_fips_source = stringr::str_pad(ids, 5, pad = "0"),
      county_fips_1990 = stringr::str_pad(ids, 5, pad = "0"),
      raw_weight = 1,
      fallback_m4_weight = 1,
      fallback_m2_weight = 1,
      raw_weight_missing_parts = 0L,
      raw_weight_parts = 1L,
      weight_sum_raw = 1,
      selected_weight_sum_raw = 1,
      crosswalk_weight = 1,
      crosswalk_weight_sum = 1,
      raw_weight_undefined = FALSE,
      raw_weight_sum_valid = TRUE,
      raw_weight_requires_policy = FALSE,
      fallback_used = FALSE,
      fallback_policy = "identity_1990",
      weight_source = "identity_1990"
    ))
  }

  file_name <- paste0("Crosswalk_", source_decade, "_1990.csv")
  source_col <- paste0("gisjoin_", source_decade)
  raw <- readr::read_csv(unz(zip_path, file_name), show_col_types = FALSE)
  required <- c(source_col, "gisjoin_1990", weight_col, "m4_weight", "m2_weight")
  missing <- setdiff(required, names(raw))
  if (length(missing) > 0) {
    stop("Missing required FTZ columns in ", file_name, ": ", paste(missing, collapse = ", "))
  }

  parts <- raw %>%
    dplyr::transmute(
      source_decade = as.integer(source_decade),
      gisjoin_source = as.character(.data[[source_col]]),
      gisjoin_1990 = as.character(.data[["gisjoin_1990"]]),
      raw_weight_part = as.numeric(.data[[weight_col]]),
      fallback_m4_part = as.numeric(.data[["m4_weight"]]),
      fallback_m2_part = as.numeric(.data[["m2_weight"]])
    ) %>%
    dplyr::mutate(
      gisjoin_source = dplyr::if_else(
        source_decade == 2020L,
        fips_to_gisjoin(stringr::str_pad(gisjoin_source, 5, pad = "0")),
        gisjoin_source
      )
    )

  out <- parts %>%
    dplyr::group_by(source_decade, gisjoin_source, gisjoin_1990) %>%
    dplyr::summarise(
      raw_weight = sum_preserve_all_na(raw_weight_part),
      fallback_m4_weight = sum_preserve_all_na(fallback_m4_part),
      fallback_m2_weight = sum_preserve_all_na(fallback_m2_part),
      raw_weight_missing_parts = sum(is.na(raw_weight_part)),
      raw_weight_parts = dplyr::n(),
      .groups = "drop"
    ) %>%
    dplyr::mutate(
      county_fips_source = gisjoin_to_fips(gisjoin_source),
      county_fips_1990 = gisjoin_to_fips(gisjoin_1990)
    ) %>%
    dplyr::group_by(source_decade, gisjoin_source) %>%
    dplyr::mutate(
      weight_sum_raw = if (all(is.na(raw_weight))) NA_real_ else sum(raw_weight, na.rm = TRUE),
      raw_weight_undefined = is.na(weight_sum_raw) | weight_sum_raw <= tolerance,
      raw_weight_sum_valid = !is.na(weight_sum_raw) & abs(weight_sum_raw - 1) <= tolerance,
      raw_weight_requires_policy = !raw_weight_sum_valid
    ) %>%
    dplyr::ungroup() %>%
    dplyr::mutate(
      selected_raw_weight = dplyr::case_when(
        !raw_weight_requires_policy ~ raw_weight,
        missing_weight_policy == "fallback_m4" ~ fallback_m4_weight,
```

```
        missing_weight_policy == "fallback_m2" ~ fallback_m2_weight,
        missing_weight_policy == "identity_if_same_fips" & county_fips_source == county_fips_1990 ~ 1,
        missing_weight_policy == "identity_if_same_fips" ~ 0,
        TRUE ~ NA_real_
      ),
      fallback_used = raw_weight_requires_policy & missing_weight_policy != "fail",
      fallback_policy = dplyr::case_when(
        !fallback_used ~ "none",
        missing_weight_policy == "fallback_m4" ~ "fallback_m4",
        missing_weight_policy == "fallback_m2" ~ "fallback_m2",
        missing_weight_policy == "identity_if_same_fips" ~ "identity_if_same_fips",
        TRUE ~ missing_weight_policy
      ),
      weight_source = dplyr::case_when(
        !fallback_used ~ weight_col,
        fallback_policy == "fallback_m4" ~ "m4_weight",
        fallback_policy == "fallback_m2" ~ "m2_weight",
        fallback_policy == "identity_if_same_fips" ~ "identity_if_same_fips",
        TRUE ~ weight_col
      )
    ) %>%
    dplyr::group_by(source_decade, gisjoin_source) %>%
    dplyr::mutate(
      selected_weight_sum_raw = if (all(is.na(selected_raw_weight))) {
        NA_real_
      } else {
        sum(selected_raw_weight, na.rm = TRUE)
      },
      crosswalk_weight = dplyr::if_else(
        !is.na(selected_weight_sum_raw) & selected_weight_sum_raw > tolerance,
        if (renormalize_weights) selected_raw_weight / selected_weight_sum_raw else selected_raw_weight,
        NA_real_
      ),
      crosswalk_weight_sum = if (all(is.na(crosswalk_weight))) NA_real_ else sum(crosswalk_weight, na.rm = TRUE)
    ) %>%
    dplyr::ungroup() %>%
    dplyr::select(
      source_decade, gisjoin_source, gisjoin_1990, county_fips_source, county_fips_1990,
      raw_weight, fallback_m4_weight, fallback_m2_weight, raw_weight_missing_parts,
      raw_weight_parts, weight_sum_raw, selected_weight_sum_raw, crosswalk_weight,
      crosswalk_weight_sum, raw_weight_undefined, raw_weight_sum_valid,
      raw_weight_requires_policy, fallback_used, fallback_policy, weight_source
    )

  out
}

fmt_num <- function(x, digits = 3) {
  ifelse(is.na(x), "", formatC(x, digits = digits, format = "f"))
}

star_code <- function(p) {
  dplyr::case_when(
    is.na(p) ~ "",
    p < 0.01 ~ "***",
    p < 0.05 ~ "**",
    p < 0.10 ~ "*",
    TRUE ~ ""
  )
}

make_interacted_control_vars <- function(data, controls, years, ref_year) {
  years_est <- setdiff(sort(unique(years)), ref_year)
  new_names <- character(0)
  for (ctrl in controls) {
    for (yr in years_est) {
      nm <- paste0("ctrl_", ctrl, "_", yr)
      data[[nm]] <- data[[ctrl]] * as.integer(data$year == yr)
      new_names <- c(new_names, nm)
    }
  }
  list(data = data, vars = new_names)
}

validation_check <- function(name, passed, fatal = TRUE, n_failed = NA_integer_, details = "") {
  tibble::tibble(
    check = name,
    status = if (isTRUE(passed)) "pass" else "fail",
    fatal = isTRUE(fatal),
    n_failed = n_failed,
    details = as.character(details)
  )
}

reported_check <- function(name, n_reported = 0L, details = "") {
  tibble::tibble(
    check = name,
    status = if (is.na(n_reported) || n_reported == 0L) "pass" else "reported",
    fatal = FALSE,
    n_failed = 0L,
```

```
    n_reported = as.integer(dplyr::coalesce(as.integer(n_reported), 0L)),
    details = as.character(details)
  )
}

has_fatal_failures <- function(checks) {
  if (nrow(checks) == 0 || !("status" %in% names(checks))) return(FALSE)
  any(checks$fatal %in% TRUE & checks$status == "fail", na.rm = TRUE)
}

has_any_failures <- function(checks) {
  if (nrow(checks) == 0 || !("status" %in% names(checks))) return(FALSE)
  any(checks$status == "fail", na.rm = TRUE)
}

has_reported_conditions <- function(checks) {
  if (nrow(checks) == 0 || !("status" %in% names(checks))) return(FALSE)
  any(checks$status == "reported", na.rm = TRUE)
}

write_dependency_manifest <- function(config) {
  core <- required_packages(config)
  optional <- optional_packages()
  pkgs <- unique(c(core, optional))
  dep <- tibble::tibble(
    package = pkgs,
    dependency_type = dplyr::case_when(
      package %in% core ~ "core",
      package %in% optional ~ "optional",
      TRUE ~ "other"
    ),
    installed = vapply(pkgs, requireNamespace, FUN.VALUE = logical(1), quietly = TRUE),
    version = vapply(pkgs, function(pkg) {
      if (!requireNamespace(pkg, quietly = TRUE)) return(NA_character_)
      as.character(utils::packageVersion(pkg))
    }, character(1))
  )
  jsonlite::write_json(
    list(
      generated_at = format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"),
      r_version = R.version.string,
      packages = dep
    ),
    config$dependency_manifest_json,
    pretty = TRUE,
    auto_unbox = TRUE
  )
  invisible(dep)
}

file_checksum <- function(path) {
  if (!file.exists(path)) return(NA_character_)
  unname(tools::md5sum(path))
}

portable_path <- function(path, config = CONFIG) {
  if (is.null(path) || length(path) == 0 || is.na(path)) return(path)
  path_chr <- as.character(path)
  root <- normalizePath(config$replication_dir, winslash = "/", mustWork = FALSE)
  norm <- normalizePath(path_chr, winslash = "/", mustWork = FALSE)
  prefix <- paste0(root, "/")
  dplyr::case_when(
    norm == root ~ ".",
    startsWith(norm, prefix) ~ substring(norm, nchar(prefix) + 1L),
    TRUE ~ path_chr
  )
}

sanitize_manifest_paths <- function(x, config = CONFIG) {
  if (is.list(x)) {
    nms <- names(x)
    out <- lapply(x, sanitize_manifest_paths, config = config)
    names(out) <- nms
    return(out)
  }
  if (is.character(x) && length(x) == 1 && (grepl("/", x) || grepl("\\\\", x))) {
    return(portable_path(x, config))
  }
  x
}

chunk_dir_for_rds <- function(path) {
  file.path(dirname(path), paste0(tools::file_path_sans_ext(basename(path)), "_rds_chunks"))
}

write_chunked_rds <- function(object, path, config = CONFIG) {
  raw <- serialize(object, connection = NULL, xdr = TRUE)
  compressed <- memCompress(raw, type = "xz")
  chunk_size <- as.integer((config$rds_chunk_size_mib %||% 45) * 1024^2)
  if (!is.finite(chunk_size) || chunk_size <= 0) chunk_size <- 45L * 1024L^2L
```

```
  chunk_dir <- chunk_dir_for_rds(path)
  if (dir.exists(chunk_dir)) unlink(chunk_dir, recursive = TRUE, force = TRUE)
  dir.create(chunk_dir, recursive = TRUE, showWarnings = FALSE)

  n <- length(compressed)
  n_chunks <- max(1L, ceiling(n / chunk_size))
  chunk_files <- character(n_chunks)
  for (i in seq_len(n_chunks)) {
    start <- (i - 1L) * chunk_size + 1L
    end <- min(i * chunk_size, n)
    chunk_files[[i]] <- file.path(chunk_dir, sprintf("part-%03d.rdsbin", i))
    writeBin(compressed[start:end], chunk_files[[i]], useBytes = TRUE)
  }

  manifest <- list(
    format = "serialized-r-object-xdr-memCompress-xz-chunked",
    original_path = portable_path(path, config),
    created_at = format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"),
    chunk_size_mib = config$rds_chunk_size_mib %||% 45,
    n_chunks = n_chunks,
    uncompressed_bytes = length(raw),
    compressed_bytes = length(compressed),
    chunks = lapply(chunk_files, function(cf) list(
      path = portable_path(cf, config),
      bytes = file.info(cf)$size,
      md5 = file_checksum(cf)
    ))
  )
  jsonlite::write_json(manifest, file.path(chunk_dir, "manifest.json"), pretty = TRUE, auto_unbox = TRUE)
  if (file.exists(path)) unlink(path)
  invisible(list(path = path, chunk_dir = chunk_dir, manifest = manifest))
}

read_chunked_rds <- function(path) {
  chunk_dir <- chunk_dir_for_rds(path)
  manifest_path <- file.path(chunk_dir, "manifest.json")
  if (!file.exists(manifest_path)) stop("Chunked RDS manifest not found: ", manifest_path, call. = FALSE)
  manifest <- jsonlite::read_json(manifest_path, simplifyVector = TRUE)
  chunk_paths <- file.path(chunk_dir, basename(manifest$chunks$path))
  if (!all(file.exists(chunk_paths))) {
    # Fall back to paths relative to the project root when the manifest was moved.
    root <- dirname(dirname(dirname(path)))
    chunk_paths <- file.path(root, manifest$chunks$path)
  }
  if (!all(file.exists(chunk_paths))) {
    missing <- chunk_paths[!file.exists(chunk_paths)]
    stop("Missing chunked RDS files: ", paste(missing, collapse = ", "), call. = FALSE)
  }
  compressed <- unlist(Map(function(pth, nbytes) {
    readBin(pth, what = "raw", n = nbytes)
  }, chunk_paths, file.info(chunk_paths)$size), use.names = FALSE)
  unserialize(memDecompress(compressed, type = "xz"))
}

write_model_object <- function(object, path, config = CONFIG) {
  if (isTRUE(config$save_single_rds)) {
    saveRDS(object, path, compress = "xz")
    chunk_dir <- chunk_dir_for_rds(path)
    if (dir.exists(chunk_dir)) unlink(chunk_dir, recursive = TRUE, force = TRUE)
    invisible(list(path = path, mode = "single_rds"))
  } else {
    out <- write_chunked_rds(object, path, config)
    invisible(c(out, list(mode = "chunked_rds")))
  }
}

read_model_object <- function(path) {
  if (file.exists(path)) return(readRDS(path))
  read_chunked_rds(path)
}

model_object_reference <- function(path, config = CONFIG) {
  if (file.exists(path)) {
    return(list(path = path, md5 = file_checksum(path), storage = "single_rds"))
  }
  chunk_dir <- chunk_dir_for_rds(path)
  manifest_path <- file.path(chunk_dir, "manifest.json")
  if (file.exists(manifest_path)) {
    return(list(
      path = path,
      chunk_manifest = manifest_path,
      chunk_dir = chunk_dir,
      manifest_md5 = file_checksum(manifest_path),
      storage = "chunked_rds"
    ))
  }
  list(path = path, md5 = NA_character_, storage = "missing")
}
```

```
output_file_category <- function(path, config = CONFIG) {
  rel <- portable_path(path, config)
  dplyr::case_when(
    startsWith(rel, "output/diagnostics/") ~ "diagnostics",
    startsWith(rel, "output/intermediate/") ~ "intermediate",
    startsWith(rel, "output/tables/") ~ "tables",
    startsWith(rel, "output/figures/") ~ "figures",
    TRUE ~ "other"
  )
}

infer_crosswalk_slug_from_path <- function(path) {
  base <- basename(path)
  hit <- stringr::str_match(base, "_(m2|m4|m5|m6)(\\.|_|$)")[, 2]
  ifelse(is.na(hit), NA_character_, hit)
}

write_output_manifest <- function(config = CONFIG) {
  config <- finalize_config(config)
  if (!dir.exists(config$output_dir)) {
    empty <- tibble::tibble(
      path = character(), category = character(), extension = character(), bytes = numeric(),
      md5 = character(), crosswalk_slug = character(), modified_time = character()
    )
    readr::write_csv(empty, config$output_manifest_csv)
    jsonlite::write_json(list(generated_at = format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"), files = empty),
                         config$output_manifest_json, pretty = TRUE, auto_unbox = TRUE)
    return(invisible(empty))
  }
  files <- list.files(config$output_dir, recursive = TRUE, full.names = TRUE, all.files = FALSE, no.. = TRUE)
  files <- files[file.exists(files) & !dir.exists(files)]
  # Avoid self-recursive checksum churn by writing the manifest after listing but
  # excluding the previous copy of itself from checksum-based reproducibility checks.
  info <- file.info(files)
  out <- tibble::tibble(
    path = vapply(files, portable_path, character(1), config = config),
    category = vapply(files, output_file_category, character(1), config = config),
    extension = tolower(tools::file_ext(files)),
    bytes = as.numeric(info$size),
    md5 = vapply(files, file_checksum, character(1)),
    crosswalk_slug = vapply(files, infer_crosswalk_slug_from_path, character(1)),
    modified_time = format(info$mtime, "%Y-%m-%d %H:%M:%S %Z")
  ) %>%
    dplyr::arrange(category, path)
  readr::write_csv(out, config$output_manifest_csv)
  jsonlite::write_json(
    list(
      generated_at = format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"),
      n_files = nrow(out),
      total_bytes = sum(out$bytes, na.rm = TRUE),
      files = out
    ),
    config$output_manifest_json,
    pretty = TRUE,
    auto_unbox = TRUE
  )
  invisible(out)
}

write_pipeline_manifest <- function(config, checks, stage, sources = list(), extra = list()) {
  sources <- sanitize_manifest_paths(sources, config)
  extra <- sanitize_manifest_paths(extra, config)
  manifest <- list(
    stage = stage,
    generated_at = format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"),
    config = config[c(
      "first_election_year", "diagnostic_first_election_year", "last_election_year",
      "reference_year", "crosswalk_weight", "crosswalk_missing_weight_policy",
      "renormalize_crosswalk_weights", "weight_sum_tolerance", "min_retained_vote_share",
      "require_balanced_panel", "require_no_duplicate_county_years",
      "require_no_vcov_repair_for_main", "allow_vcov_repair", "allow_failed_diagnostics",
      "max_abs_vote_identity_error", "max_missing_cz_year_share",
      "conley_cutoffs_km", "main_se_type", "nw_lag", "dk_lag", "interacted_controls",
      "standardize_interacted_controls", "export_crosswalk_maps", "run_crosswalk_sensitivity",
      "save_single_rds", "rds_chunk_size_mib", "county1990_shapefile_path",
      "nhgis_extract_dir", "output_manifest_csv", "output_manifest_json"
    )],
    sources = sources,
    checks = checks,
    fatal_failure = has_fatal_failures(checks),
    extra = extra
  )
  manifest <- sanitize_manifest_paths(manifest, config)
  jsonlite::write_json(manifest, config$pipeline_manifest_json, pretty = TRUE, auto_unbox = TRUE, null = "null")
  # Keep the standalone output inventory current after each manifest write.
  tryCatch(write_output_manifest(config), error = function(e) warning(
    "Could not update output manifest after writing pipeline manifest: ",
    conditionMessage(e), call. = FALSE
  ))
```

```
  invisible(manifest)
}


`%||%` <- function(x, y) if (is.null(x)) y else x

standardized_control_name <- function(ctrl) paste0("z_", ctrl)

standardize_baseline_controls <- function(data, controls) {
  out <- data
  summary <- purrr::map_dfr(controls, function(ctrl) {
    vals <- out[[ctrl]]
    mu <- mean(vals, na.rm = TRUE)
    sig <- stats::sd(vals, na.rm = TRUE)
    zname <- standardized_control_name(ctrl)
    out[[zname]] <<- if (is.finite(sig) && sig > 0) (vals - mu) / sig else NA_real_
    tibble::tibble(
      control = ctrl,
      standardized_control = zname,
      n_nonmissing = sum(!is.na(vals)),
      mean = mu,
      sd = sig,
      min = suppressWarnings(min(vals, na.rm = TRUE)),
      max = suppressWarnings(max(vals, na.rm = TRUE))
    )
  })
  list(data = out, summary = summary)
}


read_county_bridge_exceptions <- function(config) {
  empty <- tibble::tibble(
    year = integer(), county_fips = character(), county_name_pattern = character(),
    override_source_decade = integer(), override_county_fips_source = character(),
    issue_class = character(), reason = character(), apply = logical()
  )
  path <- file.path(config$src_dir, "county_bridge_exceptions.csv")
  if (!file.exists(path)) return(empty)
  out <- readr::read_csv(path, show_col_types = FALSE) %>%
    dplyr::mutate(
      year = as.integer(year),
      county_fips = stringr::str_pad(as.character(county_fips), 5, pad = "0"),
      override_county_fips_source = stringr::str_pad(as.character(override_county_fips_source), 5, pad = "0"),
      override_source_decade = as.integer(override_source_decade),
      apply = as.logical(apply)
    ) %>%
    dplyr::filter(apply %in% TRUE)
  dplyr::bind_rows(empty, out)
}

classify_unmatched_county <- function(state_fips, county_fips, county_name, year) {
  state_fips <- stringr::str_pad(as.character(state_fips), 2, pad = "0")
  county_fips <- stringr::str_pad(as.character(county_fips), 5, pad = "0")
  nm <- tolower(as.character(county_name))
  dplyr::case_when(
    state_fips %in% c("02", "15", "72") ~ "excluded_nonmainland",
    county_fips %in% c("04012", "35006", "08014", "46113", "46102", "51683", "51685", "51735") ~ "county_boundary_change_needs_bridge",
    state_fips == "51" & stringr::str_detect(nm, "south boston|clifton forge|bedford|manassas|poquoson") ~
      "county_boundary_change_needs_bridge",
    year < 1972L ~ "pre_1972_diagnostic_only",
    TRUE ~ "unexpected_unmatched"
  )
}

is_adh_mainland_source <- function(state_fips) {
  !stringr::str_pad(as.character(state_fips), 2, pad = "0") %in% c("02", "15", "72")
}

find_first_shapefile <- function(path) {
  if (is.null(path) || !nzchar(path)) return(NA_character_)
  if (file.exists(path) && grepl("\\.shp$", path, ignore.case = TRUE)) return(path)
  if (dir.exists(path)) {
    shps <- list.files(path, pattern = "\\.shp$", recursive = TRUE, full.names = TRUE, ignore.case = TRUE)
    if (length(shps) > 0) return(shps[[1]])
  }
  NA_character_
}

infer_county_fips_from_sf <- function(sf_obj) {
  nm <- names(sf_obj)
  if ("county_fips_1990" %in% nm) return(stringr::str_pad(as.character(sf_obj$county_fips_1990), 5, pad = "0"))
  if ("FIPS" %in% nm) return(stringr::str_pad(as.character(sf_obj$FIPS), 5, pad = "0"))
  if ("GEOID" %in% nm) return(stringr::str_pad(as.character(sf_obj$GEOID), 5, pad = "0"))
  if ("GISJOIN" %in% nm) return(gisjoin_to_fips(sf_obj$GISJOIN))
  if (all(c("STATEFP", "COUNTYFP") %in% nm)) return(paste0(stringr::str_pad(sf_obj$STATEFP, 2, pad = "0"),
    stringr::str_pad(sf_obj$COUNTYFP, 3, pad = "0")))
  if (all(c("STATEA", "COUNTYA") %in% nm)) return(paste0(stringr::str_pad(sf_obj$STATEA, 2, pad = "0"), stringr::str_pad(sf_obj$COUNTYA,
    3, pad = "0")))
  if (all(c("STATE_FIPS", "FIPS") %in% nm)) return(stringr::str_pad(as.character(sf_obj$FIPS), 5, pad = "0"))
```

```
  stop("Could not infer county FIPS from shapefile. Expected county_fips_1990, FIPS, GEOID, GISJOIN, STATEFP+COUNTYFP, STATEA+COUNTYA, or STATE_FIPS+FIPS.")
}

safe_read_county1990_shapefile <- function(config) {
  if (!requireNamespace("sf", quietly = TRUE)) {
    warning("Package 'sf' is not installed; skipping crosswalk support maps.", call. = FALSE)
    return(NULL)
  }

  candidate_paths <- unique(c(
    config$county1990_shapefile_path,
    config$nhgis_extract_dir,
    configured_shapefile_defaults(config)
  ))
  candidate_paths <- candidate_paths[!is.na(candidate_paths) & nzchar(candidate_paths)]
  candidate_paths <- candidate_paths[!grepl("^path/to", candidate_paths, ignore.case = TRUE)]

  shp <- NA_character_
  for (candidate in candidate_paths) {
    shp <- find_first_shapefile(candidate)
    if (!is.na(shp)) break
  }

  if (is.na(shp)) {
    warning(
      "No 1990 county shapefile found. Put co1990p020.shp (and companion files) under ",
      file.path(config$replication_dir, "spatial-data", "counties-1990"),
      ", or set COUNTY1990_SHAPEFILE to a 1990 county .shp file or folder. ",
      "Skipping map rendering but writing non-spatial diagnostics.",
      call. = FALSE
    )
    return(NULL)
  }

  sf_obj <- sf::st_read(shp, quiet = TRUE)
  sf_obj$county_fips_1990 <- infer_county_fips_from_sf(sf_obj)
  sf_obj <- sf_obj %>%
    dplyr::filter(!is.na(county_fips_1990))

  # Many historical county shapefiles store multipart counties/islands as multiple
  # rows with the same FIPS. Dissolve before any diagnostic joins so maps do not
  # trigger unexpected many-to-many warnings or over-represent multipart counties.
  sf_obj <- sf_obj %>%
    dplyr::group_by(county_fips_1990) %>%
    dplyr::summarise(geometry = sf::st_union(geometry), .groups = "drop")

  attr(sf_obj, "source_shapefile") <- shp
  sf_obj
}

haversine_km <- function(lon1, lat1, lon2, lat2) {
  rad <- pi / 180
  dlon <- (lon2 - lon1) * rad
  dlat <- (lat2 - lat1) * rad
  a <- sin(dlat / 2)^2 + cos(lat1 * rad) * cos(lat2 * rad) * sin(dlon / 2)^2
  6371.0088 * 2 * atan2(sqrt(a), sqrt(pmax(0, 1 - a)))
}

summarise_conley_neighbors <- function(panel, cutoffs_km) {
  coords <- panel %>%
    dplyr::distinct(czone, lon, lat) %>%
    dplyr::filter(is.finite(lon), is.finite(lat))
  if (nrow(coords) == 0) return(tibble::tibble())
  dist_mat <- outer(seq_len(nrow(coords)), seq_len(nrow(coords)), Vectorize(function(i, j) {
    haversine_km(coords$lon[[i]], coords$lat[[i]], coords$lon[[j]], coords$lat[[j]])
  }))
  purrr::map_dfr(cutoffs_km, function(cutoff) {
    counts <- rowSums(dist_mat <= cutoff, na.rm = TRUE) - 1L
    tibble::tibble(
      cutoff_km = cutoff,
      n_cz = nrow(coords),
      min_neighbors = min(counts),
      p25_neighbors = as.numeric(stats::quantile(counts, 0.25)),
      median_neighbors = stats::median(counts),
      mean_neighbors = mean(counts),
      p75_neighbors = as.numeric(stats::quantile(counts, 0.75)),
      max_neighbors = max(counts),
      cz_with_zero_neighbors = sum(counts == 0)
    )
  })
}

render_table_pdf <- function(fragment_tex, standalone_tex, output_pdf) {
  input_name <- basename(fragment_tex)
  lines <- c(
    "\\documentclass[11pt]{article}",
    "\\usepackage[letterpaper,landscape,margin=0.7in]{geometry}",
    "\\usepackage{array}",
    "\\usepackage{booktabs}",
```

```
    "\\usepackage{longtable}",
    "\\usepackage{threeparttablex}",
    "\\usepackage{caption}",
    "\\renewcommand{\\arraystretch}{1.08}",
    "\\newcolumntype{L}[1]{>{\\raggedright\\arraybackslash}p{#1}}",
    "\\newcolumntype{C}[1]{>{\\centering\\arraybackslash}p{#1}}",
    "\\begin{document}",
    paste0("\\input{", input_name, "}"),
    "\\end{document}"
  )
  writeLines(lines, standalone_tex)

  tex_engine <- Sys.which("pdflatex")
  if (!nzchar(tex_engine)) {
    warning("No LaTeX engine found (pdflatex not on PATH). Wrote TeX sources but did not render a PDF.")
    return(invisible(NULL))
  }

  oldwd <- getwd()
  on.exit(setwd(oldwd), add = TRUE)
  setwd(dirname(standalone_tex))

  out1 <- system2(tex_engine, c("-interaction=nonstopmode", basename(standalone_tex)), stdout = TRUE, stderr = TRUE)
  out2 <- system2(tex_engine, c("-interaction=nonstopmode", basename(standalone_tex)), stdout = TRUE, stderr = TRUE)
  pdf_path <- file.path(dirname(standalone_tex), paste0(tools::file_path_sans_ext(basename(standalone_tex)), ".pdf"))

  if (!file.exists(pdf_path)) {
    warning("pdflatex ran but no PDF was produced. Last output:\n", paste(c(out1, out2), collapse = "\n"))
    return(invisible(NULL))
  }

  file.copy(pdf_path, output_pdf, overwrite = TRUE)

  sidecars <- file.path(
    dirname(standalone_tex),
    paste0(tools::file_path_sans_ext(basename(standalone_tex)), c(".aux", ".log", ".out", ".toc"))
  )
  unlink(sidecars[file.exists(sidecars)])

  invisible(output_pdf)
}
```

3 02_build_analysis_data.R

```r
# extension-adh2013/src/02_build_analysis_data.R

preflight_ftz_builtup_support <- function(zip_path, source_decades, tolerance = 1e-6,
                                          weight_types = c("m2_weight", "m4_weight", "m5_weight", "m6_weight")) {
  # 1990 is identity-mapped to itself in this project, so M2/M4/M5/M6 support
  # is not meaningful for that decade. Omitting it avoids the misleading
  # n_source_counties = 1 synthetic row seen in earlier diagnostics.
  source_decades <- setdiff(as.integer(source_decades), 1990L)

  purrr::map_dfr(source_decades, function(source_decade) {
    file_name <- paste0("Crosswalk_", source_decade, "_1990.csv")
    source_col <- paste0("gisjoin_", source_decade)
    raw <- readr::read_csv(
      unz(zip_path, file_name),
      col_select = dplyr::all_of(c(source_col, weight_types)),
      show_col_types = FALSE
    ) %>%
      dplyr::mutate(
        gisjoin_source = as.character(.data[[source_col]])
      ) %>%
      dplyr::select(gisjoin_source, dplyr::all_of(weight_types))
    if (source_decade == 2020L) {
      raw <- raw %>%
        dplyr::mutate(gisjoin_source = fips_to_gisjoin(stringr::str_pad(gisjoin_source, 5, pad = "0")))
    }

    raw %>%
      tidyr::pivot_longer(
        cols = dplyr::all_of(weight_types),
        names_to = "weight_type",
        values_to = "weight"
      ) %>%
      dplyr::group_by(gisjoin_source, weight_type) %>%
      dplyr::summarise(
        weight_sum = sum_preserve_all_na(weight),
        n_parts = dplyr::n(),
        n_missing_parts = sum(is.na(weight)),
        .groups = "drop"
      ) %>%
      dplyr::transmute(
        source_decade = as.integer(source_decade),
        gisjoin_source,
        county_fips_source = gisjoin_to_fips(gisjoin_source),
        weight_type,
        weight_sum,
        n_parts,
        n_missing_parts,
        available = !is.na(weight_sum) & weight_sum > tolerance,
        zero_or_undefined_support = is.na(weight_sum) | weight_sum <= tolerance
      )
  })
}

read_ftz_raw_target_rows <- function(source_decade, zip_path, weight_types = c("m2_weight", "m4_weight", "m5_weight", "m6_weight")) {
  if (source_decade == 1990L) return(tibble::tibble())
  file_name <- paste0("Crosswalk_", source_decade, "_1990.csv")
  source_col <- paste0("gisjoin_", source_decade)
  raw <- readr::read_csv(
    unz(zip_path, file_name),
    col_select = dplyr::all_of(c(source_col, "gisjoin_1990", weight_types)),
    show_col_types = FALSE
  ) %>%
    dplyr::mutate(
      source_decade = as.integer(source_decade),
      gisjoin_source = as.character(.data[[source_col]]),
      gisjoin_source = dplyr::if_else(
        source_decade == 2020L,
        fips_to_gisjoin(stringr::str_pad(gisjoin_source, 5, pad = "0")),
        gisjoin_source
      ),
      county_fips_source = gisjoin_to_fips(gisjoin_source),
      county_fips_1990 = gisjoin_to_fips(gisjoin_1990)
    )
  raw %>%
    dplyr::select(source_decade, gisjoin_source, county_fips_source, gisjoin_1990, county_fips_1990, dplyr::all_of(weight_types))
}

build_target_weight_support_diagnostics <- function(zip_path, source_decades, county_pres, cz_lookup,
                                                    selected_crosswalks, tolerance = 1e-6,
                                                    adh_czones = NULL) {
  weight_types <- c("m2_weight", "m4_weight", "m5_weight", "m6_weight")
  source_support <- preflight_ftz_builtup_support(zip_path, source_decades, tolerance, weight_types)

  raw_targets <- purrr::map_dfr(setdiff(as.integer(source_decades), 1990L), read_ftz_raw_target_rows,
                                zip_path = zip_path, weight_types = weight_types)
  support_long <- source_support %>%
    dplyr::select(source_decade, gisjoin_source, county_fips_source, weight_type, weight_sum, available)
```

```
88
89    # Attach election-year vote mass so the map can indicate which 1990 counties/CZs
90    # actually receive votes from source counties whose M2/M4/M5/M6 support is unavailable.
91    source_votes <- county_pres %>%
92      dplyr::select(year, source_decade, gisjoin_source, county_fips, state_fips, state, county_name, twoparty_votes)
93
94    # Use the selected/fallback crosswalk weights as the vote-mass allocator for the
95    # demonstration map. A target is flagged only if the selected/fallback crosswalk
96    # sends positive vote mass to that 1990 target from a source county where the
97    # diagnostic weight type is unavailable. This avoids over-flagging ordinary
98    # zero-weight target rows in the raw many-to-many FTZ crosswalk.
99    selected_targets <- selected_crosswalks %>%
100     dplyr::filter(!is.na(crosswalk_weight), crosswalk_weight > tolerance) %>%
101     dplyr::select(source_decade, gisjoin_source, county_fips_1990,
102                   selected_crosswalk_weight = crosswalk_weight) %>%
103     dplyr::distinct()
104
105   target_long <- raw_targets %>%
106     tidyr::pivot_longer(
107       cols = dplyr::all_of(weight_types),
108       names_to = "weight_type",
109       values_to = "raw_weight_for_type"
110     ) %>%
111     dplyr::left_join(support_long, by = c("source_decade", "gisjoin_source", "county_fips_source", "weight_type")) %>%
112     dplyr::left_join(source_votes, by = c("source_decade", "gisjoin_source"), relationship = "many-to-many") %>%
113     dplyr::left_join(selected_targets, by = c("source_decade", "gisjoin_source", "county_fips_1990"), relationship = "many-to-many") %>%
114     dplyr::mutate(
115       unavailable = !dplyr::coalesce(available, FALSE),
116       selected_crosswalk_weight = dplyr::coalesce(selected_crosswalk_weight, 0),
117       received_vote_mass_proxy = twoparty_votes * selected_crosswalk_weight,
118       receives_positive_vote_mass = !is.na(received_vote_mass_proxy) & received_vote_mass_proxy > 0,
119       affected_received_vote_mass_proxy = dplyr::if_else(
120         unavailable & receives_positive_vote_mass,
121         received_vote_mass_proxy,
122         0
123       )
124     )
125
126   county_diag <- target_long %>%
127     dplyr::filter(!is.na(county_fips_1990), !is.na(year)) %>%
128     dplyr::group_by(county_fips_1990, weight_type) %>%
129     dplyr::summarise(
130       receives_unavailable_source = any(affected_received_vote_mass_proxy > 0, na.rm = TRUE),
131       n_unavailable_source_counties = dplyr::n_distinct(county_fips_source[affected_received_vote_mass_proxy > 0]),
132       unavailable_source_vote_mass_proxy = sum(affected_received_vote_mass_proxy, na.rm = TRUE),
133       total_received_vote_mass_proxy = sum(received_vote_mass_proxy, na.rm = TRUE),
134       share_received_from_unavailable_sources = dplyr::if_else(
135         total_received_vote_mass_proxy > 0,
136         unavailable_source_vote_mass_proxy / total_received_vote_mass_proxy,
137         NA_real_
138       ),
139       .groups = "drop"
140     ) %>%
141     dplyr::left_join(cz_lookup %>% dplyr::select(county_fips_1990, czone), by = "county_fips_1990")
142
143   if (!is.null(adh_czones)) {
144     adh_czones <- unique(as.integer(adh_czones))
145     county_diag <- county_diag %>% dplyr::filter(czone %in% adh_czones)
146   }
147
148   cz_diag <- county_diag %>%
149     dplyr::filter(!is.na(czone)) %>%
150     dplyr::group_by(czone, weight_type) %>%
151     dplyr::summarise(
152       receives_unavailable_source = any(receives_unavailable_source %in% TRUE, na.rm = TRUE),
153       n_1990_counties_receiving_unavailable_source = sum(receives_unavailable_source %in% TRUE, na.rm = TRUE),
154       n_unavailable_source_counties = sum(n_unavailable_source_counties, na.rm = TRUE),
155       unavailable_source_vote_mass_proxy = sum(unavailable_source_vote_mass_proxy, na.rm = TRUE),
156       total_received_vote_mass_proxy = sum(total_received_vote_mass_proxy, na.rm = TRUE),
157       share_received_from_unavailable_sources = dplyr::if_else(
158         total_received_vote_mass_proxy > 0,
159         unavailable_source_vote_mass_proxy / total_received_vote_mass_proxy,
160         NA_real_
161       ),
162       .groups = "drop"
163     )
164
165   list(source_support = source_support, county_diag = county_diag, cz_diag = cz_diag)
166 }
167
168 build_analysis_data <- function(config = CONFIG, stop_on_fatal = TRUE) {
169   load_required_packages(config)
170   config <- finalize_config(config)
171
172   adh_stack_path <- file.path(config$replication_dir, "adh2013", "dta", "workfile_china.dta")
173   adh_long_path <- file.path(config$replication_dir, "adh2013", "dta", "workfile_china_long.dta")
174   cz_path <- file.path(config$replication_dir, "cz-data", "cz-198090.xls")
175   centroid_path <- file.path(
176     config$replication_dir, "ftz2024", "crosswalks", "County-CD-centroid-lat-lon",
177     "lat_lon_coordinates_county_csv", "counties_1990_xy.csv"
```

```
  )
  ftz_zip <- file.path(
    config$replication_dir, "ftz2024", "crosswalks", "CountyToCounty", "1990", "1990_csv.zip"
  )
  aa_path <- file.path(
    config$replication_dir, "outcomes-data", "aa2021-nospatial",
    "County_Level_US_Elections_Data",
    "dataverse_shareable_presidential_county_returns_1868_2020.Rdata"
  )

  message("Reading ADH baseline and exposure data...")
  adh_stack <- haven::read_dta(adh_stack_path)
  adh_long <- haven::read_dta(adh_long_path)

  adh_baseline <- adh_stack %>%
    dplyr::filter(yr == 1990) %>%
    dplyr::transmute(
      czone = as.integer(czone),
      statefip = as.integer(statefip),
      dplyr::across(dplyr::all_of(config$baseline_controls), as.numeric)
    ) %>%
    dplyr::distinct()

  adh_exposure <- adh_long %>%
    dplyr::transmute(
      czone = as.integer(czone),
      statefip = as.integer(statefip),
      exposure = as.numeric(d_tradeusch_pw),
      adh_weight = as.numeric(timepwt48)
    ) %>%
    dplyr::distinct()

  adh_sample <- adh_baseline %>%
    dplyr::inner_join(adh_exposure, by = c("czone", "statefip"))

  adh_duplicate_czones <- adh_sample %>%
    dplyr::count(czone, name = "rows_per_czone") %>%
    dplyr::filter(rows_per_czone > 1)

  std_controls <- standardize_baseline_controls(adh_sample, config$baseline_controls)
  adh_sample <- std_controls$data
  readr::write_csv(std_controls$summary, config$baseline_control_summary_csv)

  control_corr <- stats::cor(
    adh_sample[, config$baseline_controls, drop = FALSE],
    use = "pairwise.complete.obs"
  )
  control_corr_long <- as.data.frame(as.table(control_corr), stringsAsFactors = FALSE) %>%
    tibble::as_tibble() %>%
    dplyr::rename(control_1 = Var1, control_2 = Var2, correlation = Freq)
  readr::write_csv(control_corr_long, config$baseline_control_correlation_csv)

  message("Reading 1990 county-to-CZ lookup and county centroids...")
  cz_lookup <- readxl::read_xls(cz_path) %>%
    dplyr::transmute(
      county_fips_1990 = stringr::str_pad(as.character(`County FIPS Code`), width = 5, side = "left", pad = "0"),
      czone = as.integer(CZ90),
      county_name_1990 = as.character(`County name`),
      pop1990 = as.numeric(`Population 1990`)
    )

  cz_lookup_duplicate_count <- cz_lookup %>%
    dplyr::count(county_fips_1990, name = "rows_per_1990_county") %>%
    dplyr::filter(rows_per_1990_county > 1) %>%
    nrow()
  cz_lookup_multi_czone_count <- cz_lookup %>%
    dplyr::distinct(county_fips_1990, czone) %>%
    dplyr::count(county_fips_1990, name = "czones_per_1990_county") %>%
    dplyr::filter(czones_per_1990_county > 1) %>%
    nrow()

  county_centroids <- readr::read_csv(centroid_path, show_col_types = FALSE) %>%
    dplyr::transmute(
      county_fips_1990 = gisjoin_to_fips(gisjoin_1990),
      lon = as.numeric(centroid_x),
      lat = as.numeric(centroid_y)
    )

  cz_centroids <- cz_lookup %>%
    dplyr::inner_join(county_centroids, by = "county_fips_1990") %>%
    dplyr::group_by(czone) %>%
    dplyr::mutate(pop_w = pop1990 / sum(pop1990, na.rm = TRUE)) %>%
    dplyr::summarise(
      lon = stats::weighted.mean(lon, pop_w, na.rm = TRUE),
      lat = stats::weighted.mean(lat, pop_w, na.rm = TRUE),
      pop1990 = sum(pop1990, na.rm = TRUE),
      counties_in_cz = dplyr::n(),
      .groups = "drop"
    ) %>%
    dplyr::inner_join(adh_sample %>% dplyr::distinct(czone), by = "czone")
```

```
  readr::write_csv(cz_centroids, config$cz_centroids_csv)

  preflight_support <- preflight_ftz_builtup_support(
    zip_path = ftz_zip,
    source_decades = config$source_decades,
    tolerance = config$weight_sum_tolerance
  )
  preflight_summary <- preflight_support %>%
    dplyr::group_by(source_decade, weight_type) %>%
    dplyr::summarise(
      n_source_counties = dplyr::n(),
      n_zero_or_undefined_support = sum(zero_or_undefined_support),
      .groups = "drop"
    )
  readr::write_csv(preflight_summary, config$crosswalk_preflight_summary_csv)
  readr::write_csv(
    preflight_support %>% dplyr::filter(zero_or_undefined_support),
    config$crosswalk_preflight_counties_csv
  )

  builtup_preflight <- preflight_support %>%
    dplyr::filter(weight_type %in% c("m5_weight", "m6_weight"))
  if (any(builtup_preflight$zero_or_undefined_support)) {
    summary_msg <- preflight_summary %>%
      dplyr::filter(weight_type %in% c("m5_weight", "m6_weight"), n_zero_or_undefined_support > 0) %>%
      dplyr::mutate(piece = paste0(weight_type, " ", source_decade, ": ", n_zero_or_undefined_support)) %>%
      dplyr::pull(piece) %>%
      paste(collapse = "; ")
    warning(
      "FTZ built-up-support preflight: strict M5/M6 weighting leaves some source counties ",
      "with zero or undefined built-up support, so a strict M5/M6 run cannot produce the balanced ",
      "ADH mainland CZ panel. Current policy is '", config$crosswalk_missing_weight_policy, "'. ",
      "Zero/undefined support counts: ", summary_msg, ". See ",
      config$crosswalk_preflight_summary_csv,
      call. = FALSE
    )
  }

  message("Reading FTZ county crosswalks with ", config$crosswalk_weight, "...")
  crosswalks <- purrr::map_dfr(
    config$source_decades,
    ~ read_ftz_crosswalk(
      source_decade = .x,
      zip_path = ftz_zip,
      county_fips_1990_identity = cz_lookup$county_fips_1990,
      weight_col = config$crosswalk_weight,
      missing_weight_policy = config$crosswalk_missing_weight_policy,
      renormalize_weights = config$renormalize_crosswalk_weights,
      tolerance = config$weight_sum_tolerance
    )
  )

  crosswalk_source_diagnostics <- crosswalks %>%
    dplyr::group_by(source_decade, gisjoin_source, county_fips_source) %>%
    dplyr::summarise(
      weight_sum_raw = dplyr::first(weight_sum_raw),
      selected_weight_sum_raw = dplyr::first(selected_weight_sum_raw),
      crosswalk_weight_sum = dplyr::first(crosswalk_weight_sum),
      raw_weight_undefined = dplyr::first(raw_weight_undefined),
      raw_weight_sum_valid = dplyr::first(raw_weight_sum_valid),
      raw_weight_requires_policy = dplyr::first(raw_weight_requires_policy),
      fallback_used = any(fallback_used),
      fallback_policy = if (any(fallback_policy != "none")) {
        dplyr::first(fallback_policy[fallback_policy != "none"])
      } else {
        "none"
      },
      n_targets = dplyr::n(),
      positive_targets = sum(!is.na(crosswalk_weight) & crosswalk_weight > 0),
      missing_raw_weight_parts = sum(raw_weight_missing_parts),
      final_weight_sum_ok = !is.na(dplyr::first(crosswalk_weight_sum)) &
        abs(dplyr::first(crosswalk_weight_sum) - 1) <= config$weight_sum_tolerance,
      .groups = "drop"
    )

  readr::write_csv(crosswalk_source_diagnostics, config$crosswalk_diagnostics_csv)
  readr::write_csv(
    crosswalk_source_diagnostics %>%
      dplyr::filter(raw_weight_undefined | !final_weight_sum_ok),
    config$crosswalk_zero_support_csv
  )
  readr::write_csv(
    crosswalks %>%
      dplyr::select(
        source_decade, county_fips_source, county_fips_1990, raw_weight,
        weight_sum_raw, crosswalk_weight, crosswalk_weight_sum,
        raw_weight_undefined, raw_weight_sum_valid, raw_weight_requires_policy,
        fallback_used, fallback_policy, weight_source
      ),
    config$crosswalk_row_diagnostics_csv
```

```
)

message("Reading and validating Amlani and Algara (2021) presidential county returns...")
aa_pres <- read_aa_rdata_object(aa_path, object_name = "pres_elections_release")

county_pres_raw <- aa_pres %>%
  dplyr::transmute(
    year = as.integer(election_year),
    county_fips = stringr::str_pad(as.character(fips), width = 5, side = "left", pad = "0"),
    state_fips = stringr::str_pad(as.character(sfips), width = 2, side = "left", pad = "0"),
    state = as.character(state),
    county_name = as.character(county_name),
    rep_votes = as.numeric(republican_raw_votes),
    dem_votes = as.numeric(democratic_raw_votes),
    twoparty_votes = as.numeric(pres_raw_county_vote_totals_two_party),
    total_votes = as.numeric(raw_county_vote_totals),
    complete_case = as.integer(complete_county_cases)
  ) %>%
  dplyr::filter(year %in% config$diagnostic_event_years)

exact_duplicate_rows <- nrow(county_pres_raw) - nrow(dplyr::distinct(county_pres_raw))
county_pres_distinct <- county_pres_raw %>% dplyr::distinct()

duplicate_county_years <- county_pres_distinct %>%
  dplyr::count(year, county_fips, name = "rows_per_county_year") %>%
  dplyr::filter(rows_per_county_year > 1) %>%
  dplyr::left_join(county_pres_distinct, by = c("year", "county_fips"))
readr::write_csv(duplicate_county_years, config$outcome_duplicate_report_csv)

county_pres_validation <- county_pres_distinct %>%
  dplyr::mutate(
    valid_fips = !is.na(county_fips) & stringr::str_detect(county_fips, "^[0-9]{5}$"),
    nonmissing_votes = !is.na(rep_votes) & !is.na(dem_votes) & !is.na(twoparty_votes),
    nonnegative_votes = rep_votes >= 0 & dem_votes >= 0 & twoparty_votes >= 0 &
      (is.na(total_votes) | total_votes >= 0),
    twoparty_matches = abs((rep_votes + dem_votes) - twoparty_votes) <= 1e-6,
    usable_for_bridge = valid_fips & nonmissing_votes & nonnegative_votes &
      twoparty_matches & twoparty_votes > 0 & complete_case == 1L
  )
readr::write_csv(county_pres_validation, config$outcome_validation_csv)

invalid_fips_count <- county_pres_validation %>%
  dplyr::filter(!valid_fips) %>%
  nrow()
max_vote_identity_error <- county_pres_validation %>%
  dplyr::filter(nonmissing_votes) %>%
  dplyr::mutate(abs_vote_identity_error = abs((rep_votes + dem_votes) - twoparty_votes)) %>%
  dplyr::pull(abs_vote_identity_error) %>%
  max(na.rm = TRUE)

bridge_exceptions <- read_county_bridge_exceptions(config)
if (nrow(bridge_exceptions) > 0) {
  readr::write_csv(bridge_exceptions, file.path(config$diagnostic_dir, "county_bridge_exceptions_applied.csv"))
}

county_pres <- county_pres_validation %>%
  dplyr::filter(usable_for_bridge) %>%
  dplyr::mutate(source_decade_default = year_to_source_decade(year)) %>%
  dplyr::left_join(
    bridge_exceptions %>%
      dplyr::select(year, county_fips, override_source_decade, override_county_fips_source, exception_issue_class = issue_class,
        ↪  exception_reason = reason),
    by = c("year", "county_fips")
  ) %>%
  dplyr::mutate(
    source_decade = dplyr::coalesce(override_source_decade, source_decade_default),
    county_fips_for_crosswalk = dplyr::coalesce(override_county_fips_source, county_fips),
    gisjoin_source = fips_to_gisjoin(county_fips_for_crosswalk),
    source_scope = dplyr::if_else(is_adh_mainland_source(state_fips), "adh_mainland_eligible", "excluded_nonmainland"),
    bridge_exception_applied = !is.na(override_county_fips_source) | !is.na(override_source_decade)
  ) %>%
  dplyr::select(
    year, county_fips, county_fips_for_crosswalk, state_fips, state, county_name, rep_votes, dem_votes,
    twoparty_votes, total_votes, source_decade, source_decade_default, gisjoin_source, source_scope,
    bridge_exception_applied, exception_issue_class, exception_reason
  )

target_support <- build_target_weight_support_diagnostics(
  zip_path = ftz_zip,
  source_decades = config$source_decades,
  county_pres = county_pres,
  cz_lookup = cz_lookup,
  selected_crosswalks = crosswalks,
  tolerance = config$weight_sum_tolerance,
  adh_czones = adh_sample$czone
)
readr::write_csv(target_support$source_support, config$all_weight_support_csv)
readr::write_csv(target_support$county_diag, config$target_weight_support_county_csv)
readr::write_csv(target_support$cz_diag, config$target_weight_support_cz_csv)
```

```
source_vote_denominators <- county_pres %>%
  dplyr::group_by(year) %>%
  dplyr::summarise(source_two_party_votes_prejoin = sum(twoparty_votes, na.rm = TRUE), .groups = "drop")

positive_target_summary <- crosswalks %>%
  dplyr::filter(!is.na(crosswalk_weight), crosswalk_weight > config$weight_sum_tolerance) %>%
  dplyr::left_join(cz_lookup %>% dplyr::select(county_fips_1990, czone), by = "county_fips_1990") %>%
  dplyr::group_by(source_decade, gisjoin_source) %>%
  dplyr::summarise(
    positive_target_1990_counties = paste(sort(unique(county_fips_1990)), collapse = ";"),
    positive_target_czones = paste(sort(unique(stats::na.omit(czone))), collapse = ";"),
    positive_target_count = dplyr::n_distinct(county_fips_1990),
    positive_target_cz_count = dplyr::n_distinct(czone, na.rm = TRUE),
    .groups = "drop"
  )

# Source-level issue report. Do not filter on row-level crosswalk_weight <= 0:
# valid source counties naturally have many zero-weight target rows in the raw
# many-to-many FTZ crosswalk. A source is an issue only if its selected raw
# weight is undefined/invalid, its final weight sum is invalid, or it has no
# positive final target weight after the chosen fallback policy.
crosswalk_source_issue_rows <- crosswalk_source_diagnostics %>%
  dplyr::mutate(
    no_positive_final_target = positive_targets == 0,
    final_weight_invalid = is.na(crosswalk_weight_sum) |
      abs(crosswalk_weight_sum - 1) > config$weight_sum_tolerance
  ) %>%
  dplyr::filter(raw_weight_undefined | raw_weight_requires_policy | final_weight_invalid | no_positive_final_target) %>%
  dplyr::left_join(positive_target_summary, by = c("source_decade", "gisjoin_source"))

crosswalk_issue_vote_report <- county_pres %>%
  dplyr::left_join(
    crosswalk_source_issue_rows,
    by = c("source_decade", "gisjoin_source")
  ) %>%
  dplyr::filter(
    raw_weight_undefined %in% TRUE |
      raw_weight_requires_policy %in% TRUE |
      final_weight_invalid %in% TRUE |
      no_positive_final_target %in% TRUE
  ) %>%
  dplyr::left_join(source_vote_denominators, by = "year") %>%
  dplyr::mutate(
    vote_share_of_year = twoparty_votes / source_two_party_votes_prejoin,
    source_issue_class = dplyr::case_when(
      no_positive_final_target %in% TRUE & fallback_used %in% TRUE ~ paste0(
        "selected_", weight_slug(config$crosswalk_weight), "_unavailable_", fallback_policy, "_used_but_no_positive_target"
      ),
      no_positive_final_target %in% TRUE & raw_weight_requires_policy %in% TRUE ~ paste0(
        "selected_", weight_slug(config$crosswalk_weight), "_unavailable_no_fallback"
      ),
      no_positive_final_target %in% TRUE ~ classify_unmatched_county(state_fips, county_fips, county_name, year),
      raw_weight_requires_policy %in% TRUE & fallback_used %in% TRUE ~ paste0(
        "selected_", weight_slug(config$crosswalk_weight), "_unavailable_", fallback_policy, "_used"
      ),
      raw_weight_requires_policy %in% TRUE & !(fallback_used %in% TRUE) ~ paste0(
        "selected_", weight_slug(config$crosswalk_weight), "_unavailable_no_fallback"
      ),
      final_weight_invalid %in% TRUE ~ "final_weight_invalid",
      TRUE ~ "reported_crosswalk_issue"
    )
  ) %>%
  dplyr::select(
    year, source_decade, source_decade_default, source_scope, source_issue_class,
    state_fips, state, county_fips, county_fips_for_crosswalk, county_name,
    bridge_exception_applied, exception_issue_class, exception_reason,
    twoparty_votes, source_two_party_votes_prejoin, vote_share_of_year,
    weight_sum_raw, selected_weight_sum_raw, crosswalk_weight_sum,
    raw_weight_undefined, raw_weight_sum_valid, raw_weight_requires_policy,
    fallback_used, fallback_policy, no_positive_final_target, final_weight_invalid,
    n_targets, positive_targets, positive_target_1990_counties, positive_target_czones,
    positive_target_count, positive_target_cz_count
  ) %>%
  dplyr::arrange(year, source_issue_class, state_fips, county_fips)
readr::write_csv(crosswalk_issue_vote_report, config$crosswalk_issue_vote_report_csv)

national_pre <- county_pres %>%
  dplyr::group_by(year) %>%
  dplyr::summarise(
    scope = "pre_bridge",
    rep_votes = sum(rep_votes, na.rm = TRUE),
    dem_votes = sum(dem_votes, na.rm = TRUE),
    twoparty_votes = sum(twoparty_votes, na.rm = TRUE),
    total_votes = sum(total_votes, na.rm = TRUE),
    source_counties = dplyr::n_distinct(county_fips),
    .groups = "drop"
  )

state_pre <- county_pres %>%
  dplyr::group_by(year, state_fips, state) %>%
```

```
    dplyr::summarise(
      scope = "pre_bridge",
      rep_votes = sum(rep_votes, na.rm = TRUE),
      dem_votes = sum(dem_votes, na.rm = TRUE),
      twoparty_votes = sum(twoparty_votes, na.rm = TRUE),
      total_votes = sum(total_votes, na.rm = TRUE),
      source_counties = dplyr::n_distinct(county_fips),
      .groups = "drop"
    )

  message("Bridging county returns to 1990 counties...")
  county_bridged <- county_pres %>%
    dplyr::left_join(
      crosswalks %>%
        dplyr::select(
          source_decade, gisjoin_source, county_fips_1990, raw_weight,
          crosswalk_weight, raw_weight_undefined, raw_weight_sum_valid,
          raw_weight_requires_policy, fallback_used, fallback_policy, weight_source
        ),
      by = c("source_decade", "gisjoin_source"),
      relationship = "many-to-many"
    )

  unmatched_source_counties <- county_bridged %>%
    dplyr::group_by(year, source_decade, source_decade_default, county_fips, county_fips_for_crosswalk, state_fips, state, county_name,
      ↪  source_scope, bridge_exception_applied, exception_issue_class, exception_reason) %>%
    dplyr::summarise(
      twoparty_votes = dplyr::first(twoparty_votes),
      has_positive_crosswalk_weight = any(!is.na(crosswalk_weight) & crosswalk_weight > 0),
      raw_weight_undefined = any(raw_weight_undefined %in% TRUE, na.rm = TRUE),
      fallback_used = any(fallback_used %in% TRUE, na.rm = TRUE),
      .groups = "drop"
    ) %>%
    dplyr::left_join(source_vote_denominators, by = "year") %>%
    dplyr::mutate(
      vote_share_of_year = twoparty_votes / source_two_party_votes_prejoin,
      source_issue_class = classify_unmatched_county(state_fips, county_fips, county_name, year)
    ) %>%
    dplyr::filter(!has_positive_crosswalk_weight)
  readr::write_csv(unmatched_source_counties, config$unmatched_source_counties_csv)
  readr::write_csv(
    unmatched_source_counties %>%
      dplyr::count(year, source_scope, source_issue_class, name = "unmatched_source_counties") %>%
      dplyr::left_join(
        unmatched_source_counties %>%
          dplyr::group_by(year, source_scope, source_issue_class) %>%
          dplyr::summarise(unmatched_two_party_votes = sum(twoparty_votes, na.rm = TRUE), .groups = "drop"),
        by = c("year", "source_scope", "source_issue_class")
      ),
    config$county_bridge_classification_csv
  )

  original_by_year <- county_pres %>%
    dplyr::group_by(year) %>%
    dplyr::summarise(
      source_counties = dplyr::n_distinct(county_fips),
      source_two_party_votes_prejoin = sum(twoparty_votes, na.rm = TRUE),
      source_rep_votes_prejoin = sum(rep_votes, na.rm = TRUE),
      source_dem_votes_prejoin = sum(dem_votes, na.rm = TRUE),
      .groups = "drop"
    )

  naive_postjoin_denominator <- county_bridged %>%
    dplyr::group_by(year) %>%
    dplyr::summarise(
      source_two_party_votes_after_join_naive_for_diagnostic_only = sum(twoparty_votes, na.rm = TRUE),
      .groups = "drop"
    )

  bridged_by_year <- county_bridged %>%
    dplyr::filter(!is.na(crosswalk_weight), crosswalk_weight > 0) %>%
    dplyr::group_by(year) %>%
    dplyr::summarise(
      matched_source_counties = dplyr::n_distinct(county_fips),
      bridged_two_party_votes = sum(twoparty_votes * crosswalk_weight, na.rm = TRUE),
      bridged_rep_votes = sum(rep_votes * crosswalk_weight, na.rm = TRUE),
      bridged_dem_votes = sum(dem_votes * crosswalk_weight, na.rm = TRUE),
      fallback_source_counties = dplyr::n_distinct(county_fips[fallback_used %in% TRUE]),
      .groups = "drop"
    )

  original_by_scope <- county_pres %>%
    dplyr::group_by(year, source_scope) %>%
    dplyr::summarise(
      source_counties = dplyr::n_distinct(county_fips),
      source_two_party_votes_prejoin = sum(twoparty_votes, na.rm = TRUE),
      .groups = "drop"
    )
  bridged_by_scope <- county_bridged %>%
    dplyr::filter(!is.na(crosswalk_weight), crosswalk_weight > 0) %>%
```

```
    dplyr::group_by(year, source_scope) %>%
    dplyr::summarise(
      matched_source_counties = dplyr::n_distinct(county_fips),
      bridged_two_party_votes = sum(twoparty_votes * crosswalk_weight, na.rm = TRUE),
      fallback_source_counties = dplyr::n_distinct(county_fips[fallback_used %in% TRUE]),
      .groups = "drop"
    )
  retention_by_scope <- original_by_scope %>%
    dplyr::left_join(bridged_by_scope, by = c("year", "source_scope")) %>%
    dplyr::mutate(
      matched_source_counties = dplyr::coalesce(matched_source_counties, 0L),
      bridged_two_party_votes = dplyr::coalesce(bridged_two_party_votes, 0),
      fallback_source_counties = dplyr::coalesce(fallback_source_counties, 0L),
      unmatched_source_counties = source_counties - matched_source_counties,
      retained_vote_share = bridged_two_party_votes / source_two_party_votes_prejoin
    )
  readr::write_csv(retention_by_scope, config$mainland_retention_csv)

  mainland_retention_wide <- retention_by_scope %>%
    dplyr::select(year, source_scope, source_two_party_votes_prejoin, bridged_two_party_votes, retained_vote_share) %>%
    tidyr::pivot_wider(
      names_from = source_scope,
      values_from = c(source_two_party_votes_prejoin, bridged_two_party_votes, retained_vote_share),
      names_sep = "_"
    )

  bridge_diagnostics <- original_by_year %>%
    dplyr::left_join(naive_postjoin_denominator, by = "year") %>%
    dplyr::left_join(bridged_by_year, by = "year") %>%
    dplyr::left_join(mainland_retention_wide, by = "year") %>%
    dplyr::mutate(
      matched_source_counties = dplyr::coalesce(matched_source_counties, 0L),
      unmatched_source_counties = source_counties - matched_source_counties,
      retained_vote_share_correct = bridged_two_party_votes / source_two_party_votes_prejoin,
      retained_vote_share = retained_vote_share_correct,
      retained_vote_share_adh_mainland_eligible = retained_vote_share_adh_mainland_eligible
    )
  readr::write_csv(bridge_diagnostics, config$bridge_diagnostics_csv)

  national_post <- county_bridged %>%
    dplyr::filter(!is.na(crosswalk_weight), crosswalk_weight > 0) %>%
    dplyr::group_by(year) %>%
    dplyr::summarise(
      scope = "post_bridge",
      rep_votes = sum(rep_votes * crosswalk_weight, na.rm = TRUE),
      dem_votes = sum(dem_votes * crosswalk_weight, na.rm = TRUE),
      twoparty_votes = sum(twoparty_votes * crosswalk_weight, na.rm = TRUE),
      total_votes = sum(total_votes * crosswalk_weight, na.rm = TRUE),
      source_counties = dplyr::n_distinct(county_fips),
      .groups = "drop"
    )
  readr::write_csv(dplyr::bind_rows(national_pre, national_post), config$national_vote_totals_csv)

  state_post <- county_bridged %>%
    dplyr::filter(!is.na(crosswalk_weight), crosswalk_weight > 0) %>%
    dplyr::group_by(year, state_fips, state) %>%
    dplyr::summarise(
      scope = "post_bridge",
      rep_votes = sum(rep_votes * crosswalk_weight, na.rm = TRUE),
      dem_votes = sum(dem_votes * crosswalk_weight, na.rm = TRUE),
      twoparty_votes = sum(twoparty_votes * crosswalk_weight, na.rm = TRUE),
      total_votes = sum(total_votes * crosswalk_weight, na.rm = TRUE),
      source_counties = dplyr::n_distinct(county_fips),
      .groups = "drop"
    )
  readr::write_csv(dplyr::bind_rows(state_pre, state_post), config$state_vote_totals_csv)

  county90_panel <- county_bridged %>%
    dplyr::filter(!is.na(county_fips_1990), !is.na(crosswalk_weight), crosswalk_weight > 0) %>%
    dplyr::mutate(
      rep_votes = rep_votes * crosswalk_weight,
      dem_votes = dem_votes * crosswalk_weight,
      twoparty_votes = twoparty_votes * crosswalk_weight,
      total_votes = total_votes * crosswalk_weight
    ) %>%
    dplyr::group_by(year, county_fips_1990) %>%
    dplyr::summarise(
      rep_votes = sum(rep_votes, na.rm = TRUE),
      dem_votes = sum(dem_votes, na.rm = TRUE),
      twoparty_votes = sum(twoparty_votes, na.rm = TRUE),
      total_votes = sum(total_votes, na.rm = TRUE),
      .groups = "drop"
    )

  message("Aggregating bridged county votes to 1990 commuting zones...")
  cz_panel <- county90_panel %>%
    dplyr::inner_join(cz_lookup, by = "county_fips_1990") %>%
    dplyr::group_by(czone, year) %>%
    dplyr::summarise(
      rep_votes = sum(rep_votes, na.rm = TRUE),
```

```
    dem_votes = sum(dem_votes, na.rm = TRUE),
    twoparty_votes = sum(twoparty_votes, na.rm = TRUE),
    total_votes = sum(total_votes, na.rm = TRUE),
    counties_contributing = dplyr::n(),
    .groups = "drop"
  ) %>%
  dplyr::mutate(rep_margin = (rep_votes - dem_votes) / twoparty_votes)

make_panel <- function(years) {
  tidyr::expand_grid(czone = sort(unique(adh_sample$czone)), year = years) %>%
    dplyr::mutate(election_index = match(year, sort(unique(years)))) %>%
    dplyr::left_join(cz_panel, by = c("czone", "year")) %>%
    dplyr::inner_join(adh_sample, by = "czone") %>%
    dplyr::inner_join(cz_centroids, by = "czone") %>%
    dplyr::arrange(czone, year)
}

analysis_panel_full <- make_panel(config$event_years)
diagnostic_panel_full <- make_panel(config$diagnostic_event_years)

analysis_panel <- analysis_panel_full %>%
  dplyr::filter(!is.na(.data[[config$outcome_var]]))
diagnostic_panel <- diagnostic_panel_full %>%
  dplyr::filter(!is.na(.data[[config$outcome_var]]))

missing_cz_years <- dplyr::bind_rows(
  analysis_panel_full %>%
    dplyr::filter(is.na(.data[[config$outcome_var]])) %>%
    dplyr::transmute(sample = "main_1972_start", czone, year),
  diagnostic_panel_full %>%
    dplyr::filter(is.na(.data[[config$outcome_var]])) %>%
    dplyr::transmute(sample = "diagnostic_1952_start", czone, year)
)
readr::write_csv(missing_cz_years, config$missing_cz_years_csv)

panel_coverage <- dplyr::bind_rows(
  analysis_panel_full %>% dplyr::mutate(sample = "main_1972_start"),
  diagnostic_panel_full %>% dplyr::mutate(sample = "diagnostic_1952_start")
) %>%
  dplyr::mutate(has_outcome = !is.na(.data[[config$outcome_var]])) %>%
  dplyr::group_by(sample, year) %>%
  dplyr::summarise(
    n_cz_total = dplyr::n(),
    n_cz_with_outcome = sum(has_outcome),
    missing_cz_years = n_cz_total - n_cz_with_outcome,
    share_cz_with_outcome = mean(has_outcome),
    .groups = "drop"
  )
readr::write_csv(panel_coverage, config$panel_coverage_csv)

saveRDS(analysis_panel, config$analysis_panel_rds)
saveRDS(analysis_panel_full, config$analysis_panel_full_rds)
saveRDS(diagnostic_panel, config$diagnostic_panel_rds)
saveRDS(diagnostic_panel_full, config$diagnostic_panel_full_rds)

main_expected_n <- length(unique(adh_sample$czone)) * length(config$event_years)
main_missing_n <- nrow(analysis_panel_full) - nrow(analysis_panel)
main_missing_share <- if (main_expected_n > 0) main_missing_n / main_expected_n else NA_real_
adh_duplicate_czone_count <- nrow(adh_duplicate_czones)
min_main_retained <- bridge_diagnostics %>%
  dplyr::filter(year %in% config$event_years) %>%
  dplyr::pull(retained_vote_share) %>%
  min(na.rm = TRUE)
duplicate_count <- duplicate_county_years %>%
  dplyr::distinct(year, county_fips) %>%
  nrow()
twoparty_mismatch_count <- county_pres_validation %>%
  dplyr::filter(complete_case == 1L, valid_fips, nonmissing_votes, !twoparty_matches) %>%
  nrow()
nonnegative_failure_count <- county_pres_validation %>%
  dplyr::filter(complete_case == 1L, valid_fips, nonmissing_votes, !nonnegative_votes) %>%
  nrow()
invalid_final_weight_count <- crosswalk_source_diagnostics %>%
  dplyr::filter(!final_weight_sum_ok) %>%
  nrow()
invalid_positive_final_weight_count <- crosswalk_source_diagnostics %>%
  dplyr::filter(positive_targets > 0, !final_weight_sum_ok) %>%
  nrow()
undefined_selected_weight_count <- crosswalk_source_diagnostics %>%
  dplyr::filter(raw_weight_undefined) %>%
  nrow()

checks <- dplyr::bind_rows(
  validation_check(
    "adh_one_row_per_czone",
    adh_duplicate_czone_count == 0,
    fatal = TRUE,
    n_failed = adh_duplicate_czone_count,
    details = "Requires one ADH baseline/exposure row per commuting zone."
  ),
```

```
    validation_check(
      "cz_lookup_unique_1990_counties",
      cz_lookup_duplicate_count == 0 && cz_lookup_multi_czone_count == 0,
      fatal = TRUE,
      n_failed = cz_lookup_duplicate_count + cz_lookup_multi_czone_count,
      details = "Requires each 1990 county FIPS to appear once and map to one CZ."
    ),
    validation_check(
      "aa_all_county_fips_valid",
      invalid_fips_count == 0,
      fatal = TRUE,
      n_failed = invalid_fips_count,
      details = "Requires all county FIPS to be five numeric digits before filtering."
    ),
    validation_check(
      "aa_vote_identity_error_within_tolerance",
      is.finite(max_vote_identity_error) && max_vote_identity_error <= config$max_abs_vote_identity_error,
      fatal = TRUE,
      n_failed = sum(!county_pres_validation$twoparty_matches, na.rm = TRUE),
      details = paste0("Max absolute vote identity error: ", signif(max_vote_identity_error, 6))
    ),
    validation_check(
      "crosswalk_positive_final_weight_sums_equal_one",
      invalid_positive_final_weight_count == 0,
      fatal = TRUE,
      n_failed = invalid_positive_final_weight_count,
      details = paste0("Tolerance: ", config$weight_sum_tolerance)
    ),
    reported_check(
      "crosswalk_all_source_weight_sums_equal_one_or_reported",
      n_reported = invalid_final_weight_count,
      details = paste0(
        "Global non-ADH or unsupported source-county rows are reported rather than treated as validation failures. ",
        "Estimation is gated by positive final weights, retained vote share, and panel balance. Tolerance: ",
        config$weight_sum_tolerance
      )
    ),
    validation_check(
      "crosswalk_undefined_selected_weights_handled",
      config$crosswalk_missing_weight_policy != "fail" || undefined_selected_weight_count == 0,
      fatal = TRUE,
      n_failed = undefined_selected_weight_count,
      details = paste0("Policy: ", config$crosswalk_missing_weight_policy)
    ),
    validation_check(
      "aa_no_duplicate_county_years_after_exact_distinct",
      duplicate_count == 0 || !isTRUE(config$require_no_duplicate_county_years),
      fatal = isTRUE(config$require_no_duplicate_county_years),
      n_failed = duplicate_count,
      details = paste0("Exact duplicate rows removed: ", exact_duplicate_rows)
    ),
    validation_check(
      "aa_two_party_votes_match_party_votes",
      twoparty_mismatch_count == 0,
      fatal = TRUE,
      n_failed = twoparty_mismatch_count,
      details = "Requires republican + democratic votes to equal two-party votes."
    ),
    validation_check(
      "aa_votes_nonnegative",
      nonnegative_failure_count == 0,
      fatal = TRUE,
      n_failed = nonnegative_failure_count,
      details = "Requires nonnegative Republican, Democratic, two-party, and total votes."
    ),
    validation_check(
      "bridged_vote_retention_above_threshold",
      is.finite(min_main_retained) && min_main_retained >= config$min_retained_vote_share,
      fatal = TRUE,
      n_failed = sum(bridge_diagnostics$retained_vote_share < config$min_retained_vote_share, na.rm = TRUE),
      details = paste0("Minimum retained share in main years: ", signif(min_main_retained, 6))
    ),
    validation_check(
      "main_panel_balanced",
      main_missing_n == 0 || (
        !isTRUE(config$require_balanced_panel) &&
          is.finite(main_missing_share) &&
          main_missing_share <= config$max_missing_cz_year_share
      ),
      fatal = isTRUE(config$require_balanced_panel) || config$max_missing_cz_year_share <= 0,
      n_failed = main_missing_n,
      details = paste0(
        "Observed complete rows: ", nrow(analysis_panel), "; expected rows: ", main_expected_n,
        "; missing share: ", signif(main_missing_share, 6)
      )
    )
  )
  readr::write_csv(checks, config$validation_checks_csv)

  sources <- list(
```

```
    adh_stack = list(path = adh_stack_path, md5 = file_checksum(adh_stack_path)),
    adh_long = list(path = adh_long_path, md5 = file_checksum(adh_long_path)),
    cz_lookup = list(path = cz_path, md5 = file_checksum(cz_path)),
    ftz_zip = list(path = ftz_zip, md5 = file_checksum(ftz_zip)),
    aa_presidential = list(path = aa_path, md5 = file_checksum(aa_path))
  )
  manifest <- write_pipeline_manifest(
    config = config,
    checks = checks,
    stage = "build_analysis_data",
    sources = sources,
    extra = list(
      main_expected_rows = main_expected_n,
      main_observed_rows = nrow(analysis_panel),
      main_missing_rows = main_missing_n,
      main_missing_share = main_missing_share,
      min_main_retained_vote_share = min_main_retained
    )
  )

  failed <- checks %>%
    dplyr::filter(fatal, status == "fail")

  if (nrow(failed) > 0 && isTRUE(stop_on_fatal) && !isTRUE(config$allow_failed_diagnostics)) {
    failed_msg <- paste(
      paste0(
        failed$check,
        " (n_failed = ", failed$n_failed, "): ",
        failed$details
      ),
      collapse = "\n"
    )

    detail_parts <- character(0)
    if ("bridged_vote_retention_above_threshold" %in% failed$check) {
      failing_years <- bridge_diagnostics %>%
        dplyr::filter(year %in% config$event_years, retained_vote_share < config$min_retained_vote_share) %>%
        dplyr::transmute(piece = paste0(year, " retained_vote_share=", signif(retained_vote_share, 6))) %>%
        dplyr::pull(piece)
      if (length(failing_years) > 0) {
        detail_parts <- c(detail_parts, paste0("Failing retention years: ", paste(failing_years, collapse = "; ")))
      }
    }
    if ("main_panel_balanced" %in% failed$check) {
      missing_main <- missing_cz_years %>%
        dplyr::filter(sample == "main_1972_start") %>%
        dplyr::mutate(piece = paste0("CZ ", czone, " in ", year)) %>%
        dplyr::pull(piece)
      if (length(missing_main) > 0) {
        shown <- head(missing_main, 30)
        suffix <- if (length(missing_main) > length(shown)) {
          paste0("; ... +", length(missing_main) - length(shown), " more")
        } else {
          ""
        }
        detail_parts <- c(detail_parts, paste0("Missing main-panel CZ-years: ", paste(shown, collapse = "; "), suffix))
      }
    }
    if ("crosswalk_positive_final_weight_sums_equal_one" %in% failed$check) {
      bad_crosswalks <- crosswalk_source_diagnostics %>%
        dplyr::filter(positive_targets > 0, !final_weight_sum_ok) %>%
        dplyr::mutate(piece = paste0(
          source_decade, " county ", county_fips_source,
          " final_sum=", signif(crosswalk_weight_sum, 6)
        )) %>%
        dplyr::pull(piece)
      if (length(bad_crosswalks) > 0) {
        shown <- head(bad_crosswalks, 30)
        suffix <- if (length(bad_crosswalks) > length(shown)) {
          paste0("; ... +", length(bad_crosswalks) - length(shown), " more")
        } else {
          ""
        }
        detail_parts <- c(detail_parts, paste0("Crosswalk source counties with invalid positive final sums: ", paste(shown, collapse = "; "), suffix))
      }
    }

    detail_msg <- if (length(detail_parts) > 0) {
      paste0("\n\nAdditional failing rows:\n", paste(detail_parts, collapse = "\n"))
    } else {
      ""
    }

    stop(
      "Fatal validation checks failed before estimation:\n",
      failed_msg,
      detail_msg,
      "\nSee ", config$pipeline_manifest_json,
      call. = FALSE
```

```
    )
  }

  invisible(list(
    config = config,
    checks = checks,
    manifest = manifest,
    analysis_panel = analysis_panel,
    analysis_panel_full = analysis_panel_full,
    diagnostic_panel = diagnostic_panel,
    diagnostic_panel_full = diagnostic_panel_full,
    crosswalks = crosswalks
  ))
}
```

4 03_estimate_event_study.R

```r
# extension-adh2013/src/03_estimate_event_study.R

add_event_study_terms <- function(data, controls, config) {
  years <- sort(unique(data$year))
  years_est <- setdiff(years, config$reference_year)
  event_vars <- paste0("es_", years_est)
  for (i in seq_along(years_est)) {
    data[[event_vars[[i]]]] <- data[[config$exposure_var]] * as.integer(data$year == years_est[[i]])
  }
  ctrl_build <- make_interacted_control_vars(data, controls, years, config$reference_year)
  list(data = ctrl_build$data, event_vars = event_vars, control_vars = ctrl_build$vars)
}

model_control_names <- function(controls, standardize = FALSE) {
  if (isTRUE(standardize)) standardized_control_name(controls) else controls
}

interacted_control_variance_diagnostics <- function(prepared_data, control_vars, sample_name, spec_name) {
  if (length(control_vars) == 0) return(tibble::tibble())
  purrr::map_dfr(control_vars, function(v) {
    vals <- prepared_data[[v]]
    tibble::tibble(
      sample = sample_name,
      spec = spec_name,
      interacted_control = v,
      n_nonmissing = sum(!is.na(vals)),
      variance = stats::var(vals, na.rm = TRUE),
      min = suppressWarnings(min(vals, na.rm = TRUE)),
      max = suppressWarnings(max(vals, na.rm = TRUE)),
      share_zero = mean(vals == 0, na.rm = TRUE)
    )
  })
}

vcov_rank_summary <- function(vcov_mat, sample_name, spec_name, se_type) {
  if (is.null(vcov_mat) || any(!is.finite(vcov_mat))) {
    return(tibble::tibble(
      sample = sample_name, spec = spec_name, se_type = se_type,
      vcov_dim = NA_integer_, vcov_rank = NA_integer_, vcov_rank_deficiency = NA_integer_,
      min_diag = NA_real_, max_diag = NA_real_, median_diag = NA_real_
    ))
  }
  q <- qr(vcov_mat, tol = 1e-10)
  d <- diag(vcov_mat)
  tibble::tibble(
    sample = sample_name, spec = spec_name, se_type = se_type,
    vcov_dim = nrow(vcov_mat), vcov_rank = q$rank,
    vcov_rank_deficiency = nrow(vcov_mat) - q$rank,
    min_diag = suppressWarnings(min(d, na.rm = TRUE)),
    max_diag = suppressWarnings(max(d, na.rm = TRUE)),
    median_diag = suppressWarnings(stats::median(d, na.rm = TRUE))
  )
}

residualized_design_diagnostics <- function(data, rhs_terms, config, sample_name, spec_name) {
  x <- as.matrix(data[, rhs_terms, drop = FALSE])
  fe <- stats::model.matrix(~ factor(czone) + factor(year) - 1, data = data)
  w <- data[[config$weight_var]]
  w <- ifelse(is.finite(w) & w > 0, sqrt(w), 1)
  xw <- x * w
  few <- fe * w
  fit <- stats::lm.fit(few, xw)
  resid_x <- fit$residuals
  qr_obj <- qr(resid_x, tol = 1e-7)
  singular_values <- tryCatch(svd(resid_x, nu = 0, nv = 0)$d, error = function(e) numeric(0))
  positive_sv <- singular_values[singular_values > 1e-10]
  condition_number <- if (length(positive_sv) == 0) NA_real_ else max(positive_sv) / min(positive_sv)
  tibble::tibble(
    sample = sample_name,
    spec = spec_name,
    n_obs = nrow(data),
    n_cz = dplyr::n_distinct(data$czone),
    n_years = dplyr::n_distinct(data$year),
    n_rhs_terms = length(rhs_terms),
    residualized_rank = qr_obj$rank,
    residualized_rank_deficiency = length(rhs_terms) - qr_obj$rank,
    condition_number = condition_number,
    min_singular_value = if (length(singular_values) == 0) NA_real_ else min(singular_values),
    max_singular_value = if (length(singular_values) == 0) NA_real_ else max(singular_values)
  )
}

vcov_eigen_summary <- function(vcov_mat) {
  if (is.null(vcov_mat) || any(!is.finite(vcov_mat))) {
    return(list(
      eigenvalues = NA_real_, min_eigenvalue = NA_real_,
      max_eigenvalue = NA_real_, repair_required = TRUE,
```

```
      n_negative_eigenvalues = NA_integer_,
      eigenvalue_ratio_abs_min_to_max = NA_real_,
      nonfinite = TRUE
    ))
  }
  ev <- eigen((vcov_mat + t(vcov_mat)) / 2, symmetric = TRUE, only.values = TRUE)$values
  min_ev <- min(ev)
  max_ev <- max(ev)
  list(
    eigenvalues = ev,
    min_eigenvalue = min_ev,
    max_eigenvalue = max_ev,
    repair_required = min_ev < -1e-8,
    n_negative_eigenvalues = sum(ev < 0),
    eigenvalue_ratio_abs_min_to_max = if (is.finite(max_ev) && max_ev > 0) abs(min_ev) / max_ev else NA_real_,
    nonfinite = FALSE
  )
}

extract_event_coefficients <- function(model, vcov_mat, spec_name, sample_name, se_type,
                                       repair_required, vcov_repaired) {
  ct <- fixest::coeftable(model, vcov = vcov_mat)
  ct_df <- data.frame(term = rownames(ct), ct, row.names = NULL, check.names = FALSE)
  names(ct_df)[names(ct_df) == "Estimate"] <- "estimate"
  names(ct_df)[names(ct_df) == "Std. Error"] <- "std.error"
  if ("Pr(>|t|)" %in% names(ct_df)) names(ct_df)[names(ct_df) == "Pr(>|t|)"] <- "p.value"
  if ("Pr(>|z|)" %in% names(ct_df)) names(ct_df)[names(ct_df) == "Pr(>|z|)"] <- "p.value"
  if ("t value" %in% names(ct_df)) names(ct_df)[names(ct_df) == "t value"] <- "statistic"
  if ("z value" %in% names(ct_df)) names(ct_df)[names(ct_df) == "z value"] <- "statistic"
  if (!("p.value" %in% names(ct_df))) ct_df$p.value <- NA_real_
  if (!("statistic" %in% names(ct_df))) ct_df$statistic <- NA_real_

  tibble::as_tibble(ct_df) %>%
    dplyr::filter(grepl("^es_", term)) %>%
    dplyr::mutate(
      sample = sample_name,
      spec = spec_name,
      se_type = se_type,
      year = as.integer(sub("^es_", "", term)),
      conf.low = estimate - stats::qnorm(0.975) * std.error,
      conf.high = estimate + stats::qnorm(0.975) * std.error,
      vcov_repair_required = repair_required,
      vcov_repaired = vcov_repaired
    ) %>%
    dplyr::select(
      sample, spec, se_type, term, year, estimate, std.error, conf.low, conf.high,
      statistic, p.value, vcov_repair_required, vcov_repaired
    ) %>%
    dplyr::arrange(sample, spec, se_type, year)
}

vcov_specifications <- function(config) {
  conley_specs <- purrr::map(config$conley_cutoffs_km, function(cutoff) {
    list(
      se_type = paste0("conley_", cutoff, "km"),
      compute = function(model, repair) {
        fixest::vcov_conley(
          model,
          lat = "lat",
          lon = "lon",
          cutoff = cutoff,
          distance = config$conley_distance,
          vcov_fix = repair
        )
      }
    )
  })
  c(
    list(
      list(
        se_type = "heteroskedastic",
        compute = function(model, repair) fixest::vcov_hetero(model, vcov_fix = repair)
      ),
      list(
        se_type = "cluster_cz",
        compute = function(model, repair) {
          fixest::vcov_cluster(model, cluster = stats::as.formula("~czone"), vcov_fix = repair)
        }
      ),
      list(
        se_type = "cluster_state",
        compute = function(model, repair) {
          fixest::vcov_cluster(model, cluster = stats::as.formula("~statefip"), vcov_fix = repair)
        }
      ),
      list(
        se_type = "cluster_cz_year",
        compute = function(model, repair) {
          fixest::vcov_cluster(model, cluster = stats::as.formula("~czone + year"), vcov_fix = repair)
        }
```

```
      ),
      list(
        se_type = paste0("newey_west_lag", config$nw_lag),
        compute = function(model, repair) {
          fixest::vcov_NW(
            model, unit = "czone", time = "election_index",
            lag = config$nw_lag, vcov_fix = repair
          )
        }
      ),
      list(
        se_type = paste0("driscoll_kraay_lag", config$dk_lag),
        compute = function(model, repair) {
          fixest::vcov_DK(model, time = "election_index", lag = config$dk_lag, vcov_fix = repair)
        }
      )
    ),
    conley_specs
  )
}

compute_vcov_outputs <- function(model, spec_name, sample_name, config) {
  specs <- vcov_specifications(config)
  coefficients <- list()
  diagnostics <- list()
  eigenvalues <- list()
  vcov_mats <- list()

  for (s in specs) {
    se_type <- s$se_type
    raw <- tryCatch(s$compute(model, FALSE), error = function(e) e)
    if (inherits(raw, "error")) {
      diagnostics[[se_type]] <- tibble::tibble(
        sample = sample_name, spec = spec_name, se_type = se_type,
        status = "failed", error_message = conditionMessage(raw),
        min_eigenvalue = NA_real_, max_eigenvalue = NA_real_,
        n_negative_eigenvalues = NA_integer_,
        eigenvalue_ratio_abs_min_to_max = NA_real_,
        repair_required = TRUE, vcov_repaired = FALSE
      )
      next
    }

    eig <- vcov_eigen_summary(raw)
    eigenvalues[[se_type]] <- tibble::tibble(
      sample = sample_name,
      spec = spec_name,
      se_type = se_type,
      eigen_index = seq_along(eig$eigenvalues),
      eigenvalue = eig$eigenvalues
    )

    use_vcov <- raw
    repaired <- FALSE
    status <- "ok"
    err <- NA_character_

    if (isTRUE(eig$repair_required)) {
      status <- "repair_required"
      err <- "Unrepaired VCOV is not positive semidefinite."
      if (isTRUE(config$allow_vcov_repair)) {
        repaired_try <- tryCatch(s$compute(model, TRUE), error = function(e) e)
        if (inherits(repaired_try, "error")) {
          status <- "failed"
          err <- conditionMessage(repaired_try)
        } else {
          use_vcov <- repaired_try
          repaired <- TRUE
          status <- "repaired"
        }
      }
    }

    diagnostics[[se_type]] <- tibble::tibble(
      sample = sample_name, spec = spec_name, se_type = se_type,
      status = status, error_message = err,
      min_eigenvalue = eig$min_eigenvalue, max_eigenvalue = eig$max_eigenvalue,
      n_negative_eigenvalues = eig$n_negative_eigenvalues,
      eigenvalue_ratio_abs_min_to_max = eig$eigenvalue_ratio_abs_min_to_max,
      repair_required = eig$repair_required, vcov_repaired = repaired
    )

    # Write coefficient rows for every successfully computed VCOV, including
    # unrepaired VCOVs flagged as repair_required. This keeps the all-spec CSV
    # complete for diagnostics (especially all Conley cutoffs/specifications),
    # while the vcov_repair_required/vcov_repaired flags make clear which rows
    # should not be treated as publication-ready inference.
    if (status %in% c("ok", "repaired", "repair_required")) {
      coef_tbl <- tryCatch(
        extract_event_coefficients(
```

```
          model = model, vcov_mat = use_vcov, spec_name = spec_name,
          sample_name = sample_name, se_type = se_type,
          repair_required = eig$repair_required, vcov_repaired = repaired
        ),
        error = function(e) e
      )
      if (inherits(coef_tbl, "error")) {
        diagnostics[[se_type]]$status <- "coeftable_failed"
        diagnostics[[se_type]]$error_message <- conditionMessage(coef_tbl)
      } else {
        coefficients[[se_type]] <- coef_tbl
        vcov_mats[[se_type]] <- use_vcov
      }
    }
  }

  list(
    coefficients = dplyr::bind_rows(coefficients),
    diagnostics = dplyr::bind_rows(diagnostics),
    eigenvalues = dplyr::bind_rows(eigenvalues),
    vcov = vcov_mats
  )
}

controls_manifest_for_spec <- function(sample_name, spec_name, controls, model_controls, standardize_controls, config, model) {
  dropped <- model$collin.var %||% character(0)
  tibble::tibble(control = config$baseline_controls) %>%
    dplyr::mutate(
      sample = sample_name,
      spec = spec_name,
      requested = control %in% controls,
      standardize_controls = isTRUE(standardize_controls),
      model_control = dplyr::if_else(requested, model_control_names(control, standardize_controls), NA_character_),
      status = dplyr::case_when(
        !requested ~ "omitted_from_spec",
        vapply(model_control, function(ctrl) !is.na(ctrl) && any(startsWith(dropped, paste0("ctrl_", ctrl, "_"))), logical(1)) ~
          "dropped_by_fixest",
        TRUE ~ "included"
      )
    ) %>%
    dplyr::select(sample, spec, control, requested, standardize_controls, model_control, status)
}

estimate_one_spec <- function(data, sample_name, spec_name, controls, controls_label, config, standardize_controls = FALSE) {
  model_controls <- model_control_names(controls, standardize_controls)
  missing_model_controls <- setdiff(model_controls, names(data))
  if (length(missing_model_controls) > 0) {
    stop("Missing model control variables: ", paste(missing_model_controls, collapse = ", "))
  }
  prepared <- add_event_study_terms(data, model_controls, config)
  rhs_terms <- c(prepared$event_vars, prepared$control_vars)
  formula_chr <- paste0(config$outcome_var, " ~ ", paste(rhs_terms, collapse = " + "), " | czone + year")
  design_diag <- residualized_design_diagnostics(prepared$data, rhs_terms, config, sample_name, spec_name)

  tryCatch({
    message("Estimating ", sample_name, " / ", spec_name)
    model <- fixest::feols(
      fml = stats::as.formula(formula_chr),
      data = prepared$data,
      weights = stats::as.formula(paste0("~", config$weight_var)),
      panel.id = stats::as.formula("~czone + election_index"),
      warn = TRUE,
      notes = FALSE
    )
    vcovs <- compute_vcov_outputs(model, spec_name, sample_name, config)
    main_vcov_diag <- vcovs$diagnostics %>% dplyr::filter(se_type == config$main_se_type)
    main_vcov_failed <- nrow(main_vcov_diag) == 0 ||
      main_vcov_diag$status %in% c("failed", "repair_required") ||
      (isTRUE(config$require_no_vcov_repair_for_main) && isTRUE(main_vcov_diag$repair_required[[1]]))
    status <- if (main_vcov_failed) "failed" else "ok"
    error_message <- if (main_vcov_failed) {
      paste0("Main SE ", config$main_se_type, " failed or required VCOV repair.")
    } else {
      NA_character_
    }

    model_stats <- list(
      "Observations" = format(nrow(prepared$data), big.mark = ","),
      "Commuting zones" = format(dplyr::n_distinct(prepared$data$czone), big.mark = ","),
      "Election years" = paste0(min(prepared$data$year), "--", max(prepared$data$year)),
      "Reference election year" = as.character(config$reference_year),
      "CZ fixed effects" = "Yes",
      "Election-year fixed effects" = "Yes",
      "Controls" = controls_label,
      "Controls standardized" = if (isTRUE(standardize_controls)) "Yes" else "No",
      "Sample" = sample_name,
      "ADH weights" = "timepwt48",
      "Main SE type" = config$main_se_type
    )

```

```
    list(
      sample = sample_name, spec = spec_name, status = status, error_message = error_message,
      formula_chr = formula_chr, model = model, vcov = vcovs$vcov,
      coefficients = vcovs$coefficients, vcov_diagnostics = vcovs$diagnostics,
      vcov_eigenvalues = vcovs$eigenvalues, model_diagnostics = design_diag,
      interacted_control_variance = interacted_control_variance_diagnostics(prepared$data, prepared$control_vars, sample_name, spec_name),
      controls_manifest = controls_manifest_for_spec(sample_name, spec_name, controls, model_controls, standardize_controls, config,
      ↪  model),
      model_stats = model_stats, n_obs = nrow(prepared$data),
      n_cz = dplyr::n_distinct(prepared$data$czone), min_year = min(prepared$data$year),
      max_year = max(prepared$data$year)
    )
  }, error = function(e) {
    list(
      sample = sample_name, spec = spec_name, status = "failed",
      error_message = conditionMessage(e), formula_chr = formula_chr,
      model = NULL, vcov = list(), coefficients = tibble::tibble(),
      vcov_diagnostics = tibble::tibble(), vcov_eigenvalues = tibble::tibble(),
      model_diagnostics = design_diag, interacted_control_variance = tibble::tibble(), controls_manifest = tibble::tibble(),
      model_stats = list(
        "Observations" = format(nrow(prepared$data), big.mark = ","),
        "Commuting zones" = format(dplyr::n_distinct(prepared$data$czone), big.mark = ","),
        "Election years" = paste0(min(prepared$data$year), "--", max(prepared$data$year)),
        "Reference election year" = as.character(config$reference_year),
        "CZ fixed effects" = "Yes",
        "Election-year fixed effects" = "Yes",
        "Controls" = controls_label,
        "Controls standardized" = if (isTRUE(standardize_controls)) "Yes" else "No",
        "Sample" = sample_name,
        "ADH weights" = "timepwt48",
        "Main SE type" = "Failed"
      ),
      n_obs = nrow(prepared$data), n_cz = dplyr::n_distinct(prepared$data$czone),
      min_year = min(prepared$data$year), max_year = max(prepared$data$year)
    )
  })
}

run_pretrend_tests <- function(results, config) {
  purrr::map_dfr(results, function(res) {
    if (is.null(res$model) || res$sample != "main_1972_start") return(tibble::tibble())
    main_vcov <- res$vcov[[config$main_se_type]]
    if (is.null(main_vcov)) return(tibble::tibble())
    pre_years <- sort(unique(res$coefficients$year[res$coefficients$year < config$reference_year]))
    if (length(pre_years) == 0) return(tibble::tibble())
    keep_pattern <- paste0("^es_(", paste(pre_years, collapse = "|"), ")$")
    wt <- tryCatch(
      fixest::wald(res$model, keep = keep_pattern, vcov = main_vcov, print = FALSE),
      error = function(e) e
    )
    if (inherits(wt, "error")) {
      tibble::tibble(
        sample = res$sample, spec = res$spec, se_type = config$main_se_type,
        pre_years = paste(pre_years, collapse = ","),
        statistic = NA_real_, p.value = NA_real_, df1 = NA_real_, df2 = NA_real_,
        status = "failed", error_message = conditionMessage(wt)
      )
    } else {
      tibble::tibble(
        sample = res$sample, spec = res$spec, se_type = config$main_se_type,
        pre_years = paste(pre_years, collapse = ","),
        statistic = wt$stat, p.value = wt$p, df1 = wt$df1, df2 = wt$df2,
        status = "ok", error_message = NA_character_
      )
    }
  })
}

estimate_event_study <- function(config = CONFIG, stop_on_fatal = TRUE) {
  load_required_packages(config)
  config <- finalize_config(config)

  existing_checks <- if (file.exists(config$validation_checks_csv)) {
    readr::read_csv(config$validation_checks_csv, show_col_types = FALSE)
  } else {
    tibble::tibble()
  }
  if (nrow(existing_checks) > 0 && has_fatal_failures(existing_checks) && isTRUE(stop_on_fatal)) {
    stop("Fatal build-stage validation checks are present. Rebuild with a nonfatal policy before estimating.", call. = FALSE)
  }

  main_panel <- readRDS(config$analysis_panel_rds)
  diagnostic_panel <- readRDS(config$diagnostic_panel_rds)

  spec_definitions <- list(
    minimal_full_panel = list(
      controls = character(0),
      standardize_controls = FALSE,
      label = "None"
    ),
```

```
  interacted_core_controls_diagnostic = list(
    controls = config$core_interacted_controls,
    standardize_controls = isTRUE(config$standardize_interacted_controls),
    label = "Three 1990 ADH controls interacted with election-year indicators"
  ),
  interacted_controls_full_panel = list(
    controls = config$interacted_controls,
    standardize_controls = isTRUE(config$standardize_interacted_controls),
    label = "All configured 1990 ADH baseline controls interacted with election-year indicators"
  ),
  interacted_controls_full_panel_raw_controls_diagnostic = list(
    controls = config$interacted_controls,
    standardize_controls = FALSE,
    label = "All configured 1990 ADH baseline controls interacted with election-year indicators (raw-control diagnostic)"
  )
)

samples <- list(
  main_1972_start = main_panel,
  diagnostic_1952_start = diagnostic_panel
)

all_results <- list()
for (sample_name in names(samples)) {
  for (spec_name in names(spec_definitions)) {
    def <- spec_definitions[[spec_name]]
    key <- paste(sample_name, spec_name, sep = "__")
    all_results[[key]] <- estimate_one_spec(
      data = samples[[sample_name]],
      sample_name = sample_name,
      spec_name = spec_name,
      controls = def$controls,
      controls_label = def$label,
      config = config,
      standardize_controls = isTRUE(def$standardize_controls)
    )
  }
}

status_tbl <- purrr::map_dfr(all_results, ~ tibble::tibble(
  sample = .x$sample, spec = .x$spec, status = .x$status, error_message = .x$error_message,
  n_obs = .x$n_obs, n_cz = .x$n_cz, min_year = .x$min_year, max_year = .x$max_year,
  formula_chr = .x$formula_chr
))
coef_tbl_all <- purrr::map_dfr(all_results, ~ .x$coefficients)
model_diagnostics <- purrr::map_dfr(all_results, ~ {
  dropped <- if (is.null(.x$model)) character(0) else (.x$model$collin.var %||% character(0))
  .x$model_diagnostics %>%
    dplyr::mutate(
      fixest_dropped_variables = paste(dropped, collapse = "; "),
      fixest_dropped_count = length(dropped)
    )
})
controls_manifest <- purrr::map_dfr(all_results, ~ .x$controls_manifest)
interacted_control_variance <- purrr::map_dfr(all_results, ~ .x$interacted_control_variance %||% tibble::tibble())
vcov_diagnostics <- purrr::map_dfr(all_results, ~ .x$vcov_diagnostics)
vcov_eigenvalues <- purrr::map_dfr(all_results, ~ .x$vcov_eigenvalues)
vcov_rank_diagnostics <- purrr::imap_dfr(all_results, function(res, nm) {
  if (is.null(res$vcov) || length(res$vcov) == 0) return(tibble::tibble())
  purrr::imap_dfr(res$vcov, ~ vcov_rank_summary(.x, res$sample, res$spec, .y))
})
se_warning_diagnostics <- coef_tbl_all %>%
  dplyr::group_by(sample, spec, se_type) %>%
  dplyr::summarise(
    n_event_coefficients = dplyr::n(),
    min_std_error = min(std.error, na.rm = TRUE),
    median_std_error = stats::median(std.error, na.rm = TRUE),
    max_std_error = max(std.error, na.rm = TRUE),
    near_zero_se_count = sum(abs(std.error) < 1e-10, na.rm = TRUE),
    nonfinite_se_count = sum(!is.finite(std.error)),
    .groups = "drop"
  ) %>%
  dplyr::left_join(
    vcov_rank_diagnostics %>%
      dplyr::select(sample, spec, se_type, vcov_dim, vcov_rank, vcov_rank_deficiency),
    by = c("sample", "spec", "se_type")
  ) %>%
  dplyr::left_join(
    vcov_diagnostics %>%
      dplyr::select(sample, spec, se_type, vcov_status = status, repair_required, vcov_repaired, min_eigenvalue),
    by = c("sample", "spec", "se_type")
  ) %>%
  dplyr::mutate(
    diagnostic_note = dplyr::case_when(
      grepl("driscoll", se_type) & near_zero_se_count > 0 & dplyr::coalesce(vcov_rank_deficiency, 0L) > 0 ~
        "Suspicious near-zero Driscoll-Kraay SEs and rank-deficient VCOV; do not report without further diagnosis.",
      grepl("driscoll", se_type) & near_zero_se_count > 0 ~
        "Suspicious near-zero Driscoll-Kraay SEs; do not report without further diagnosis.",
      grepl("conley", se_type) & (dplyr::coalesce(repair_required, FALSE) | dplyr::coalesce(vcov_repaired, FALSE)) ~
        "Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.",
```

```
      dplyr::coalesce(vcov_repaired, FALSE) ~
        "VCOV required numerical repair; diagnostic only unless separately justified.",
      dplyr::coalesce(vcov_rank_deficiency, 0L) > 0 ~
        "VCOV is rank deficient; use as diagnostic only unless justified.",
      grepl("cluster_cz_year", se_type) ~
        "Two-way clustering includes very few year clusters; use as diagnostic only.",
      TRUE ~ NA_character_
    )
  )
pretrend_tests <- run_pretrend_tests(all_results, config)
pretrend_coefficients <- coef_tbl_all %>%
  dplyr::filter(sample == "main_1972_start", se_type == config$main_se_type, year < config$reference_year)

rank_failures <- model_diagnostics %>%
  dplyr::filter(residualized_rank_deficiency > 0)
main_vcov_problem_count <- vcov_diagnostics %>%
  dplyr::filter(
    sample == "main_1972_start",
    spec %in% c("minimal_full_panel", "interacted_controls_full_panel"),
    se_type == config$main_se_type,
    !status %in% c("ok", "repaired")
  ) %>%
  nrow()
main_vcov_repair_count <- vcov_diagnostics %>%
  dplyr::filter(
    sample == "main_1972_start",
    spec %in% c("minimal_full_panel", "interacted_controls_full_panel"),
    se_type == config$main_se_type,
    repair_required %in% TRUE
  ) %>%
  nrow()
conley_repair_count <- vcov_diagnostics %>%
  dplyr::filter(grepl("^conley_", se_type), repair_required %in% TRUE) %>%
  nrow()

build_checks <- if (file.exists(config$validation_checks_csv)) {
  readr::read_csv(config$validation_checks_csv, show_col_types = FALSE)
} else {
  tibble::tibble()
}
estimation_checks <- dplyr::bind_rows(
  validation_check(
    "event_study_residualized_design_full_rank",
    nrow(rank_failures) == 0,
    fatal = TRUE,
    n_failed = nrow(rank_failures),
    details = "Requires no residualized RHS rank deficiency in 1952-start or 1972-start specifications."
  ),
  validation_check(
    "event_study_main_vcov_available_without_repair",
    main_vcov_problem_count == 0 &&
      (!isTRUE(config$require_no_vcov_repair_for_main) || main_vcov_repair_count == 0),
    fatal = TRUE,
    n_failed = main_vcov_problem_count + main_vcov_repair_count,
    details = paste0("Main SE type: ", config$main_se_type)
  ),
  validation_check(
    "event_study_conley_repair_diagnostics_recorded",
    TRUE,
    fatal = FALSE,
    n_failed = conley_repair_count,
    details = paste0("Conley VCOVs requiring repair across all samples/specs/cutoffs: ", conley_repair_count)
  )
)
combined_checks <- dplyr::bind_rows(build_checks, estimation_checks)
readr::write_csv(combined_checks, config$validation_checks_csv)

write_model_object(all_results, config$all_specs_rds, config)
readr::write_csv(status_tbl, config$spec_status_csv)
readr::write_csv(coef_tbl_all, config$all_specs_csv)
readr::write_csv(model_diagnostics, config$model_diagnostics_csv)
readr::write_csv(controls_manifest, config$controls_manifest_csv)
readr::write_csv(interacted_control_variance, config$interacted_control_variance_csv)
readr::write_csv(vcov_diagnostics, config$vcov_diagnostics_csv)
readr::write_csv(vcov_eigenvalues, config$vcov_eigenvalues_csv)
readr::write_csv(vcov_rank_diagnostics, config$vcov_rank_csv)
readr::write_csv(se_warning_diagnostics, config$se_warning_diagnostics_csv)
readr::write_csv(summarise_conley_neighbors(main_panel, config$conley_cutoffs_km), config$conley_neighbor_counts_csv)
readr::write_csv(pretrend_tests, config$pretrend_tests_csv)
readr::write_csv(pretrend_coefficients, config$pretrend_coefficients_csv)

main_display_specs <- c("minimal_full_panel", "interacted_core_controls_diagnostic", "interacted_controls_full_panel")
main_coefficients <- coef_tbl_all %>%
  dplyr::filter(
    sample == "main_1972_start",
    se_type == config$main_se_type,
    spec %in% main_display_specs
  ) %>%
  dplyr::arrange(factor(spec, levels = main_display_specs), year)
main_models <- all_results[grepl("^main_1972_start__", names(all_results))]
```

```
  main_out <- list(
    coefficients = main_coefficients,
    main_se_type = config$main_se_type,
    display_specs = main_display_specs,
    models = main_models
  )
  write_model_object(main_out, config$event_study_rds, config)
  readr::write_csv(main_coefficients, config$event_study_csv)

  failed_main_vcov <- vcov_diagnostics %>%
    dplyr::filter(
      sample == "main_1972_start",
      spec %in% c("minimal_full_panel", "interacted_controls_full_panel"),
      se_type == config$main_se_type,
      !status %in% c("ok", "repaired")
    )

  write_pipeline_manifest(
    config = config,
    checks = combined_checks,
    stage = "estimate_event_study",
    sources = list(
      analysis_panel = list(path = config$analysis_panel_rds, md5 = file_checksum(config$analysis_panel_rds)),
      diagnostic_panel = list(path = config$diagnostic_panel_rds, md5 = file_checksum(config$diagnostic_panel_rds))
    ),
    extra = list(
      specs = status_tbl,
      conley_repair_count = conley_repair_count,
      main_vcov_problem_count = main_vcov_problem_count,
      main_vcov_repair_count = main_vcov_repair_count
    )
  )

  if (nrow(failed_main_vcov) > 0 && isTRUE(stop_on_fatal) && !isTRUE(config$allow_failed_diagnostics)) {
    stop(
      "Main event-study VCOV failed or required repair. See ",
      config$vcov_diagnostics_csv,
      call. = FALSE
    )
  }

  if (has_fatal_failures(combined_checks) && isTRUE(stop_on_fatal) && !isTRUE(config$allow_failed_diagnostics)) {
    stop("Fatal event-study validation checks failed. See ", config$validation_checks_csv, call. = FALSE)
  }

  invisible(list(config = config, results = all_results, status = status_tbl, coefficients = coef_tbl_all))
}
```

5 04_export_outputs.R

```
# extension-adh2013/src/04_export_outputs.R

export_event_study_outputs <- function(config = CONFIG) {
  load_required_packages(config)
  config <- finalize_config(config)

  checks <- if (file.exists(config$validation_checks_csv)) {
    readr::read_csv(config$validation_checks_csv, show_col_types = FALSE)
  } else {
    tibble::tibble()
  }
  if (nrow(checks) > 0 && has_fatal_failures(checks) && !isTRUE(config$allow_failed_diagnostics)) {
    stop("Refusing to export figures/tables because fatal validation checks are present.", call. = FALSE)
  }
  has_validation_failures <- nrow(checks) > 0 && has_any_failures(checks)
  has_reported_nonfatal_conditions <- nrow(checks) > 0 && has_reported_conditions(checks)

  res_all <- read_model_object(config$all_specs_rds)
  spec_order <- c("minimal_full_panel", "interacted_core_controls_diagnostic", "interacted_controls_full_panel")
  pretty_spec_names <- c(
    minimal_full_panel = "Minimal",
    interacted_core_controls_diagnostic = "Core controls",
    interacted_controls_full_panel = "Full controls"
  )

  ok_keys <- names(res_all)[vapply(res_all, function(x) {
    x$sample == "main_1972_start" && x$spec %in% spec_order && x$status == "ok"
  }, logical(1))]
  if (length(ok_keys) == 0) stop("No successful main specifications found in ", config$all_specs_rds)

  coef_tbl_all <- purrr::map_dfr(ok_keys, function(k) {
    res_all[[k]]$coefficients %>%
      dplyr::filter(se_type == config$main_se_type) %>%
      dplyr::mutate(spec_label = pretty_spec_names[spec])
  })

  table_csv <- coef_tbl_all %>%
    dplyr::transmute(
      sample, spec, spec_label, se_type, year, term, estimate, std.error,
      conf.low, conf.high, statistic, p.value, vcov_repair_required, vcov_repaired
    ) %>%
    dplyr::arrange(factor(spec, levels = spec_order), year)
  readr::write_csv(table_csv, config$table_csv)
  uses_repaired_main_vcov <- any(table_csv$vcov_repaired %in% TRUE, na.rm = TRUE)

  event_years_display <- sort(unique(c(coef_tbl_all$year, config$reference_year)))
  coef_cell <- function(df, year_value) {
    row <- df %>% dplyr::filter(year == year_value)
    if (year_value == config$reference_year) return("--")
    if (nrow(row) == 0) return("")
    paste0(fmt_num(row$estimate[[1]]), star_code(row$p.value[[1]]), " (", fmt_num(row$std.error[[1]]), ")")
  }

  table_body <- tibble::tibble(`Election year` = as.character(event_years_display))
  for (s in spec_order) {
    df_s <- coef_tbl_all %>% dplyr::filter(spec == s)
    if (nrow(df_s) > 0) {
      table_body[[pretty_spec_names[[s]]]] <- vapply(event_years_display, function(y) {
        coef_cell(df_s, y)
      }, character(1))
    }
  }

  stat_names <- c(
    "Observations", "Commuting zones", "Election years", "Reference election year",
    "CZ fixed effects", "Election-year fixed effects", "Controls", "Sample",
    "ADH weights", "Controls standardized", "Main SE type"
  )
  get_stat_value <- function(ms, nm) {
    if (is.null(ms) || is.null(ms[[nm]]) || length(ms[[nm]]) == 0) return("")
    as.character(ms[[nm]][1])
  }
  stat_rows <- tibble::tibble(`Election year` = stat_names)
  for (s in spec_order) {
    key <- ok_keys[vapply(ok_keys, function(k) res_all[[k]]$spec == s, logical(1))]
    if (length(key) > 0) {
      vals <- vapply(stat_names, function(nm) get_stat_value(res_all[[key[[1]]]]$model_stats, nm), character(1))
      stat_rows[[pretty_spec_names[[s]]]] <- vals
    }
  }

  table_for_tex <- dplyr::bind_rows(table_body, stat_rows)
  fallback_label <- dplyr::case_when(
    config$crosswalk_missing_weight_policy == "fallback_m2" ~ "M2 fallback",
    config$crosswalk_missing_weight_policy == "fallback_m4" ~ "M4 fallback",
    config$crosswalk_missing_weight_policy == "identity_if_same_fips" ~ "same-FIPS identity fallback",
    TRUE ~ config$crosswalk_missing_weight_policy
```

```
)
crosswalk_note <- if (identical(config$crosswalk_missing_weight_policy, "fail")) {
  paste0(toupper(config$crosswalk_weight_slug), " weights with no fallback")
} else {
  paste0(
    toupper(config$crosswalk_weight_slug),
    " weights where valid, with ", fallback_label,
    " for source counties whose selected weights are undefined or invalid"
  )
}

se_label <- dplyr::case_when(
  config$main_se_type == "cluster_cz" ~ "CZ-clustered",
  config$main_se_type == "heteroskedastic" ~ "heteroskedastic",
  config$main_se_type == "cluster_state" ~ "state-clustered",
  config$main_se_type == "newey_west" ~ "Newey-West",
  TRUE ~ config$main_se_type
)

notes_text <- paste0(
  if (uses_repaired_main_vcov || has_validation_failures) {
    "PROVISIONAL: at least one fatal/non-handled validation check failed or a main numerical repair/warning was recorded; treat these as diagnostic output until reviewed. "
  } else {
    ""
  },
  if (has_reported_nonfatal_conditions) {
    "Handled nonfatal crosswalk-support conditions are reported in the diagnostics and do not block the ADH-mainland estimating sample. "
  } else {
    ""
  },
  "Each coefficient is the interaction between ADH's 1990-2007 import-exposure measure and the indicated presidential-election year, ",
  "with ", config$reference_year, " omitted as the reference year. Exposure is measured in ",
  config$exposure_units, ". The dependent variable is the commuting-zone Republican presidential vote margin, ",
  "defined as Republican minus Democratic votes divided by two-party votes. County presidential returns are from Amlani and Algara (2021), ",
  "bridged to 1990 counties with the Ferrara, Testa, and Zhou (2024) ",
  crosswalk_note, ", then aggregated to 1990 commuting zones. ",
  "Standard errors use ", se_label, ". Significance stars: *** p<0.01, ** p<0.05, * p<0.10."
)

latex_tbl <- knitr::kable(
  table_for_tex,
  format = "latex",
  booktabs = TRUE,
  longtable = TRUE,
  escape = TRUE,
  caption = paste0(
    if (uses_repaired_main_vcov || has_validation_failures) "Provisional " else "",
    "event-study estimates: ADH China exposure and Republican presidential vote margin"
  ),
  align = c("l", rep("c", ncol(table_for_tex) - 1))
) %>%
  kableExtra::kable_styling(
    latex_options = c("repeat_header", "hold_position"),
    position = "center",
    font_size = 9
  ) %>%
  kableExtra::footnote(general = notes_text, threeparttable = TRUE, escape = TRUE)

writeLines(as.character(latex_tbl), config$table_tex)
render_table_pdf(config$table_tex, config$table_standalone_tex, config$table_pdf)

plot_tbl <- coef_tbl_all %>%
  dplyr::mutate(
    spec_label = factor(spec_label, levels = c("Minimal", "Core controls", "Full controls")),
    line_group = ifelse(year < config$reference_year, "pre", "post")
  )
ref_rows <- plot_tbl %>%
  dplyr::distinct(sample, spec, se_type, spec_label) %>%
  dplyr::mutate(
    term = paste0("es_", config$reference_year),
    year = config$reference_year,
    estimate = 0,
    std.error = NA_real_,
    conf.low = 0,
    conf.high = 0,
    statistic = NA_real_,
    p.value = NA_real_,
    vcov_repair_required = FALSE,
    vcov_repaired = FALSE,
    line_group = "reference"
  )
plot_with_ref <- dplyr::bind_rows(plot_tbl, ref_rows)
line_tbl <- dplyr::bind_rows(
  plot_with_ref %>%
    dplyr::filter(year <= config$reference_year) %>%
    dplyr::mutate(line_group = "pre"),
  plot_with_ref %>%
```

```
      dplyr::filter(year >= config$reference_year) %>%
      dplyr::mutate(line_group = "post")
  )

  period_bands <- tibble::tibble(
    xmin = c(1990, 2008, 2016, 2020),
    xmax = c(2007, 2012, 2019.8, 2020.8),
    label = c("ADH exposure", "Great Recession / Obama", "Trump era", "COVID era"),
    fill = c("#9ecae1", "#fdd0a2", "#c7e9c0", "#dadaeb")
  )

  event_plot <- ggplot2::ggplot() +
    ggplot2::geom_rect(
      data = period_bands,
      ggplot2::aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf, fill = label),
      alpha = 0.16,
      inherit.aes = FALSE
    ) +
    ggplot2::geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.35) +
    ggplot2::geom_vline(xintercept = config$reference_year, linetype = "dotted", linewidth = 0.35) +
    ggplot2::geom_errorbar(
      data = plot_tbl,
      ggplot2::aes(x = year, ymin = conf.low, ymax = conf.high),
      width = 0.6,
      linewidth = 0.35
    ) +
    ggplot2::geom_line(
      data = line_tbl,
      ggplot2::aes(x = year, y = estimate, group = interaction(spec_label, line_group)),
      linewidth = 0.4
    ) +
    ggplot2::geom_point(
      data = plot_with_ref,
      ggplot2::aes(x = year, y = estimate),
      size = 1.8
    ) +
    ggplot2::facet_wrap(~ spec_label, ncol = 1) +
    ggplot2::scale_fill_manual(values = stats::setNames(period_bands$fill, period_bands$label), name = NULL) +
    ggplot2::scale_x_continuous(
      breaks = seq(min(plot_with_ref$year), max(plot_with_ref$year), by = 8),
      minor_breaks = seq(min(plot_with_ref$year), max(plot_with_ref$year), by = 4)
    ) +
    ggplot2::labs(
      title = "China exposure and Republican presidential vote margin, 1972-2020",
      subtitle = paste0("Event-study coefficients with 95% ", se_label, " CIs; ", config$reference_year, " normalized to zero."),
      x = "Presidential election year",
      y = paste0("Coefficient on ADH China exposure (per ", config$exposure_units, ")")
    ) +
    ggplot2::theme_bw(base_size = 11) +
    ggplot2::theme(
      plot.title = ggplot2::element_text(face = "bold"),
      panel.grid.minor = ggplot2::element_blank(),
      panel.grid.major.x = ggplot2::element_blank(),
      legend.position = "bottom"
    )

  ggplot2::ggsave(filename = config$figure_pdf, plot = event_plot, width = 8.5, height = 7.5, bg = "white")
  ggplot2::ggsave(filename = config$figure_png, plot = event_plot, width = 8.5, height = 7.5, dpi = 320, bg = "white")

  output_inventory <- write_output_manifest(config)

  write_pipeline_manifest(
    config = config,
    checks = checks,
    stage = "export_event_study_outputs",
    sources = list(
      all_specs = model_object_reference(config$all_specs_rds, config)
    ),
    extra = list(
      outputs = list(
        table_csv = list(path = config$table_csv, md5 = file_checksum(config$table_csv)),
        table_tex = list(path = config$table_tex, md5 = file_checksum(config$table_tex)),
        table_pdf = list(path = config$table_pdf, md5 = file_checksum(config$table_pdf)),
        figure_pdf = list(path = config$figure_pdf, md5 = file_checksum(config$figure_pdf)),
        figure_png = list(path = config$figure_png, md5 = file_checksum(config$figure_png)),
        output_manifest_csv = list(path = config$output_manifest_csv, md5 = file_checksum(config$output_manifest_csv)),
        output_manifest_json = list(path = config$output_manifest_json, md5 = file_checksum(config$output_manifest_json))
      ),
      output_manifest_n_files = nrow(output_inventory),
      validation_failures = has_validation_failures,
      reported_nonfatal_conditions = has_reported_nonfatal_conditions
    )
  )

  invisible(list(table_csv = table_csv, table_tex = config$table_tex, figure = event_plot))
}
```

6 05_export_diagnostic_maps.R

```
# extension-adh2013/src/05_export_diagnostic_maps.R

crosswalk_sensitivity_plan <- function() {
  tibble::tribble(
    ~crosswalk_weight, ~crosswalk_missing_weight_policy, ~label, ~render_support_maps,
    "m5_weight", "fallback_m2", "M5 + fallback M2", TRUE,
    "m6_weight", "fallback_m2", "M6 + fallback M2", FALSE,
    "m2_weight", "fallback_m2", "Pure M2", FALSE,
    "m4_weight", "fallback_m2", "M4 + fallback M2", FALSE
  )
}

label_from_crosswalk_slug <- function(slug) {
  dplyr::case_when(
    slug == "m5" ~ "M5 + fallback M2",
    slug == "m6" ~ "M6 + fallback M2",
    slug == "m2" ~ "Pure M2",
    slug == "m4" ~ "M4 + fallback M2",
    TRUE ~ toupper(slug)
  )
}

export_crosswalk_specification_comparison <- function(config = CONFIG) {
  load_required_packages(config)
  config <- finalize_config(config)

  coef_files <- list.files(
    config$intermediate_dir,
    pattern = "^event_study_coefficients_all_specs_aa2021_rep_margin_.*\\.csv$",
    full.names = TRUE
  )
  if (length(coef_files) == 0) {
    warning("No all-spec coefficient CSVs found; skipping crosswalk specification comparison plot.", call. = FALSE)
    return(invisible(NULL))
  }

  crosswalk_levels <- c("M5 + fallback M2", "M6 + fallback M2", "Pure M2", "M4 + fallback M2")
  display_specs <- c("minimal_full_panel", "interacted_core_controls_diagnostic", "interacted_controls_full_panel")

  coef_long <- purrr::map_dfr(coef_files, function(path) {
    slug <- sub("^event_study_coefficients_all_specs_aa2021_rep_margin_", "", basename(path))
    slug <- sub("\\.csv$", "", slug)
    readr::read_csv(path, show_col_types = FALSE) %>%
      dplyr::mutate(
        crosswalk_specification = label_from_crosswalk_slug(slug),
        crosswalk_slug = slug
      )
  }) %>%
    dplyr::filter(
      sample == "main_1972_start",
      se_type == config$main_se_type,
      spec %in% display_specs
    ) %>%
    dplyr::mutate(
      spec_label = dplyr::recode(
        spec,
        minimal_full_panel = "Minimal",
        interacted_core_controls_diagnostic = "Core controls",
        interacted_controls_full_panel = "Full controls"
      ),
      spec_label = factor(spec_label, levels = c("Minimal", "Core controls", "Full controls")),
      crosswalk_specification = factor(crosswalk_specification, levels = crosswalk_levels)
    ) %>%
    dplyr::filter(!is.na(crosswalk_specification))

  if (nrow(coef_long) == 0) {
    warning("Crosswalk comparison inputs exist but no matching main-spec coefficients were found.", call. = FALSE)
    return(invisible(NULL))
  }

  # Draw thicker baseline/early layers first and thinner layers last. When the
  # four crosswalk estimates are nearly identical, this makes the overlap show
  # up as concentric colored strokes rather than one hidden line.
  draw_order_tbl <- tibble::tibble(
    crosswalk_specification = factor(crosswalk_levels, levels = crosswalk_levels),
    draw_order = c(4L, 3L, 1L, 2L),
    line_width = c(0.75, 0.95, 1.45, 1.20),
    point_size = c(1.25, 1.45, 2.05, 1.75),
    alpha_value = c(0.95, 0.90, 0.70, 0.82)
  )

  coef_long <- coef_long %>%
    dplyr::left_join(draw_order_tbl, by = "crosswalk_specification")

  readr::write_csv(coef_long, config$crosswalk_comparison_csv)

  n_specs <- dplyr::n_distinct(coef_long$crosswalk_specification)
```

```
plot_title <- if (n_specs > 1) {
  "Crosswalk-specification sensitivity"
} else {
  paste0("Event-study coefficients: ", as.character(unique(coef_long$crosswalk_specification)))
}
plot_subtitle <- if (n_specs > 1) {
  paste0("Main 1972-2020 event-study coefficients; colored/width-varied lines reveal nearly overlapping estimates. SE type = ",
    ↪ config$main_se_type)
} else {
  paste0("Only one crosswalk specification was found; rerun sensitivity mode to add M6, M2, and M4. SE type = ", config$main_se_type)
}

# Colors are intentionally specified here because the plot is meant for a
# final appendix figure where four nearly identical series must be visually
# distinguishable.
comparison_colors <- c(
  "M5 + fallback M2" = "#1b9e77",
  "M6 + fallback M2" = "#d95f02",
  "Pure M2" = "#7570b3",
  "M4 + fallback M2" = "#e7298a"
)
comparison_linetypes <- c(
  "M5 + fallback M2" = "solid",
  "M6 + fallback M2" = "22",
  "Pure M2" = "longdash",
  "M4 + fallback M2" = "dotdash"
)
comparison_linewidths <- c(
  "M5 + fallback M2" = 0.75,
  "M6 + fallback M2" = 0.95,
  "Pure M2" = 1.45,
  "M4 + fallback M2" = 1.20
)
comparison_alphas <- c(
  "M5 + fallback M2" = 0.95,
  "M6 + fallback M2" = 0.90,
  "Pure M2" = 0.70,
  "M4 + fallback M2" = 0.82
)
comparison_shapes <- c(
  "M5 + fallback M2" = 16,
  "M6 + fallback M2" = 17,
  "Pure M2" = 15,
  "M4 + fallback M2" = 18
)

coef_plot_data <- coef_long %>%
  dplyr::arrange(spec_label, draw_order, year)

comp_plot <- ggplot2::ggplot(
  coef_plot_data,
  ggplot2::aes(
    x = year, y = estimate,
    color = crosswalk_specification,
    linetype = crosswalk_specification,
    linewidth = crosswalk_specification,
    alpha = crosswalk_specification,
    group = crosswalk_specification
  )
) +
  ggplot2::geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.3, color = "grey45") +
  ggplot2::geom_line(lineend = "round") +
  ggplot2::geom_point(ggplot2::aes(shape = crosswalk_specification, size = crosswalk_specification), stroke = 0.2) +
  ggplot2::scale_color_manual(values = comparison_colors, drop = FALSE) +
  ggplot2::scale_linetype_manual(values = comparison_linetypes, drop = FALSE) +
  ggplot2::scale_linewidth_manual(values = comparison_linewidths, drop = FALSE, guide = "none") +
  ggplot2::scale_alpha_manual(values = comparison_alphas, drop = FALSE, guide = "none") +
  ggplot2::scale_shape_manual(values = comparison_shapes, drop = FALSE) +
  ggplot2::scale_size_manual(values = c(
    "M5 + fallback M2" = 1.25,
    "M6 + fallback M2" = 1.45,
    "Pure M2" = 2.05,
    "M4 + fallback M2" = 1.75
  ), drop = FALSE, guide = "none") +
  ggplot2::facet_wrap(~ spec_label, ncol = 1) +
  ggplot2::labs(
    title = plot_title,
    subtitle = plot_subtitle,
    x = "Presidential election year",
    y = paste0("Coefficient on ADH China exposure (per ", config$exposure_units, ")"),
    color = "Crosswalk",
    linetype = "Crosswalk",
    shape = "Crosswalk"
  ) +
  ggplot2::theme_bw(base_size = 11) +
  ggplot2::theme(legend.position = "bottom", panel.grid.minor = ggplot2::element_blank())

ggplot2::ggsave(config$crosswalk_comparison_pdf, comp_plot, width = 8.5, height = 7.5, bg = "white")
ggplot2::ggsave(config$crosswalk_comparison_png, comp_plot, width = 8.5, height = 7.5, dpi = 320, bg = "white")
```

```
  # Appendix companion: differences relative to Pure M2. This makes the actual
  # magnitude of crosswalk sensitivity visible when the level plot overlaps.
  pure_m2 <- coef_long %>%
    dplyr::filter(crosswalk_specification == "Pure M2") %>%
    dplyr::select(spec_label, year, pure_m2_estimate = estimate)

  delta_long <- coef_long %>%
    dplyr::left_join(pure_m2, by = c("spec_label", "year")) %>%
    dplyr::mutate(delta_vs_pure_m2 = estimate - pure_m2_estimate) %>%
    dplyr::arrange(spec_label, draw_order, year)

  readr::write_csv(delta_long, config$crosswalk_comparison_delta_csv)

  if (n_specs > 1 && any(is.finite(delta_long$delta_vs_pure_m2))) {
    delta_plot <- ggplot2::ggplot(
      delta_long,
      ggplot2::aes(
        x = year, y = delta_vs_pure_m2,
        color = crosswalk_specification,
        linetype = crosswalk_specification,
        linewidth = crosswalk_specification,
        alpha = crosswalk_specification,
        group = crosswalk_specification
      )
    ) +
      ggplot2::geom_hline(yintercept = 0, linetype = "dashed", linewidth = 0.3, color = "grey45") +
      ggplot2::geom_line(lineend = "round") +
      ggplot2::geom_point(ggplot2::aes(shape = crosswalk_specification, size = crosswalk_specification), stroke = 0.2) +
      ggplot2::scale_color_manual(values = comparison_colors, drop = FALSE) +
      ggplot2::scale_linetype_manual(values = comparison_linetypes, drop = FALSE) +
      ggplot2::scale_linewidth_manual(values = comparison_linewidths, drop = FALSE, guide = "none") +
      ggplot2::scale_alpha_manual(values = comparison_alphas, drop = FALSE, guide = "none") +
      ggplot2::scale_shape_manual(values = comparison_shapes, drop = FALSE) +
      ggplot2::scale_size_manual(values = c(
        "M5 + fallback M2" = 1.25,
        "M6 + fallback M2" = 1.45,
        "Pure M2" = 2.05,
        "M4 + fallback M2" = 1.75
      ), drop = FALSE, guide = "none") +
      ggplot2::facet_wrap(~ spec_label, ncol = 1, scales = "free_y") +
      ggplot2::labs(
        title = "Crosswalk-specification sensitivity relative to Pure M2",
        subtitle = paste0("Main 1972-2020 event-study coefficients; SE type = ", config$main_se_type),
        x = "Presidential election year",
        y = paste0("Coefficient difference vs. Pure M2 (per ", config$exposure_units, ")"),
        color = "Crosswalk",
        linetype = "Crosswalk",
        shape = "Crosswalk"
      ) +
      ggplot2::theme_bw(base_size = 11) +
      ggplot2::theme(legend.position = "bottom", panel.grid.minor = ggplot2::element_blank())

    ggplot2::ggsave(config$crosswalk_comparison_delta_pdf, delta_plot, width = 8.5, height = 7.5, bg = "white")
    ggplot2::ggsave(config$crosswalk_comparison_delta_png, delta_plot, width = 8.5, height = 7.5, dpi = 320, bg = "white")
  }

  invisible(comp_plot)
}

simplify_sf_for_plot <- function(sf_obj, tolerance = 0.03) {
  if (is.null(sf_obj)) return(sf_obj)
  out <- tryCatch(
    sf::st_simplify(sf_obj, dTolerance = tolerance, preserveTopology = TRUE),
    error = function(e) sf_obj
  )
  out
}

save_crosswalk_support_plot_pair <- function(plot, simplified_plot, pdf_path, png_path, simplified_pdf_path, simplified_png_path,
 ↪  small_png_path = NULL) {
  ggplot2::ggsave(pdf_path, plot, width = 10, height = 7.5, bg = "white")
  ggplot2::ggsave(png_path, plot, width = 10, height = 7.5, dpi = 320, bg = "white")
  ggplot2::ggsave(simplified_pdf_path, simplified_plot, width = 10, height = 7.5, bg = "white")
  ggplot2::ggsave(simplified_png_path, simplified_plot, width = 10, height = 7.5, dpi = 260, bg = "white")
  if (!is.null(small_png_path)) {
    ggplot2::ggsave(small_png_path, simplified_plot, width = 10, height = 7.5, dpi = 160, bg = "white")
  }
}

make_support_map_plot <- function(map_data, title, subtitle, caption, boundary_color = NA) {
  ggplot2::ggplot(map_data) +
    ggplot2::geom_sf(ggplot2::aes(fill = map_fill), color = boundary_color, linewidth = 0.05) +
    ggplot2::facet_wrap(~ weight_type, ncol = 2) +
    ggplot2::scale_fill_manual(
      values = c(
        "No positive affected vote mass" = "grey90",
        "Receives positive affected vote mass" = "grey25"
      ),
      name = NULL
    ) +
```

```
    ggplot2::labs(title = title, subtitle = subtitle, caption = caption) +
    ggplot2::theme_void(base_size = 10) +
    ggplot2::theme(legend.position = "bottom", plot.title = ggplot2::element_text(face = "bold"))
}

export_crosswalk_support_maps <- function(config = CONFIG) {
  load_required_packages(config)
  config <- finalize_config(config)
  if (!isTRUE(config$export_crosswalk_maps)) return(invisible(NULL))

  if (!file.exists(config$target_weight_support_county_csv) || !file.exists(config$target_weight_support_cz_csv)) {
    warning("Crosswalk support diagnostics are missing; skipping support maps.", call. = FALSE)
    return(invisible(NULL))
  }
  if (!requireNamespace("sf", quietly = TRUE)) {
    warning("Package 'sf' is not installed; skipping support maps.", call. = FALSE)
    return(invisible(NULL))
  }

  county_sf <- safe_read_county1990_shapefile(config)
  if (is.null(county_sf)) return(invisible(NULL))

  county_diag <- readr::read_csv(config$target_weight_support_county_csv, show_col_types = FALSE) %>%
    dplyr::mutate(
      weight_type = factor(
        weight_type,
        levels = c("m2_weight", "m4_weight", "m5_weight", "m6_weight"),
        labels = c("M2 unavailable", "M4 unavailable", "M5 unavailable", "M6 unavailable")
      )
    )
  county_map <- county_sf %>%
    dplyr::left_join(county_diag, by = "county_fips_1990") %>%
    dplyr::filter(!is.na(weight_type)) %>%
    dplyr::mutate(
      receives_unavailable_source = dplyr::coalesce(receives_unavailable_source, FALSE),
      map_fill = dplyr::if_else(receives_unavailable_source, "Receives positive affected vote mass", "No positive affected vote mass")
    )

  county_plot <- make_support_map_plot(
    county_map,
    title = "ADH-mainland 1990 counties receiving affected bridged vote mass",
    subtitle = "Demonstration map: flags counties receiving positive vote mass from a source county where the indicated FTZ weight type
    ↪  has zero/undefined support.",
    caption = "Uses the selected/fallback crosswalk as the vote-mass allocator and the 1990 county shapefile under
    ↪  spatial-data/counties-1990 unless overridden.",
    boundary_color = NA
  )
  county_map_simple <- simplify_sf_for_plot(county_map, tolerance = 0.03)
  county_plot_simple <- make_support_map_plot(
    county_map_simple,
    title = "ADH-mainland 1990 counties receiving affected bridged vote mass",
    subtitle = "Simplified-geometry version for smaller appendix files.",
    caption = "Uses the selected/fallback crosswalk as the vote-mass allocator and simplified 1990 county geometry.",
    boundary_color = NA
  )
  save_crosswalk_support_plot_pair(
    county_plot, county_plot_simple,
    config$crosswalk_support_county_map_pdf, config$crosswalk_support_county_map_png,
    config$crosswalk_support_county_map_simplified_pdf, config$crosswalk_support_county_map_simplified_png,
    config$crosswalk_support_county_map_small_png
  )

  cz_lookup <- readxl::read_xls(file.path(config$replication_dir, "cz-data", "cz-198090.xls")) %>%
    dplyr::transmute(
      county_fips_1990 = stringr::str_pad(as.character(`County FIPS Code`), width = 5, side = "left", pad = "0"),
      czone = as.integer(CZ90)
    )

  cz_diag <- readr::read_csv(config$target_weight_support_cz_csv, show_col_types = FALSE) %>%
    dplyr::mutate(
      weight_type = factor(
        weight_type,
        levels = c("m2_weight", "m4_weight", "m5_weight", "m6_weight"),
        labels = c("M2 unavailable", "M4 unavailable", "M5 unavailable", "M6 unavailable")
      )
    )

  cz_sf <- county_sf %>%
    dplyr::inner_join(cz_lookup, by = "county_fips_1990") %>%
    dplyr::filter(czone %in% unique(cz_diag$czone)) %>%
    dplyr::group_by(czone) %>%
    dplyr::summarise(geometry = sf::st_union(geometry), .groups = "drop")

  cz_map <- cz_sf %>%
    dplyr::left_join(cz_diag, by = "czone") %>%
    dplyr::filter(!is.na(weight_type)) %>%
    dplyr::mutate(
      receives_unavailable_source = dplyr::coalesce(receives_unavailable_source, FALSE),
      map_fill = dplyr::if_else(receives_unavailable_source, "Receives positive affected vote mass", "No positive affected vote mass")
    )
```

```
  cz_plot <- make_support_map_plot(
    cz_map,
    title = "ADH-mainland 1990 CZs receiving affected bridged vote mass",
    subtitle = "Demonstration map: flags CZs receiving positive vote mass from a source county where the indicated FTZ weight type has zero/undefined support.",
    caption = "CZ shapes are dissolved from 1990 counties using cz-198090.xls and limited to the ADH diagnostic CZ universe.",
    boundary_color = "white"
  )
  cz_map_simple <- simplify_sf_for_plot(cz_map, tolerance = 0.05)
  cz_plot_simple <- make_support_map_plot(
    cz_map_simple,
    title = "ADH-mainland 1990 CZs receiving affected bridged vote mass",
    subtitle = "Simplified-geometry version for smaller appendix files.",
    caption = "CZ shapes are dissolved from simplified 1990 counties and limited to the ADH diagnostic CZ universe.",
    boundary_color = "white"
  )
  save_crosswalk_support_plot_pair(
    cz_plot, cz_plot_simple,
    config$crosswalk_support_cz_map_pdf, config$crosswalk_support_cz_map_png,
    config$crosswalk_support_cz_map_simplified_pdf, config$crosswalk_support_cz_map_simplified_png,
    config$crosswalk_support_cz_map_small_png
  )

  invisible(list(county_plot = county_plot, cz_plot = cz_plot))
}

copy_if_exists <- function(from, to_dir) {
  if (is.null(from) || length(from) == 0 || is.na(from) || !file.exists(from)) return(invisible(FALSE))
  dir.create(to_dir, recursive = TRUE, showWarnings = FALSE)
  file.copy(from, file.path(to_dir, basename(from)), overwrite = TRUE)
  invisible(TRUE)
}

archive_crosswalk_run_diagnostics <- function(config, label, status = "completed", error_message = NA_character_) {
  config <- finalize_config(config)
  slug <- config$crosswalk_weight_slug
  out_dir <- file.path(config$diagnostic_dir, "by_crosswalk", slug)
  dir.create(out_dir, recursive = TRUE, showWarnings = FALSE)

  files <- unique(stats::na.omit(c(
    config$validation_checks_csv, config$pipeline_manifest_json, config$crosswalk_diagnostics_csv,
    config$crosswalk_zero_support_csv, config$crosswalk_row_diagnostics_csv,
    config$crosswalk_issue_vote_report_csv, config$bridge_diagnostics_csv,
    config$mainland_retention_csv, config$unmatched_source_counties_csv,
    config$county_bridge_classification_csv, config$missing_cz_years_csv,
    config$panel_coverage_csv, config$spec_status_csv, config$model_diagnostics_csv,
    config$vcov_diagnostics_csv, config$se_warning_diagnostics_csv, config$event_study_csv,
    config$all_specs_csv, config$table_csv
  )))
  invisible(lapply(files, copy_if_exists, to_dir = out_dir))

  tibble::tibble(
    crosswalk_slug = slug,
    crosswalk_specification = label,
    crosswalk_weight = config$crosswalk_weight,
    crosswalk_missing_weight_policy = config$crosswalk_missing_weight_policy,
    status = status,
    error_message = error_message,
    diagnostics_dir = portable_path(out_dir, config),
    archived_at = format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z")
  ) %>%
    readr::write_csv(file.path(out_dir, "run_status.csv"))

  invisible(out_dir)
}

write_crosswalk_sensitivity_status <- function(status_rows, config) {
  config <- finalize_config(config)
  status_tbl <- dplyr::bind_rows(status_rows)
  path <- file.path(config$diagnostic_dir, "crosswalk_sensitivity_run_status.csv")
  readr::write_csv(status_tbl, path)
  invisible(path)
}

export_diagnostic_maps_and_comparisons <- function(config = CONFIG) {
  config <- finalize_config(config)
  export_crosswalk_specification_comparison(config)
  export_crosswalk_support_maps(config)

  checks <- if (file.exists(config$validation_checks_csv)) {
    readr::read_csv(config$validation_checks_csv, show_col_types = FALSE)
  } else {
    tibble::tibble()
  }
  output_inventory <- write_output_manifest(config)

  write_pipeline_manifest(
    config = config,
    checks = checks,
```

```
    stage = "export_diagnostic_maps_and_comparisons",
    sources = list(
      county_support = list(path = config$target_weight_support_county_csv, md5 = file_checksum(config$target_weight_support_county_csv)),
      cz_support = list(path = config$target_weight_support_cz_csv, md5 = file_checksum(config$target_weight_support_cz_csv))
    ),
    extra = list(
      outputs = list(
        crosswalk_comparison_png = list(path = config$crosswalk_comparison_png, md5 = file_checksum(config$crosswalk_comparison_png)),
        crosswalk_comparison_delta_png = list(path = config$crosswalk_comparison_delta_png, md5 =
        ↪  file_checksum(config$crosswalk_comparison_delta_png)),
        crosswalk_comparison_delta_csv = list(path = config$crosswalk_comparison_delta_csv, md5 =
        ↪  file_checksum(config$crosswalk_comparison_delta_csv)),
        crosswalk_support_county_map_png = list(path = config$crosswalk_support_county_map_png, md5 =
        ↪  file_checksum(config$crosswalk_support_county_map_png)),
        crosswalk_support_county_map_simplified_png = list(path = config$crosswalk_support_county_map_simplified_png, md5 =
        ↪  file_checksum(config$crosswalk_support_county_map_simplified_png)),
        crosswalk_support_county_map_small_png = list(path = config$crosswalk_support_county_map_small_png, md5 =
        ↪  file_checksum(config$crosswalk_support_county_map_small_png)),
        crosswalk_support_cz_map_png = list(path = config$crosswalk_support_cz_map_png, md5 =
        ↪  file_checksum(config$crosswalk_support_cz_map_png)),
        crosswalk_support_cz_map_simplified_png = list(path = config$crosswalk_support_cz_map_simplified_png, md5 =
        ↪  file_checksum(config$crosswalk_support_cz_map_simplified_png)),
        crosswalk_support_cz_map_small_png = list(path = config$crosswalk_support_cz_map_small_png, md5 =
        ↪  file_checksum(config$crosswalk_support_cz_map_small_png)),
        output_manifest_csv = list(path = config$output_manifest_csv, md5 = file_checksum(config$output_manifest_csv)),
        output_manifest_json = list(path = config$output_manifest_json, md5 = file_checksum(config$output_manifest_json))
      ),
      output_manifest_n_files = nrow(output_inventory)
    )
  )
  invisible(NULL)
}

run_crosswalk_sensitivity_pipeline <- function(config = CONFIG) {
  base_config <- finalize_config(config)
  plan <- crosswalk_sensitivity_plan()
  ensure_output_dirs(base_config)
  write_dependency_manifest(base_config)

  status_rows <- list()
  for (i in seq_len(nrow(plan))) {
    local_config <- base_config
    local_config$crosswalk_weight <- plan$crosswalk_weight[[i]]
    local_config$crosswalk_missing_weight_policy <- plan$crosswalk_missing_weight_policy[[i]]
    local_config$export_crosswalk_maps <- isTRUE(base_config$export_crosswalk_maps) && isTRUE(plan$render_support_maps[[i]])
    local_config <- finalize_config(local_config)

    label <- plan$label[[i]]
    message("Running crosswalk sensitivity specification: ", label, " (", local_config$crosswalk_weight, ", ",
    ↪  local_config$crosswalk_missing_weight_policy, ")")

    step_status <- "completed"
    step_error <- NA_character_
    started_at <- Sys.time()
    tryCatch({
      build_analysis_data(local_config, stop_on_fatal = TRUE)
      estimate_event_study(local_config, stop_on_fatal = TRUE)
      export_event_study_outputs(local_config)
      if (isTRUE(local_config$export_crosswalk_maps)) export_crosswalk_support_maps(local_config)
    }, error = function(e) {
      step_status <<- "failed"
      step_error <<- conditionMessage(e)
      message("Crosswalk sensitivity specification failed: ", label, ": ", step_error)
    })

    archive_crosswalk_run_diagnostics(local_config, label, status = step_status, error_message = step_error)
    status_rows[[length(status_rows) + 1L]] <- tibble::tibble(
      crosswalk_slug = local_config$crosswalk_weight_slug,
      crosswalk_specification = label,
      crosswalk_weight = local_config$crosswalk_weight,
      crosswalk_missing_weight_policy = local_config$crosswalk_missing_weight_policy,
      status = step_status,
      error_message = step_error,
      started_at = format(started_at, "%Y-%m-%d %H:%M:%S %Z"),
      finished_at = format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z")
    )
    write_crosswalk_sensitivity_status(status_rows, base_config)
  }

  # After all slug-specific coefficient files exist, create the appendix-ready comparison figure.
  comparison_config <- finalize_config(base_config)
  export_crosswalk_specification_comparison(comparison_config)

  if (isTRUE(comparison_config$run_sixth_extensions)) {
    # Sixth-section extensions are estimated only once for the preferred M5 + fallback-M2 design.
    run_sixth_extensions(comparison_config)
  }

  checks <- if (file.exists(comparison_config$validation_checks_csv)) {
```

```
    readr::read_csv(comparison_config$validation_checks_csv, show_col_types = FALSE)
  } else {
    tibble::tibble()
  }
  status_tbl <- dplyr::bind_rows(status_rows)
  write_crosswalk_sensitivity_status(status_rows, comparison_config)
  output_inventory <- write_output_manifest(comparison_config)
  write_pipeline_manifest(
    config = comparison_config,
    checks = checks,
    stage = "crosswalk_sensitivity_complete",
    sources = list(
      sensitivity_plan = list(specifications = plan),
      sensitivity_status = list(path = file.path(comparison_config$diagnostic_dir, "crosswalk_sensitivity_run_status.csv"), md5 =
        ↪ file_checksum(file.path(comparison_config$diagnostic_dir, "crosswalk_sensitivity_run_status.csv")))
    ),
    extra = list(
      any_failed_specifications = any(status_tbl$status != "completed"),
      completed_specifications = status_tbl %>% dplyr::filter(status == "completed") %>% dplyr::pull(crosswalk_specification),
      failed_specifications = status_tbl %>% dplyr::filter(status != "completed") %>% dplyr::pull(crosswalk_specification),
      comparison_csv = list(path = comparison_config$crosswalk_comparison_csv, md5 =
        ↪ file_checksum(comparison_config$crosswalk_comparison_csv)),
      comparison_png = list(path = comparison_config$crosswalk_comparison_png, md5 =
        ↪ file_checksum(comparison_config$crosswalk_comparison_png)),
      comparison_delta_csv = list(path = comparison_config$crosswalk_comparison_delta_csv, md5 =
        ↪ file_checksum(comparison_config$crosswalk_comparison_delta_csv)),
      comparison_delta_png = list(path = comparison_config$crosswalk_comparison_delta_png, md5 =
        ↪ file_checksum(comparison_config$crosswalk_comparison_delta_png)),
      output_manifest_csv = list(path = comparison_config$output_manifest_csv, md5 =
        ↪ file_checksum(comparison_config$output_manifest_csv)),
      output_manifest_json = list(path = comparison_config$output_manifest_json, md5 =
        ↪ file_checksum(comparison_config$output_manifest_json)),
      output_manifest_n_files = nrow(output_inventory)
    )
  )
  invisible(status_tbl)
}
```

7 06_sixth_extensions.R

```
# extension-adh2013/src/06_sixth_extensions.R

# Highest-priority Sixth-section extensions. These modules intentionally run once
# for the preferred M5 + fallback-M2 design, even when the crosswalk-sensitivity
# wrapper is active. The mechanism/identification diagnostics are meant to be
# interpreted for the preferred analysis sample rather than repeated for every
# crosswalk robustness specification.

extension_primary_config <- function(config = CONFIG) {
  out <- config
  out$crosswalk_weight <- config$extension_crosswalk_weight %||% "m5_weight"
  out$crosswalk_missing_weight_policy <- config$extension_crosswalk_missing_weight_policy %||% "fallback_m2"
  out <- finalize_config(out)
  current <- finalize_config(config)
  if (!file.exists(out$analysis_panel_rds) && file.exists(current$analysis_panel_rds)) out <- current
  out
}

safe_num <- function(x) suppressWarnings(as.numeric(x))

safe_divide <- function(num, den) {
  num <- safe_num(num); den <- safe_num(den)
  out <- ifelse(is.na(num) | is.na(den) | den <= 0, NA_real_, num / den)
  out[!is.finite(out)] <- NA_real_
  out
}

scalar_or_na <- function(x) {
  if (is.null(x) || length(x) == 0) return(NA_real_)
  as.numeric(x[[1]])
}

weighted_mean_safe <- function(x, w = NULL) {
  x <- safe_num(x)
  if (is.null(w)) w <- rep(1, length(x)) else w <- safe_num(w)
  ok <- !is.na(x) & !is.na(w) & w > 0
  if (!any(ok)) return(NA_real_)
  stats::weighted.mean(x[ok], w[ok], na.rm = TRUE)
}

extension_spec_label <- function(x) {
  dplyr::recode(as.character(x),
    minimal = "Minimal",
    core_controls = "Core controls",
    full_controls = "Full controls",
    .default = as.character(x)
  )
}

extension_outcome_label <- function(x) {
  dplyr::recode(as.character(x),
    rep_margin = "Republican margin",
    rep_2party_share = "Republican two-party share",
    dem_2party_share = "Democratic two-party share",
    two_party_votes_per_1990_pop = "Two-party votes / 1990 population",
    total_votes_per_1990_pop = "Total votes / 1990 population",
    rep_votes_per_1990_pop = "Republican votes / 1990 population",
    dem_votes_per_1990_pop = "Democratic votes / 1990 population",
    rep_margin_swing = "Republican-margin swing",
    rep_2party_share_swing = "Republican-share swing",
    reg_voters_pct = "Registered voters / CVAP",
    voter_turnout_pct = "Ballots cast / CVAP",
    reg_voter_turnout_pct = "Ballots cast / registered voters",
    pres_rep_ratio = "Presidential Republican ratio",
    pres_dem_ratio = "Presidential Democratic ratio",
    partisan_index_dem = "Democratic partisanship index",
    partisan_index_rep = "Republican partisanship index",
    .default = as.character(x)
  )
}

adhm_outcome_label <- function(x) {
  dplyr::recode(as.character(x),
    d2_rwin_2002_2016 = "Republican House win share, 2002-2016",
    d2_cfavg_repcon_2002_2016 = "Conservative Republican CF-score share",
    d2_cfavg_repmod_2002_2016 = "Moderate Republican CF-score share",
    d2_cfavg_demmod_2002_2016 = "Moderate Democratic CF-score share",
    d2_cfavg_demlib_2002_2016 = "Liberal Democratic CF-score share",
    d2_teaparty_2002_2010 = "Tea Party / Freedom Caucus outcome",
    d_shnr_pres_2000_2016 = "Republican presidential share, 2000-2016",
    d_shnr_pres_2000_2008 = "Republican presidential share, 2000-2008",
    .default = as.character(x)
  )
}

plot_pointrange_event <- function(df, title, subtitle, y_lab = "Coefficient on ADH exposure × year") {
  ggplot2::ggplot(df, ggplot2::aes(x = year, y = estimate, ymin = estimate - 1.96 * std.error, ymax = estimate + 1.96 * std.error)) +
```

```
    ggplot2::geom_hline(yintercept = 0, linewidth = 0.25) +
    ggplot2::geom_pointrange(linewidth = 0.25) +
    ggplot2::labs(title = title, subtitle = subtitle, x = NULL, y = y_lab) +
    ggplot2::theme_minimal(base_size = 10)
}

first_existing_col <- function(df, candidates) {
  nm <- names(df)
  hit <- candidates[candidates %in% nm]
  if (length(hit) == 0) NA_character_ else hit[[1]]
}

read_first_rdata_frame <- function(path) {
  e <- new.env(parent = emptyenv())
  loaded <- load(path, envir = e)
  candidates <- loaded[vapply(loaded, function(nm) is.data.frame(e[[nm]]), logical(1))]
  if (length(candidates) == 0) stop("No data.frame object found in ", path)
  e[[candidates[[1]]]]
}

coeftable_to_tibble <- function(model, vcov = NULL, term_prefix = NULL) {
  ct <- tryCatch({
    if (is.null(vcov)) fixest::coeftable(model) else fixest::coeftable(model, vcov = vcov)
  }, error = function(e) NULL)
  if (is.null(ct)) return(tibble::tibble())
  out <- as.data.frame(ct) %>% tibble::rownames_to_column("term") %>% tibble::as_tibble()
  names(out) <- c("term", "estimate", "std.error", "statistic", "p.value")
  if (!is.null(term_prefix)) out <- out %>% dplyr::filter(startsWith(term, term_prefix))
  out
}

fit_weighted_feols <- function(formula, data, weight_var = "adh_weight", cluster_var = "czone") {
  weights_fml <- if (!is.null(weight_var) && weight_var %in% names(data)) stats::as.formula(paste0("~", weight_var)) else NULL
  cluster_fml <- if (!is.null(cluster_var) && cluster_var %in% names(data)) stats::as.formula(paste0("~", cluster_var)) else NULL
  fixest::feols(formula, data = data, weights = weights_fml, cluster = cluster_fml, warn = FALSE, notes = FALSE)
}

write_plot_safely <- function(plot, png_path, pdf_path, width = 10, height = 6) {
  try(ggplot2::ggsave(png_path, plot, width = width, height = height, dpi = 300, bg = "white"), silent = TRUE)
  try(ggplot2::ggsave(pdf_path, plot, width = width, height = height, bg = "white"), silent = TRUE)
  invisible(NULL)
}

load_adh_extension_data <- function(config = CONFIG) {
  adh_stack_path <- file.path(config$replication_dir, "adh2013", "dta", "workfile_china.dta")
  adh_long_path <- file.path(config$replication_dir, "adh2013", "dta", "workfile_china_long.dta")
  stack <- haven::read_dta(adh_stack_path)
  long <- haven::read_dta(adh_long_path)
  baseline <- stack %>%
    dplyr::filter(yr == 1990) %>%
    dplyr::transmute(
      czone = as.integer(czone),
      statefip = as.integer(statefip),
      dplyr::across(dplyr::all_of(config$baseline_controls), safe_num)
    ) %>%
    dplyr::distinct()
  exposure <- long %>%
    dplyr::transmute(
      czone = as.integer(czone),
      statefip = as.integer(statefip),
      exposure_1990_2007 = safe_num(d_tradeusch_pw),
      instrument_1990_2007 = safe_num(d_tradeotch_pw_lag),
      adh_weight = safe_num(timepwt48)
    ) %>%
    dplyr::distinct()
  baseline %>%
    dplyr::inner_join(exposure, by = c("czone", "statefip")) %>%
    standardize_baseline_controls(config$baseline_controls) %>%
    purrr::pluck("data")
}

make_single_exposure_event_terms <- function(data, exposure_var, years, ref_year, prefix = "es") {
  years <- sort(unique(years))
  for (yr in setdiff(years, ref_year)) {
    nm <- paste0(prefix, "_", yr)
    data[[nm]] <- data[[exposure_var]] * as.integer(data$year == yr)
  }
  data
}

apply_control_interactions <- function(data, controls, years, ref_year, standardize = TRUE) {
  controls_use <- model_control_names(controls, standardize = standardize)
  controls_use <- controls_use[controls_use %in% names(data)]
  if (length(controls_use) == 0) return(list(data = data, controls = controls_use, terms = character(0)))
  built <- make_interacted_control_vars(data, controls_use, years, ref_year)
  list(data = built$data, controls = controls_use, terms = built$vars)
}

fit_single_outcome_event_study <- function(panel, outcome_var, controls, spec_name, config,
                                           ref_year = config$reference_year,
```

```
                                        standardize_controls = TRUE,
                                        sample_label = "main_1972_start") {
  yrs <- sort(unique(panel$year[!is.na(panel[[outcome_var]])]))
  if (length(yrs) == 0) return(list(status = "empty", coefficients = tibble::tibble()))
  if (!ref_year %in% yrs) ref_year <- min(yrs, na.rm = TRUE)
  dat <- panel %>% dplyr::filter(year %in% yrs, !is.na(.data[[outcome_var]]), !is.na(exposure))
  if (nrow(dat) == 0) return(list(status = "empty", coefficients = tibble::tibble()))
  dat <- make_single_exposure_event_terms(dat, "exposure", yrs, ref_year, prefix = "es")
  event_terms <- paste0("es_", setdiff(yrs, ref_year))
  ctrl_build <- apply_control_interactions(dat, controls, yrs, ref_year, standardize = standardize_controls)
  dat <- ctrl_build$data
  rhs <- c(event_terms, ctrl_build$terms)
  if (length(rhs) == 0) return(list(status = "no_terms", coefficients = tibble::tibble()))
  fml <- stats::as.formula(paste(outcome_var, "~", paste(rhs, collapse = " + "), "| czone + year"))
  tryCatch({
    model <- fixest::feols(fml, data = dat, weights = ~adh_weight, cluster = ~czone, panel.id = ~czone + election_index, warn = FALSE,
     ↪  notes = FALSE)
    coef <- coeftable_to_tibble(model) %>%
      dplyr::filter(term %in% event_terms) %>%
      dplyr::mutate(
        outcome = outcome_var,
        spec = spec_name,
        sample = sample_label,
        se_type = "cluster_cz",
        reference_year = ref_year,
        year = as.integer(stringr::str_extract(term, "[0-9]{4}")),
        n_obs = stats::nobs(model),
        n_cz = dplyr::n_distinct(dat$czone),
        controls = if (length(ctrl_build$controls) == 0) "none" else paste(ctrl_build$controls, collapse = "; ")
      ) %>%
      dplyr::select(outcome, spec, sample, se_type, term, year, reference_year, estimate, std.error, statistic, p.value, n_obs, n_cz,
       ↪  controls)
    list(status = "ok", coefficients = coef)
  }, error = function(e) {
    list(status = paste0("error: ", conditionMessage(e)), coefficients = tibble::tibble())
  })
}

# ------------------------------------------------------------------------------
# 1. Bartik identification diagnostics
# ------------------------------------------------------------------------------

extract_trade_shift_summary <- function(config) {
  trade_path <- file.path(config$replication_dir, "adh2013", "dta", "sic87dd_trade_data.dta")
  if (!file.exists(trade_path)) return(tibble::tibble(status = "missing", path = trade_path))
  tr <- haven::read_dta(trade_path)
  names(tr) <- tolower(names(tr))
  req <- c("year", "importer", "exporter", "imports", "sic87dd")
  if (!all(req %in% names(tr))) {
    return(tibble::tibble(status = "missing_required_columns", columns_found = paste(names(tr), collapse = ";")))
  }
  tr <- tr %>%
    dplyr::mutate(
      year = as.integer(year),
      importer = toupper(trimws(as.character(importer))),
      exporter = toupper(trimws(as.character(exporter))),
      imports = safe_num(imports),
      sic87dd = as.character(sic87dd)
    )
  periods <- tibble::tibble(start_year = c(1991L, 2000L, 1991L), end_year = c(1999L, 2007L, 2007L), period = c("1991-1999", "2000-2007",
   ↪  "1991-2007"))
  purrr::pmap_dfr(periods, function(start_year, end_year, period) {
    base <- tr %>% dplyr::filter(year %in% c(start_year, end_year), exporter %in% c("CHN", "CHINA"), importer %in% c("USA", "OTH",
     ↪  "OTHER", "OTHCAN"))
    if (nrow(base) == 0) return(tibble::tibble(period = period, status = "no_matching_trade_rows"))
    wide <- base %>%
      dplyr::group_by(sic87dd, importer, year) %>%
      dplyr::summarise(imports = sum(imports, na.rm = TRUE), .groups = "drop") %>%
      tidyr::pivot_wider(names_from = year, values_from = imports, names_prefix = "imports_")
    start_col <- paste0("imports_", start_year)
    end_col <- paste0("imports_", end_year)
    if (!all(c(start_col, end_col) %in% names(wide))) return(tibble::tibble(period = period, status = "missing_period_endpoints"))
    wide %>%
      dplyr::mutate(
        start_imports = .data[[start_col]],
        end_imports = .data[[end_col]],
        import_growth = end_imports - start_imports
      ) %>%
      dplyr::group_by(period = period, importer) %>%
      dplyr::slice_max(order_by = abs(import_growth), n = 15, with_ties = FALSE) %>%
      dplyr::ungroup() %>%
      dplyr::mutate(status = "ok") %>%
      dplyr::select(status, period, importer, sic87dd, start_imports, end_imports, import_growth)
  })
}

run_bartik_identification_diagnostics <- function(config = CONFIG) {
  config <- finalize_config(config)
  adh <- load_adh_extension_data(config)
  controls_core <- model_control_names(config$core_interacted_controls, standardize = TRUE)
```

```
controls_full <- model_control_names(config$interacted_controls, standardize = TRUE)
controls_core <- controls_core[controls_core %in% names(adh)]
controls_full <- controls_full[controls_full %in% names(adh)]
fs_specs <- list(minimal = character(0), core_controls = controls_core, full_controls = controls_full)
fs_tbl <- purrr::imap_dfr(fs_specs, function(ctrls, spec) {
  rhs <- c("instrument_1990_2007", ctrls)
  fml <- stats::as.formula(paste("exposure_1990_2007 ~", paste(rhs, collapse = " + ")))
  model <- fit_weighted_feols(fml, adh, weight_var = "adh_weight", cluster_var = "statefip")
  ct <- coeftable_to_tibble(model) %>% dplyr::filter(term == "instrument_1990_2007")
  tibble::tibble(
    spec = spec,
    endogenous_variable = "ADH exposure, 1990-2007",
    instrument = "ADH other-high-income-country import growth instrument",
    estimate = scalar_or_na(ct$estimate),
    std.error = scalar_or_na(ct$std.error),
    statistic = scalar_or_na(ct$statistic),
    p.value = scalar_or_na(ct$p.value),
    first_stage_F_t2 = (scalar_or_na(ct$statistic))^2,
    n_obs = stats::nobs(model),
    r2 = tryCatch(fixest::r2(model, type = "r2"), error = function(e) NA_real_),
    controls = if (length(ctrls) == 0) "none" else paste(ctrls, collapse = "; ")
  )
})
readr::write_csv(fs_tbl, config$bartik_first_stage_csv)
p_fs <- ggplot2::ggplot(adh, ggplot2::aes(x = instrument_1990_2007, y = exposure_1990_2007)) +
  ggplot2::geom_point(ggplot2::aes(size = adh_weight), alpha = 0.35) +
  ggplot2::geom_smooth(method = "lm", se = TRUE, linewidth = 0.5) +
  ggplot2::guides(size = "none") +
  ggplot2::labs(title = "ADH exposure first stage", subtitle = "CZ-level China exposure and ADH other-country instrument", x =
    ↪  "Instrument", y = "Exposure") +
  ggplot2::theme_minimal(base_size = 11)
write_plot_safely(p_fs, config$bartik_first_stage_plot_png, config$bartik_first_stage_plot_pdf, width = 8, height = 5)

balance_vars <- unique(c(config$baseline_controls, controls_full, "statefip"))
balance_vars <- balance_vars[balance_vars %in% names(adh)]
balance_tbl <- purrr::map_dfr(balance_vars, function(v) {
  x <- adh[[v]]
  tibble::tibble(
    variable = v,
    correlation_with_exposure = suppressWarnings(stats::cor(x, adh$exposure_1990_2007, use = "pairwise.complete.obs")),
    correlation_with_instrument = suppressWarnings(stats::cor(x, adh$instrument_1990_2007, use = "pairwise.complete.obs")),
    weighted_mean = weighted_mean_safe(x, adh$adh_weight),
    sd = stats::sd(safe_num(x), na.rm = TRUE)
  )
})
readr::write_csv(balance_tbl, config$bartik_balance_csv)

pretrend_tbl <- tibble::tibble()
if (file.exists(config$diagnostic_panel_rds)) {
  panel <- readRDS(config$diagnostic_panel_rds) %>% dplyr::filter(year < config$reference_year) %>% dplyr::arrange(czone, year)
  if (nrow(panel) > 0) {
    slope_tbl <- panel %>%
      dplyr::group_by(czone) %>%
      dplyr::summarise(
        pretrend_slope_rep_margin = tryCatch(stats::coef(stats::lm(rep_margin ~ year))[2], error = function(e) NA_real_),
        pretrend_change_first_to_last = rep_margin[which.max(year)] - rep_margin[which.min(year)],
        n_pre_years = sum(!is.na(rep_margin)),
        .groups = "drop"
      ) %>% dplyr::left_join(adh, by = "czone")
    for (outcome in c("pretrend_slope_rep_margin", "pretrend_change_first_to_last")) {
      for (rhs_var in c("exposure_1990_2007", "instrument_1990_2007")) {
        for (spec_name in names(fs_specs)) {
          ctrls <- fs_specs[[spec_name]]
          dat <- slope_tbl %>% dplyr::filter(!is.na(.data[[outcome]]), !is.na(.data[[rhs_var]]), n_pre_years >= 2)
          if (nrow(dat) > 0) {
            rhs <- c(rhs_var, ctrls)
            fml <- stats::as.formula(paste(outcome, "~", paste(rhs, collapse = " + ")))
            model <- tryCatch(fit_weighted_feols(fml, dat, weight_var = "adh_weight", cluster_var = "statefip"), error = function(e)
              ↪  NULL)
            if (!is.null(model)) {
              ct <- coeftable_to_tibble(model) %>% dplyr::filter(term == rhs_var)
              pretrend_tbl <- dplyr::bind_rows(pretrend_tbl, tibble::tibble(
                outcome = outcome, spec = spec_name, regressor = rhs_var,
                estimate = scalar_or_na(ct$estimate), std.error = scalar_or_na(ct$std.error),
                statistic = scalar_or_na(ct$statistic), p.value = scalar_or_na(ct$p.value),
                n_obs = stats::nobs(model), note = "CZ-level pre-1988 presidential Republican-margin pretrend diagnostic; not a causal
                  ↪  test."
              ))
            }
          }
        }
      }
    }
  }
}
readr::write_csv(pretrend_tbl, config$bartik_pretrend_csv)

preperiod_path <- file.path(config$replication_dir, "adh2013", "dta", "workfile_china_preperiod.dta")
preperiod_tbl <- tibble::tibble()
if (file.exists(preperiod_path)) {
```

```
  pre_raw <- haven::read_dta(preperiod_path)
  names(pre_raw) <- make.names(names(pre_raw), unique = TRUE)
  names(pre_raw) <- gsub("\\.", "_", names(pre_raw))
  pre_raw <- pre_raw %>% dplyr::mutate(czone = as.integer(czone))
  candidate_outcomes <- names(pre_raw)[grepl("^d_", names(pre_raw))]
  candidate_outcomes <- setdiff(candidate_outcomes, c("d_tradeusch_pw", "d_tradeotch_pw_lag", "d_tradeusch_pw_future",
   ↪  "d_tradeotch_pw_lag_future"))
  # The preperiod workfile can contain multiple rows per CZ. Aggregate to one
  # row per CZ before running diagnostics so the output is comparable to the
  # main CZ-level exposure regressions.
  pre <- pre_raw %>%
    dplyr::group_by(czone) %>%
    dplyr::summarise(
      dplyr::across(dplyr::all_of(candidate_outcomes), ~ mean(safe_num(.x), na.rm = TRUE)),
      preperiod_rows_per_cz = dplyr::n(),
      .groups = "drop"
    ) %>%
    dplyr::mutate(dplyr::across(dplyr::all_of(candidate_outcomes), ~ dplyr::if_else(is.nan(.x), NA_real_, .x))) %>%
    dplyr::left_join(adh, by = "czone")
  for (outcome in candidate_outcomes) {
    for (rhs_var in c("exposure_1990_2007", "instrument_1990_2007")) {
      dat <- pre %>% dplyr::filter(!is.na(.data[[outcome]]), !is.na(.data[[rhs_var]]))
      if (nrow(dat) >= 50) {
        rhs <- c(rhs_var, controls_core)
        fml <- stats::as.formula(paste(outcome, "~", paste(rhs, collapse = " + ")))
        mod <- tryCatch(fit_weighted_feols(fml, dat, weight_var = "adh_weight", cluster_var = "statefip"), error = function(e) NULL)
        if (!is.null(mod)) {
          ct <- coeftable_to_tibble(mod) %>% dplyr::filter(term == rhs_var)
          preperiod_tbl <- dplyr::bind_rows(preperiod_tbl, tibble::tibble(
            outcome = outcome, regressor = rhs_var, estimate = scalar_or_na(ct$estimate), std.error = scalar_or_na(ct$std.error),
            statistic = scalar_or_na(ct$statistic), p.value = scalar_or_na(ct$p.value), n_obs = stats::nobs(mod),
            median_source_rows_per_cz = stats::median(dat$preperiod_rows_per_cz, na.rm = TRUE),
            note = "ADH preperiod-workfile diagnostic: source rows aggregated to one CZ row before regression; future
             ↪  exposure/instrument predicting preperiod/local-labor-market changes."
          ))
        }
      }
    }
  }
}
readr::write_csv(preperiod_tbl, config$bartik_preperiod_workfile_csv)

trade_summary <- tryCatch(extract_trade_shift_summary(config), error = function(e) tibble::tibble(status = "error", message =
 ↪  conditionMessage(e)))
readr::write_csv(trade_summary, config$bartik_industry_shift_summary_csv)

rot_tbl <- tibble::tibble(
  diagnostic = c("rotemberg_weights_exact", "industry_by_cz_shares_status", "industry_shift_summary_status", "next_step"),
  status = c("not_yet_computed", "requires_reconstructing_CZ_by_industry_shares", ifelse(nrow(trade_summary) > 0, "written",
   ↪  "not_written"), "feasible_with_ADH_raw_CBP_scripts_but_not_a_one-file_join"),
  details = c(
    "Exact GPSS Rotemberg weights require a CZ-by-industry baseline share matrix matched to industry-level shifts. This patch adds
     ↪  industry-shift summaries and documents the reconstruction path, but does not yet run the full ADH CBP-to-CZ industry-share
     ↪  reconstruction.",
    "Relevant public ADH files include other/cbp1980_to_cz.do, cbp1990_to_cz.do, cbp2000_to_cz.do, cbp*_imputations.do, cw_n97_s87.dta,
     ↪  and sic87dd_trade_data.dta.",
    "Industry-level China import-growth summaries are exported from sic87dd_trade_data.dta where columns are available.",
    "To compute exact GPSS weights: reconstruct 1990 CZ x SIC87DD employment shares, create shift vectors from Chinese import growth in
     ↪  the instrument country group, residualize by controls, and compute Rotemberg weights."
  )
)
readr::write_csv(rot_tbl, config$bartik_data_availability_csv)

memo <- c(
  "# Bartik / shift-share identification memo", "",
  "This project estimates reduced-form event-study relationships between CZ-level ADH China import exposure and political outcomes. The
   ↪  ADH exposure variable is a shift-share object: local baseline industry composition is combined with national or foreign
   ↪  import-growth shocks.", "",
  "## Required assumptions", "",
  "A causal reading requires high-exposure CZs not to have been on different political trajectories for reasons unrelated to the China
   ↪  shock after conditioning on fixed effects and baseline-control-by-year interactions. For the ADH-style instrument, one can
   ↪  emphasize exogeneity of foreign import-growth shifts or exogeneity of baseline industry shares; both routes require substantive
   ↪  defense.", "",
  "## Project-specific threats", "",
  "- Initial industry shares may be correlated with prior political realignment, union decline, automation, racial composition,
   ↪  religiosity, education, or local media markets.",
  "- If only a few high-shock industries drive exposure, the estimates may capture politics of those industries rather than a generic
   ↪  China-shock effect.",
  "- Migration and turnout can make CZ-level political outcomes change even if individual beliefs do not change.",
  "- Candidate supply and elite rhetoric may translate material shocks into political outcomes; the ADHM 2020 mechanism outputs are
   ↪  diagnostic for this channel.", "",
  "## Diagnostics written by the pipeline", "",
  "- `bartik_first_stage_diagnostics.csv` and `fig_bartik_first_stage.png`: first-stage strength of the ADH other-country instrument.",
  "- `bartik_balance_correlations.csv`: exposure/instrument correlations with baseline controls.",
  "- `bartik_pretrend_placebos.csv`: whether exposure/instrument predict pre-1988 Republican-margin trends.",
  "- `bartik_preperiod_workfile_diagnostics.csv`: diagnostics using ADH's preperiod workfile.",
  "- `bartik_industry_shift_summary.csv`: industry-level import-shift summaries from ADH public trade data.",
  "- `bartik_rotemberg_data_availability.csv`: exact Rotemberg weights are a feasible future reconstruction, not a completed diagnostic
   ↪  here.", ""
```

```
  )
  writeLines(memo, config$bartik_identification_memo_md)
  invisible(list(first_stage = fs_tbl, balance = balance_tbl, pretrend = pretrend_tbl, preperiod = preperiod_tbl, trade = trade_summary))
}

# ------------------------------------------------------------------------------
# 2. Alternative political outcomes
# ------------------------------------------------------------------------------

add_alternative_political_outcomes <- function(panel) {
  panel %>%
    dplyr::arrange(czone, year) %>%
    dplyr::group_by(czone) %>%
    dplyr::mutate(
      rep_2party_share = safe_divide(rep_votes, twoparty_votes),
      dem_2party_share = safe_divide(dem_votes, twoparty_votes),
      two_party_votes_per_1990_pop = safe_divide(twoparty_votes, pop1990),
      total_votes_per_1990_pop = safe_divide(total_votes, pop1990),
      rep_votes_per_1990_pop = safe_divide(rep_votes, pop1990),
      dem_votes_per_1990_pop = safe_divide(dem_votes, pop1990),
      rep_margin_swing = rep_margin - dplyr::lag(rep_margin),
      rep_2party_share_swing = rep_2party_share - dplyr::lag(rep_2party_share)
    ) %>%
    dplyr::ungroup()
}

run_alternative_political_outcome_event_studies <- function(config = CONFIG) {
  config <- finalize_config(config)
  if (!file.exists(config$analysis_panel_rds)) {
    readr::write_csv(tibble::tibble(module = "alternative_political_outcomes", status = "missing_panel", detail =
      ↪ config$analysis_panel_rds), config$alternative_outcomes_status_csv)
    return(invisible(NULL))
  }
  panel <- readRDS(config$analysis_panel_rds) %>% add_alternative_political_outcomes()
  outcomes <- c("rep_margin", "rep_2party_share", "dem_2party_share", "two_party_votes_per_1990_pop", "total_votes_per_1990_pop",
   ↪ "rep_votes_per_1990_pop", "dem_votes_per_1990_pop", "rep_margin_swing", "rep_2party_share_swing")
  outcome_labels <- tibble::tibble(outcome = outcomes, label = c("Republican margin", "Republican two-party share", "Democratic two-party
   ↪ share", "Two-party votes / 1990 population", "Total votes / 1990 population", "Republican votes / 1990 population", "Democratic
   ↪ votes / 1990 population", "Republican-margin swing since previous election", "Republican-share swing since previous election"))
  specs <- list(minimal = list(controls = character(0), standardize = FALSE), core_controls = list(controls =
   ↪ config$core_interacted_controls, standardize = TRUE), full_controls = list(controls = config$interacted_controls, standardize =
   ↪ TRUE))
  coef_tbl <- tibble::tibble(); status_tbl <- tibble::tibble()
  for (outcome in outcomes) {
    for (spec_name in names(specs)) {
      spec <- specs[[spec_name]]
      fit <- fit_single_outcome_event_study(panel, outcome, spec$controls, spec_name, config, standardize_controls = spec$standardize)
      coef_tbl <- dplyr::bind_rows(coef_tbl, fit$coefficients)
      status_tbl <- dplyr::bind_rows(status_tbl, tibble::tibble(module = "alternative_political_outcomes", outcome = outcome, spec =
        ↪ spec_name, status = fit$status, n_coefficients = nrow(fit$coefficients)))
    }
  }
  coef_tbl <- coef_tbl %>% dplyr::left_join(outcome_labels, by = "outcome") %>% dplyr::mutate(spec_label = extension_spec_label(spec))
  readr::write_csv(coef_tbl, config$alternative_outcomes_coefficients_csv)
  readr::write_csv(status_tbl, config$alternative_outcomes_status_csv)
  summary_tbl <- panel %>%
    dplyr::summarise(dplyr::across(dplyr::all_of(outcomes), list(n = ~sum(!is.na(.x)), mean = ~mean(.x, na.rm = TRUE), sd =
      ↪ ~stats::sd(.x, na.rm = TRUE), min = ~min(.x, na.rm = TRUE), max = ~max(.x, na.rm = TRUE)), .names = "{.col}_{.fn}")) %>%
    tidyr::pivot_longer(dplyr::everything(), names_to = "stat", values_to = "value")
  readr::write_csv(summary_tbl, config$alternative_outcomes_summary_csv)
  if (nrow(coef_tbl) > 0) {
    plot_df <- coef_tbl %>% dplyr::filter(spec %in% c("minimal", "core_controls", "full_controls"), outcome %in% outcomes[1:7])
    p <- ggplot2::ggplot(plot_df, ggplot2::aes(x = year, y = estimate, ymin = estimate - 1.96 * std.error, ymax = estimate + 1.96 *
      ↪ std.error)) +
      ggplot2::geom_hline(yintercept = 0, linewidth = 0.25) + ggplot2::geom_pointrange(linewidth = 0.25) +
      ggplot2::facet_grid(label ~ spec_label, scales = "free_y") +
      ggplot2::labs(title = "Alternative political outcome event studies", subtitle = "CZ-clustered 95% intervals; 1988 reference where
       ↪ available", x = NULL, y = "Coefficient on ADH exposure × election year") +
      ggplot2::theme_minimal(base_size = 10)
    write_plot_safely(p, config$alternative_outcomes_plot_png, config$alternative_outcomes_plot_pdf, width = 12, height = 10)
    vote_choice <- coef_tbl %>% dplyr::filter(spec == "full_controls", outcome %in% c("rep_margin", "rep_2party_share",
     ↪ "dem_2party_share"))
    if (nrow(vote_choice) > 0) {
      p_choice <- plot_pointrange_event(vote_choice, "Vote-choice outcome diagnostics", "Full-control specification; CZ-clustered 95%
       ↪ intervals") +
        ggplot2::facet_wrap(~label, scales = "free_y", ncol = 1)
      write_plot_safely(p_choice, config$alternative_outcomes_vote_choice_plot_png, config$alternative_outcomes_vote_choice_plot_pdf,
       ↪ width = 8.5, height = 7)
    }
    vote_volume <- coef_tbl %>% dplyr::filter(spec == "full_controls", outcome %in% c("two_party_votes_per_1990_pop",
     ↪ "total_votes_per_1990_pop", "rep_votes_per_1990_pop", "dem_votes_per_1990_pop"))
    if (nrow(vote_volume) > 0) {
      p_volume <- plot_pointrange_event(vote_volume, "Vote-volume outcome diagnostics", "Full-control specification; CZ-clustered 95%
       ↪ intervals") +
        ggplot2::facet_wrap(~label, scales = "free_y", ncol = 2)
      write_plot_safely(p_volume, config$alternative_outcomes_vote_volume_plot_png, config$alternative_outcomes_vote_volume_plot_pdf,
       ↪ width = 9.5, height = 7)
    }
```

```
    decomp <- coef_tbl %>% dplyr::filter(spec == "full_controls", outcome %in% c("rep_margin", "rep_votes_per_1990_pop",
     ↪  "dem_votes_per_1990_pop", "two_party_votes_per_1990_pop"))
    p2 <- ggplot2::ggplot(decomp, ggplot2::aes(x = year, y = estimate, ymin = estimate - 1.96 * std.error, ymax = estimate + 1.96 *
     ↪  std.error)) +
      ggplot2::geom_hline(yintercept = 0, linewidth = 0.25) + ggplot2::geom_pointrange(linewidth = 0.25) +
      ggplot2::facet_wrap(~label, scales = "free_y", ncol = 2) +
      ggplot2::labs(title = "Vote-margin decomposition diagnostics", subtitle = "Full-control specification; CZ-clustered 95% intervals",
       ↪  x = NULL, y = "Coefficient") +
      ggplot2::theme_minimal(base_size = 10)
    write_plot_safely(p2, config$alternative_outcomes_decomposition_plot_png, config$alternative_outcomes_decomposition_plot_pdf, width =
     ↪  10, height = 7)
  }
  invisible(coef_tbl)
}

# ------------------------------------------------------------------------------
# Shared county-to-CZ bridge for county-year auxiliary outcomes
# ------------------------------------------------------------------------------

build_generic_county_to_cz_panel <- function(config, county_year_data, value_cols) {
  config <- finalize_config(config)
  cz_path <- file.path(config$replication_dir, "cz-data", "cz-198090.xls")
  ftz_zip <- file.path(config$replication_dir, "ftz2024", "crosswalks", "CountyToCounty", "1990", "1990_csv.zip")
  cz_lookup <- readxl::read_xls(cz_path) %>%
    dplyr::transmute(county_fips_1990 = stringr::str_pad(as.character(`County FIPS Code`), width = 5, side = "left", pad = "0"), czone =
     ↪  as.integer(CZ90), pop1990 = safe_num(`Population 1990`))
  source_decades <- sort(unique(year_to_source_decade(county_year_data$year)))
  xwalk <- purrr::map_dfr(source_decades, ~ read_ftz_crosswalk(.x, ftz_zip, cz_lookup$county_fips_1990, config$crosswalk_weight,
   ↪  config$crosswalk_missing_weight_policy, config$renormalize_crosswalk_weights, config$weight_sum_tolerance)) %>%
    dplyr::filter(!is.na(crosswalk_weight), crosswalk_weight > 0)
  bridge_exceptions <- read_county_bridge_exceptions(config)
  county_year_data %>%
    dplyr::mutate(source_decade_default = year_to_source_decade(year)) %>%
    dplyr::left_join(
      bridge_exceptions %>% dplyr::select(year, county_fips, override_source_decade, override_county_fips_source),
      by = c("year", "county_fips")
    ) %>%
    dplyr::mutate(
      source_decade = dplyr::coalesce(override_source_decade, source_decade_default),
      county_fips_source = dplyr::coalesce(override_county_fips_source, county_fips)
    ) %>%
    dplyr::left_join(xwalk, by = c("source_decade", "county_fips_source"), relationship = "many-to-many") %>%
    dplyr::filter(!is.na(county_fips_1990), !is.na(crosswalk_weight), crosswalk_weight > 0) %>%
    dplyr::mutate(dplyr::across(dplyr::all_of(value_cols), ~ safe_num(.x) * crosswalk_weight)) %>%
    dplyr::group_by(year, county_fips_1990) %>%
    dplyr::summarise(dplyr::across(dplyr::all_of(value_cols), ~ sum(.x, na.rm = TRUE)), .groups = "drop") %>%
    dplyr::inner_join(cz_lookup, by = "county_fips_1990") %>%
    dplyr::group_by(czone, year) %>%
    dplyr::summarise(dplyr::across(dplyr::all_of(value_cols), ~ sum(.x, na.rm = TRUE)), pop1990 = sum(pop1990, na.rm = TRUE), .groups =
     ↪  "drop")
}

# ------------------------------------------------------------------------------
# 3. NaNDA turnout and partisanship
# ------------------------------------------------------------------------------

run_nanda_outcome_event_studies <- function(config = CONFIG) {
  config <- finalize_config(config)
  nanda_path <- file.path(config$replication_dir, "outcomes-data", "nanda-2004-2022", "DS0001", "38506-0001-Data.rda")
  if (!file.exists(nanda_path)) {
    readr::write_csv(tibble::tibble(status = "missing", path = nanda_path), config$nanda_outcomes_summary_csv)
    return(invisible(NULL))
  }
  raw <- read_first_rdata_frame(nanda_path)
  names(raw) <- toupper(names(raw))
  fips_col <- first_existing_col(raw, c("STCOFIPS10", "STCOFIPS20", "FIPS", "COUNTY_FIPS"))
  if (is.na(fips_col) || !"YEAR" %in% names(raw)) stop("NaNDA data must contain county FIPS and YEAR columns.")
  fips_source <- if (all(c("STCOFIPS10", "STCOFIPS20") %in% names(raw))) {
    dplyr::if_else(as.integer(raw$YEAR) >= 2020L, as.character(raw$STCOFIPS20), as.character(raw$STCOFIPS10))
  } else {
    as.character(raw[[fips_col]])
  }
  numeric_candidates <- c("REG_VOTERS", "BALLOTS_CAST", "CVAP", "PRES_DEM_VOTES", "PRES_REP_VOTES", "SEN_DEM_VOTES", "SEN_REP_VOTES")
  count_cols <- numeric_candidates[numeric_candidates %in% names(raw)]
  index_cols <- c("PARTISAN_INDEX_DEM", "PARTISAN_INDEX_REP")
  index_cols <- index_cols[index_cols %in% names(raw)]
  raw$COUNTY_FIPS_SELECTED_FOR_BRIDGE <- fips_source
  nanda <- raw %>%
    dplyr::transmute(county_fips = stringr::str_pad(as.character(COUNTY_FIPS_SELECTED_FOR_BRIDGE), 5, pad = "0"), year =
     ↪  as.integer(YEAR), dplyr::across(dplyr::all_of(c(count_cols, index_cols)), safe_num)) %>%
    dplyr::mutate(dplyr::across(dplyr::all_of(c(count_cols, index_cols)), ~ dplyr::na_if(.x, -1)))
  # Restrict to documented NaNDA availability; avoid legacy/structural zeros.
  nanda <- nanda %>% dplyr::filter(year >= 2004, year <= 2022)
  county_validation <- nanda %>%
    dplyr::mutate(
      reg_voters_pct_raw = if (all(c("REG_VOTERS", "CVAP") %in% names(nanda))) safe_divide(REG_VOTERS, CVAP) else NA_real_,
      voter_turnout_pct_raw = if (all(c("BALLOTS_CAST", "CVAP") %in% names(nanda))) safe_divide(BALLOTS_CAST, CVAP) else NA_real_,
      reg_voter_turnout_pct_raw = if (all(c("BALLOTS_CAST", "REG_VOTERS") %in% names(nanda))) safe_divide(BALLOTS_CAST, REG_VOTERS) else
       ↪  NA_real_,
```

```
    county_rate_flag = dplyr::if_else(reg_voters_pct_raw > 1.5 | voter_turnout_pct_raw > 1.5 | reg_voter_turnout_pct_raw > 1.5, TRUE, FALSE, missing = FALSE)
  )
readr::write_csv(county_validation %>% dplyr::select(county_fips, year, dplyr::ends_with("_raw"), county_rate_flag), config$nanda_county_validation_csv)
# Convert index shares to CVAP-weighted pseudo-counts before bridging.
for (idx in index_cols) nanda[[paste0(idx, "_NUM")]] <- nanda[[idx]] * if ("CVAP" %in% count_cols) nanda$CVAP else 1
bridge_cols <- unique(c(count_cols, paste0(index_cols, "_NUM")))
bridge_cols <- bridge_cols[bridge_cols %in% names(nanda)]
cz_panel <- build_generic_county_to_cz_panel(config, nanda, bridge_cols)
if ("CVAP" %in% names(cz_panel)) {
  if ("REG_VOTERS" %in% names(cz_panel)) cz_panel$reg_voters_pct <- safe_divide(cz_panel$REG_VOTERS, cz_panel$CVAP)
  if ("BALLOTS_CAST" %in% names(cz_panel)) cz_panel$voter_turnout_pct <- safe_divide(cz_panel$BALLOTS_CAST, cz_panel$CVAP)
  if (all(c("BALLOTS_CAST", "REG_VOTERS") %in% names(cz_panel))) cz_panel$reg_voter_turnout_pct <- safe_divide(cz_panel$BALLOTS_CAST, cz_panel$REG_VOTERS)
  if (all(c("PRES_REP_VOTES", "PRES_DEM_VOTES") %in% names(cz_panel))) {
    denom <- cz_panel$PRES_REP_VOTES + cz_panel$PRES_DEM_VOTES
    cz_panel$pres_rep_ratio <- safe_divide(cz_panel$PRES_REP_VOTES, denom)
    cz_panel$pres_dem_ratio <- safe_divide(cz_panel$PRES_DEM_VOTES, denom)
  }
  for (idx in index_cols) {
    num <- paste0(idx, "_NUM")
    if (num %in% names(cz_panel)) cz_panel[[tolower(idx)]] <- safe_divide(cz_panel[[num]], cz_panel$CVAP)
  }
}
rate_cols <- c("reg_voters_pct", "voter_turnout_pct", "reg_voter_turnout_pct", "pres_rep_ratio", "pres_dem_ratio", "partisan_index_dem", "partisan_index_rep")
rate_cols <- rate_cols[rate_cols %in% names(cz_panel)]
cz_initial <- cz_panel %>%
  dplyr::mutate(dplyr::across(dplyr::all_of(rate_cols), ~ dplyr::if_else(.x < 0 | .x > 1.5 | !is.finite(.x), NA_real_, .x)))
outlier_rows <- cz_panel %>%
  dplyr::filter(dplyr::if_any(dplyr::all_of(rate_cols), ~ !is.na(.x) & (!is.finite(.x) | .x < 0 | .x > 1.5))) %>%
  dplyr::select(czone, year, dplyr::all_of(rate_cols))
readr::write_csv(outlier_rows, config$nanda_outlier_rates_csv)

availability_long <- cz_initial %>%
  dplyr::select(czone, year, dplyr::all_of(rate_cols)) %>%
  tidyr::pivot_longer(dplyr::all_of(rate_cols), names_to = "outcome", values_to = "value") %>%
  dplyr::group_by(outcome, year) %>%
  dplyr::summarise(
    n_cz = dplyr::n(),
    n_nonmissing = sum(!is.na(value)),
    n_zero = sum(!is.na(value) & abs(value) < 1e-12),
    n_nonzero = sum(!is.na(value) & abs(value) >= 1e-12),
    all_nonmissing_values_zero = n_nonmissing > 0 && n_nonzero == 0,
    min = if (n_nonmissing > 0) min(value, na.rm = TRUE) else NA_real_,
    max = if (n_nonmissing > 0) max(value, na.rm = TRUE) else NA_real_,
    mean = if (n_nonmissing > 0) mean(value, na.rm = TRUE) else NA_real_,
    .groups = "drop"
  ) %>%
  dplyr::mutate(available_for_event_study = n_nonmissing >= 50 & !all_nonmissing_values_zero)
readr::write_csv(availability_long, config$nanda_outcome_availability_csv)

unavailable_years <- availability_long %>%
  dplyr::filter(!available_for_event_study) %>%
  dplyr::select(outcome, year)
cz_val <- cz_initial
if (nrow(unavailable_years) > 0) {
  for (i in seq_len(nrow(unavailable_years))) {
    outcome_i <- unavailable_years$outcome[[i]]
    year_i <- unavailable_years$year[[i]]
    cz_val[[outcome_i]][cz_val$year == year_i] <- NA_real_
  }
}

cz_summary <- cz_val %>%
  dplyr::summarise(n_cz = dplyr::n_distinct(czone), min_year = min(year, na.rm = TRUE), max_year = max(year, na.rm = TRUE), dplyr::across(dplyr::all_of(rate_cols), list(n_nonmissing = ~sum(!is.na(.x)), min = ~min(.x, na.rm = TRUE), max = ~max(.x, na.rm = TRUE)), .names = "{.col}_{.fn}"))
readr::write_csv(cz_summary, config$nanda_cz_validation_csv)
readr::write_csv(cz_val, config$nanda_cz_panel_csv)
adh <- load_adh_extension_data(config) %>% dplyr::select(czone, statefip, exposure = exposure_1990_2007, adh_weight, dplyr::all_of(config$baseline_controls), dplyr::starts_with("z_"))
nanda_panel <- cz_val %>% dplyr::inner_join(adh, by = "czone") %>% dplyr::arrange(czone, year) %>% dplyr::group_by(czone) %>% dplyr::mutate(election_index = dplyr::dense_rank(year)) %>% dplyr::ungroup()
outcomes <- rate_cols
specs <- list(minimal = list(controls = character(0), standardize = FALSE), core_controls = list(controls = config$core_interacted_controls, standardize = TRUE), full_controls = list(controls = config$interacted_controls, standardize = TRUE))
coef_tbl <- tibble::tibble(); status_tbl <- tibble::tibble()
for (outcome in outcomes) {
  outcome_i <- outcome
  valid_years <- availability_long %>% dplyr::filter(.data$outcome == outcome_i, available_for_event_study) %>% dplyr::pull(year) %>% sort()
  ref <- if (outcome_i %in% c("partisan_index_dem", "partisan_index_rep")) {
    if (2006L %in% valid_years) 2006L else if (length(valid_years) > 0) min(valid_years) else NA_integer_
  } else if (outcome_i %in% c("pres_rep_ratio", "pres_dem_ratio")) {
    if (2004L %in% valid_years) 2004L else if (length(valid_years) > 0) min(valid_years) else NA_integer_
  } else {
```

```
      if (2004L %in% valid_years) 2004L else if (length(valid_years) > 0) min(valid_years) else NA_integer_
    }
    for (spec_name in names(specs)) {
      spec <- specs[[spec_name]]
      fit <- if (length(valid_years) >= 2 && !is.na(ref)) {
        panel_subset <- nanda_panel %>% dplyr::filter(year %in% valid_years)
        fit_single_outcome_event_study(panel_subset, outcome_i, spec$controls, spec_name, config, ref_year = ref, standardize_controls = spec$standardize, sample_label = "nanda_2004_2022")
      } else {
        list(status = "insufficient_valid_years", coefficients = tibble::tibble())
      }
      coef_tbl <- dplyr::bind_rows(coef_tbl, fit$coefficients)
      status_tbl <- dplyr::bind_rows(status_tbl, tibble::tibble(module = "nanda_turnout_partisanship", outcome = outcome_i, outcome_label = extension_outcome_label(outcome_i), spec = spec_name, spec_label = extension_spec_label(spec_name), reference_year = ref, valid_years = paste(valid_years, collapse = ";"), status = fit$status, n_coefficients = nrow(fit$coefficients)))
    }
  }
  coef_tbl <- coef_tbl %>% dplyr::mutate(outcome_label = extension_outcome_label(.data$outcome), spec_label = extension_spec_label(.data$spec))
  readr::write_csv(coef_tbl, config$nanda_outcomes_coefficients_csv)
  readr::write_csv(status_tbl, config$nanda_outcomes_status_csv)
  readr::write_csv(availability_long, config$nanda_outcomes_summary_csv)
  if (nrow(coef_tbl) > 0) {
    p <- plot_pointrange_event(coef_tbl, "NaNDA turnout and partisanship outcomes", "Outcome-specific valid years and reference years; rates outside [0, 1.5] and all-zero structural years set missing; CZ-clustered intervals") +
      ggplot2::facet_grid(outcome_label ~ spec_label, scales = "free_y")
    write_plot_safely(p, config$nanda_outcomes_plot_png, config$nanda_outcomes_plot_pdf, width = 12, height = 10)
    plot_family <- function(family_outcomes, title, png_path, pdf_path, height = 6) {
      df <- coef_tbl %>% dplyr::filter(outcome %in% family_outcomes, spec == "full_controls")
      if (nrow(df) > 0) {
        pp <- plot_pointrange_event(df, title, "Full-control specification; outcome-specific reference years; CZ-clustered intervals") +
          ggplot2::facet_wrap(~outcome_label, scales = "free_y", ncol = 1)
        write_plot_safely(pp, png_path, pdf_path, width = 8.5, height = height)
      }
    }
    plot_family(c("reg_voters_pct", "voter_turnout_pct", "reg_voter_turnout_pct"), "NaNDA turnout and registration diagnostics", config$nanda_turnout_plot_png, config$nanda_turnout_plot_pdf, height = 7)
    plot_family(c("pres_rep_ratio", "pres_dem_ratio"), "NaNDA presidential vote-ratio diagnostics", config$nanda_vote_plot_png, config$nanda_vote_plot_pdf, height = 5.5)
    plot_family(c("partisan_index_dem", "partisan_index_rep"), "NaNDA partisanship-index diagnostics", config$nanda_partisanship_plot_png, config$nanda_partisanship_plot_pdf, height = 5.5)
  }
  invisible(coef_tbl)
}

# ---------------------------------------------------------------------------
# 4. ADHM 2020 political-supply/mechanism diagnostics
# ---------------------------------------------------------------------------

run_adhm2020_mechanism_diagnostics <- function(config = CONFIG) {
  config <- finalize_config(config)
  adhm_dir <- file.path(config$replication_dir, "adhm2020", "dta")
  house_path <- file.path(adhm_dir, "house_2002_2016.dta")
  pres_path <- file.path(adhm_dir, "president_2000_2016.dta")
  if (!file.exists(house_path) && !file.exists(pres_path)) {
    readr::write_csv(tibble::tibble(status = "missing", path = adhm_dir), config$adhm2020_summary_csv)
    return(invisible(NULL))
  }
  adh <- load_adh_extension_data(config)
  spec_controls <- list(minimal = character(0), core_controls = model_control_names(config$core_interacted_controls, standardize = TRUE), full_controls = model_control_names(config$interacted_controls, standardize = TRUE))
  spec_controls <- purrr::map(spec_controls, ~ .x[.x %in% names(adh)])
  regressions <- tibble::tibble(); status <- tibble::tibble()
  outcome_dict <- tibble::tibble(
    outcome = c("d2_rwin_2002_2016", "d2_cfavg_repcon_2002_2016", "d2_cfavg_repmod_2002_2016", "d2_cfavg_demmod_2002_2016", "d2_cfavg_demlib_2002_2016", "d2_teaparty_2002_2010", "d_shnr_pres_2000_2016", "d_shnr_pres_2000_2008"),
    outcome_label = adhm_outcome_label(outcome),
    interpretation = c("Change in Republican House win indicator/share", "Change in conservative Republican CF-score share", "Change in moderate Republican CF-score share", "Change in moderate Democratic CF-score share", "Change in liberal Democratic CF-score share", "Change in Tea Party outcome", "Change in Republican presidential vote share, 2000-2016", "Change in Republican presidential vote share, 2000-2008"),
    report_family = c("House control", "House ideology", "House ideology", "House ideology", "House ideology", "Tea Party", "Presidential", "Presidential")
  )
  readr::write_csv(outcome_dict, config$adhm2020_outcome_dictionary_csv)
  fit_cross_section <- function(df, outcomes, source, weight_var = NULL) {
    purrr::map_dfr(outcomes[outcomes %in% names(df)], function(outcome) {
      purrr::imap_dfr(spec_controls, function(ctrls, spec_name) {
        dat <- df %>% dplyr::inner_join(adh, by = "czone") %>% dplyr::filter(!is.na(.data[[outcome]]), !is.na(exposure_1990_2007))
        if (nrow(dat) == 0) {
          status <<- dplyr::bind_rows(status, tibble::tibble(source = source, outcome = outcome, spec = spec_name, status = "empty", n_obs = 0L, note = NA_character_))
          return(tibble::tibble())
        }
        rhs <- c("exposure_1990_2007", ctrls)
        fml <- stats::as.formula(paste(outcome, "~", paste(rhs, collapse = " + ")))
        mod <- tryCatch(fit_weighted_feols(fml, dat, weight_var = if (!is.null(weight_var) && weight_var %in% names(dat)) weight_var else "adh_weight", cluster_var = "statefip"), error = function(e) e)
        if (inherits(mod, "error")) {
```

```
          status <<- dplyr::bind_rows(status, tibble::tibble(source = source, outcome = outcome, spec = spec_name, status = "error", n_obs = nrow(dat), note = conditionMessage(mod)))
          return(tibble::tibble())
        }
        ct <- coeftable_to_tibble(mod) %>% dplyr::filter(term == "exposure_1990_2007")
        status <<- dplyr::bind_rows(status, tibble::tibble(source = source, outcome = outcome, spec = spec_name, status = "ok", n_obs = stats::nobs(mod), note = NA_character_))
        tibble::tibble(source = source, outcome = outcome, outcome_label = adhm_outcome_label(outcome), spec = spec_name, spec_label = extension_spec_label(spec_name), estimate = scalar_or_na(ct$estimate), std.error = scalar_or_na(ct$std.error), statistic = scalar_or_na(ct$statistic), p.value = scalar_or_na(ct$p.value), n_obs = stats::nobs(mod), controls = if (length(ctrls) == 0) "none" else paste(ctrls, collapse = "; "))
      })
    })
  }
  if (file.exists(house_path)) {
    house <- haven::read_dta(house_path) %>% dplyr::mutate(czone = as.integer(czone), sh_district_2002 = dplyr::coalesce(safe_num(sh_district_2002), 1))
    house_outcomes <- c("d2_rwin_2002_2016", "d2_cfavg_repcon_2002_2016", "d2_cfavg_repmod_2002_2016", "d2_cfavg_demmod_2002_2016", "d2_cfavg_demlib_2002_2016", "d2_teaparty_2002_2010")
    house_cz <- house %>% dplyr::group_by(czone) %>% dplyr::summarise(dplyr::across(dplyr::all_of(house_outcomes[house_outcomes %in% names(house)]), ~ weighted_mean_safe(.x, sh_district_2002)), .groups = "drop")
    regressions <- dplyr::bind_rows(regressions, fit_cross_section(house_cz, house_outcomes, "adhm2020_house_2002_2016"))
  }
  if (file.exists(pres_path)) {
    pres <- haven::read_dta(pres_path) %>% dplyr::mutate(czone = as.integer(czone), totvote_2000pres = dplyr::coalesce(safe_num(totvote_2000pres), 1))
    pres_cz <- pres %>%
      dplyr::group_by(czone) %>%
      dplyr::summarise(shnr_pres2000 = weighted_mean_safe(shnr_pres2000, totvote_2000pres), shnr_pres2008 = weighted_mean_safe(shnr_pres2008, totvote_2000pres), shnr_pres2016 = weighted_mean_safe(shnr_pres2016, totvote_2000pres), .groups = "drop") %>%
      dplyr::mutate(d_shnr_pres_2000_2016 = shnr_pres2016 - shnr_pres2000, d_shnr_pres_2000_2008 = shnr_pres2008 - shnr_pres2000)
    regressions <- dplyr::bind_rows(regressions, fit_cross_section(pres_cz, c("d_shnr_pres_2000_2016", "d_shnr_pres_2000_2008"), "adhm2020_president_2000_2016"))
  }
  readr::write_csv(regressions, config$adhm2020_regressions_csv)
  readr::write_csv(status, config$adhm2020_status_csv)
  hetero_vars <- c("l_sh_pop_white", "l_sh_popwht", "sh_pop_white", "l_sh_pop_black", "l_sh_pop_hispanic")
  avail <- tibble::tibble(candidate_variable = hetero_vars, available_in_current_ADH_extension_data = hetero_vars %in% names(adh), note = "If a majority-white or race-composition variable is available, add exposure × baseline-race heterogeneity to compare with ADHM 2020 heterogeneity results.")
  readr::write_csv(avail, config$adhm2020_heterogeneity_availability_csv)
  summary <- tibble::tibble(source = c("house_2002_2016", "president_2000_2016"), file_present = c(file.exists(house_path), file.exists(pres_path)), purpose = c("Political-supply and House ideology outcomes from ADHM public replication data", "Presidential-vote benchmark outcomes from ADHM public replication data"))
  readr::write_csv(summary, config$adhm2020_summary_csv)
  if (nrow(regressions) > 0) {
    plot_dat <- regressions %>%
      dplyr::mutate(outcome_label = factor(outcome_label, levels = rev(unique(outcome_label))))
    p <- ggplot2::ggplot(plot_dat, ggplot2::aes(x = estimate, y = outcome_label, xmin = estimate - 1.96 * std.error, xmax = estimate + 1.96 * std.error)) +
      ggplot2::geom_vline(xintercept = 0, linewidth = 0.25) + ggplot2::geom_pointrange(linewidth = 0.25) +
      ggplot2::facet_grid(source ~ spec_label, scales = "free_y", space = "free_y") +
      ggplot2::labs(title = "ADHM 2020 political-supply diagnostics", subtitle = "Cross-sectional CZ regressions on ADH exposure; public ADHM outcomes aggregated to CZs", x = "Coefficient on ADH exposure", y = NULL) +
      ggplot2::theme_minimal(base_size = 10)
    write_plot_safely(p, config$adhm2020_plot_png, config$adhm2020_plot_pdf, width = 12, height = 7)
  }
  invisible(regressions)
}

# ---------------------------------------------------------------------------
# 6. Subperiod exposure
# ---------------------------------------------------------------------------

run_subperiod_exposure_event_studies <- function(config = CONFIG) {
  config <- finalize_config(config)
  adh_stack_path <- file.path(config$replication_dir, "adh2013", "dta", "workfile_china.dta")
  if (!file.exists(adh_stack_path) || !file.exists(config$analysis_panel_rds)) return(invisible(NULL))
  stack <- haven::read_dta(adh_stack_path)
  exp_period <- stack %>%
    dplyr::filter(yr %in% c(1990, 2000)) %>%
    dplyr::transmute(czone = as.integer(czone), period = dplyr::case_when(yr == 1990 ~ "1990_2000", yr == 2000 ~ "2000_2007", TRUE ~ as.character(yr)), exposure = safe_num(d_tradeusch_pw), instrument = safe_num(d_tradeotch_pw_lag)) %>%
    tidyr::pivot_wider(names_from = period, values_from = c(exposure, instrument), names_sep = "_")
  readr::write_csv(exp_period, config$subperiod_exposure_csv)
  readr::write_csv(exp_period %>% dplyr::summarise(dplyr::across(-czone, list(mean = ~mean(.x, na.rm = TRUE), sd = ~stats::sd(.x, na.rm = TRUE), min = ~min(.x, na.rm = TRUE), max = ~max(.x, na.rm = TRUE)), .names = "{.col}_{.fn}")), config$subperiod_exposure_summary_csv)
  adh <- load_adh_extension_data(config)
  fs <- exp_period %>% dplyr::inner_join(adh %>% dplyr::select(czone, adh_weight, statefip, dplyr::starts_with("z_")), by = "czone")
  fs_tbl <- tibble::tibble()
  for (pair in list(c("exposure_1990_2000", "instrument_1990_2000"), c("exposure_2000_2007", "instrument_2000_2007"))) {
    if (all(pair %in% names(fs))) {
      for (spec_name in c("minimal", "core_controls", "full_controls")) {
        ctrls <- if (spec_name == "minimal") character(0) else if (spec_name == "core_controls") model_control_names(config$core_interacted_controls, TRUE) else model_control_names(config$interacted_controls, TRUE)
        ctrls <- ctrls[ctrls %in% names(fs)]
        fml <- stats::as.formula(paste(pair[1], "~", paste(c(pair[2], ctrls), collapse = " + ")))
```

```
      mod <- tryCatch(fit_weighted_feols(fml, fs, weight_var = "adh_weight", cluster_var = "statefip"), error = function(e) NULL)
      if (!is.null(mod)) {
        ct <- coeftable_to_tibble(mod) %>% dplyr::filter(term == pair[2])
        fs_tbl <- dplyr::bind_rows(fs_tbl, tibble::tibble(exposure_period = gsub("exposure_", "", pair[1]), spec = spec_name,
          ↪ instrument = pair[2], estimate = scalar_or_na(ct$estimate), std.error = scalar_or_na(ct$std.error), statistic =
          ↪ scalar_or_na(ct$statistic), p.value = scalar_or_na(ct$p.value), first_stage_F_t2 = scalar_or_na(ct$statistic)^2, n_obs =
          ↪ stats::nobs(mod)))
      }
    }
  }
}
readr::write_csv(fs_tbl, config$subperiod_first_stage_csv)

panel <- readRDS(config$analysis_panel_rds) %>% dplyr::left_join(exp_period, by = "czone")
yrs <- config$event_years; ref <- config$reference_year
for (ev in c("exposure_1990_2000", "exposure_2000_2007")) for (yr in setdiff(yrs, ref)) panel[[paste0(ev, "_x_", yr)]] <- panel[[ev]] *
 ↪ as.integer(panel$year == yr)
specs <- list(minimal = list(controls = character(0), standardize = FALSE), core_controls = list(controls =
 ↪ config$core_interacted_controls, standardize = TRUE), full_controls = list(controls = config$interacted_controls, standardize =
 ↪ TRUE))
coef_tbl <- tibble::tibble(); status_tbl <- tibble::tibble()
for (spec_name in names(specs)) {
  spec <- specs[[spec_name]]
  dat <- panel
  ctrl_build <- apply_control_interactions(dat, spec$controls, yrs, ref, standardize = spec$standardize)
  dat <- ctrl_build$data
  event_terms <- unlist(lapply(c("exposure_1990_2000", "exposure_2000_2007"), function(ev) paste0(ev, "_x_", setdiff(yrs, ref))))
  fml <- stats::as.formula(paste("rep_margin ~", paste(c(event_terms, ctrl_build$terms), collapse = " + "), "| czone + year"))
  fit <- tryCatch(fixest::feols(fml, data = dat, weights = ~adh_weight, cluster = ~czone, panel.id = ~czone + election_index, warn =
   ↪ FALSE, notes = FALSE), error = function(e) e)
  if (inherits(fit, "error")) {
    status_tbl <- dplyr::bind_rows(status_tbl, tibble::tibble(spec = spec_name, status = "error", error_message =
     ↪ conditionMessage(fit), n_coefficients = 0L))
  } else {
    ct <- coeftable_to_tibble(fit) %>%
      dplyr::filter(term %in% event_terms) %>%
      dplyr::mutate(
        spec = spec_name,
        exposure_period = dplyr::case_when(startsWith(term, "exposure_1990_2000") ~ "1990-2000", startsWith(term, "exposure_2000_2007")
         ↪ ~ "2000-2007", TRUE ~ NA_character_),
        year = as.integer(stringr::str_extract(term, "[0-9]{4}$")),
        period_type = dplyr::case_when(year < 1990 ~ "placebo_pre_shock", year <= 2008 ~ "during_or_immediate_post_shock", year <= 2016
         ↪ ~ "medium_run", TRUE ~ "long_run"),
        reference_year = ref,
        n_obs = stats::nobs(fit), n_cz = dplyr::n_distinct(dat$czone)
      ) %>%
      dplyr::select(spec, exposure_period, period_type, term, year, reference_year, estimate, std.error, statistic, p.value, n_obs,
       ↪ n_cz)
    coef_tbl <- dplyr::bind_rows(coef_tbl, ct)
    status_tbl <- dplyr::bind_rows(status_tbl, tibble::tibble(spec = spec_name, status = "ok", error_message = NA_character_,
     ↪ n_coefficients = nrow(ct)))
  }
}
readr::write_csv(coef_tbl, config$subperiod_event_study_coefficients_csv)
readr::write_csv(status_tbl, config$subperiod_event_study_status_csv)
equality <- tibble::tibble()
if (nrow(coef_tbl) > 0) {
  wide <- coef_tbl %>% dplyr::select(spec, year, exposure_period, estimate, std.error) %>% tidyr::pivot_wider(names_from =
   ↪ exposure_period, values_from = c(estimate, std.error), names_sep = "_")
  if (all(c("estimate_1990-2000", "estimate_2000-2007") %in% names(wide))) {
    equality <- wide %>% dplyr::mutate(difference_early_minus_late = `estimate_1990-2000` - `estimate_2000-2007`, approx_se_difference =
     ↪ sqrt((`std.error_1990-2000`)^2 + (`std.error_2000-2007`)^2), approx_t_difference = difference_early_minus_late /
     ↪ approx_se_difference, approx_p_difference = 2 * stats::pnorm(-abs(approx_t_difference)), note = "Approximate test ignores
     ↪ covariance between early and late exposure coefficients; use as descriptive diagnostic.")
  }
}
readr::write_csv(equality, config$subperiod_equality_tests_csv)
selected_horizons <- tibble::tibble()
if (nrow(equality) > 0) {
  selected_horizons <- equality %>%
    dplyr::filter(year %in% c(1996L, 2000L, 2008L, 2016L, 2020L)) %>%
    dplyr::transmute(
      spec, year,
      early_exposure_coefficient = `estimate_1990-2000`,
      late_exposure_coefficient = `estimate_2000-2007`,
      early_minus_late = difference_early_minus_late,
      approx_se_difference, approx_t_difference, approx_p_difference,
      interpretation_note = "Approximate early-vs-late comparison; covariance ignored. Use for descriptive horizon interpretation."
    )
}
readr::write_csv(selected_horizons, config$subperiod_selected_horizons_csv)
if (nrow(coef_tbl) > 0) {
  p <- ggplot2::ggplot(coef_tbl, ggplot2::aes(x = year, y = estimate, ymin = estimate - 1.96 * std.error, ymax = estimate + 1.96 *
   ↪ std.error, color = exposure_period)) +
    ggplot2::geom_hline(yintercept = 0, linewidth = 0.25) + ggplot2::geom_pointrange(position = ggplot2::position_dodge(width = 0.6),
     ↪ linewidth = 0.25) +
    ggplot2::facet_wrap(~spec, ncol = 1, scales = "free_y") +
    ggplot2::labs(title = "Subperiod ADH exposure event study", subtitle = "1990-2000 and 2000-2007 exposure terms entered jointly", x =
     ↪ NULL, y = "Coefficient", color = "Exposure period") +
```

```
      ggplot2::theme_minimal(base_size = 11)
    write_plot_safely(p, config$subperiod_event_study_plot_png, config$subperiod_event_study_plot_pdf, width = 10, height = 8)
  }
  invisible(coef_tbl)
}

run_sixth_extensions <- function(config = CONFIG) {
  config <- extension_primary_config(config)
  ensure_output_dirs(config)
  load_required_packages(config)
  message("Running Sixth-section extension diagnostics for primary design: ", toupper(config$crosswalk_weight_slug), " + ",
   ↪  config$crosswalk_missing_weight_policy)
  results <- list()
  results$bartik <- tryCatch(run_bartik_identification_diagnostics(config), error = function(e) { warning("Bartik diagnostics failed: ",
   ↪  conditionMessage(e), call. = FALSE); NULL })
  results$alternative_outcomes <- tryCatch(run_alternative_political_outcome_event_studies(config), error = function(e) {
   ↪  warning("Alternative political outcomes failed: ", conditionMessage(e), call. = FALSE); NULL })
  results$nanda <- tryCatch(run_nanda_outcome_event_studies(config), error = function(e) { warning("NaNDA outcomes failed: ",
   ↪  conditionMessage(e), call. = FALSE); NULL })
  results$adhm2020 <- tryCatch(run_adhm2020_mechanism_diagnostics(config), error = function(e) { warning("ADHM 2020 diagnostics failed:
   ↪  ", conditionMessage(e), call. = FALSE); NULL })
  results$subperiod <- tryCatch(run_subperiod_exposure_event_studies(config), error = function(e) { warning("Subperiod exposure
   ↪  diagnostics failed: ", conditionMessage(e), call. = FALSE); NULL })
  checks <- if (file.exists(config$validation_checks_csv)) readr::read_csv(config$validation_checks_csv, show_col_types = FALSE) else
   ↪  tibble::tibble()
  write_pipeline_manifest(config = config, checks = checks, stage = "sixth_extensions", sources = list(adh2013 = list(path =
   ↪  file.path(config$replication_dir, "adh2013")), nanda = list(path = file.path(config$replication_dir, "outcomes-data",
   ↪  "nanda-2004-2022")), adhm2020 = list(path = file.path(config$replication_dir, "adhm2020"))), extra = list(extension_modules =
   ↪  c("bartik_identification", "alternative_political_outcomes", "nanda_turnout_partisanship", "adhm2020_political_supply",
   ↪  "subperiod_exposure"), crosswalk_used_for_extensions = paste0(config$crosswalk_weight, " / ",
   ↪  config$crosswalk_missing_weight_policy)))
  invisible(results)
}
```

8 run_pipeline.R

```
# extension-adh2013/src/run_pipeline.R

if (!requireNamespace("here", quietly = TRUE)) {
  stop("Package 'here' is required. Install it before running the pipeline.", call. = FALSE)
}
library(here)
here::i_am("src/run_pipeline.R")

source(here::here("src", "00_config.R"))
source(here::here("src", "01_helpers.R"))
source(here::here("src", "02_build_analysis_data.R"))
source(here::here("src", "03_estimate_event_study.R"))
source(here::here("src", "04_export_outputs.R"))
source(here::here("src", "05_export_diagnostic_maps.R"))
source(here::here("src", "06_sixth_extensions.R"))

read_bool_env <- function(name, default) {
  value <- Sys.getenv(name, unset = "")
  if (!nzchar(value)) return(default)
  tolower(value) %in% c("1", "true", "t", "yes", "y")
}

run_single_pipeline <- function(config) {
  config <- finalize_config(config)
  ensure_output_dirs(config)
  write_dependency_manifest(config)

  message("Building analysis data...")
  build_analysis_data(config, stop_on_fatal = TRUE)

  message("Estimating event-study specifications...")
  estimate_event_study(config, stop_on_fatal = TRUE)

  message("Exporting tables and figures...")
  export_event_study_outputs(config)

  message("Exporting diagnostic maps and crosswalk comparisons when inputs are available...")
  export_diagnostic_maps_and_comparisons(config)

  if (isTRUE(config$run_sixth_extensions)) {
    message("Running Sixth-section extensions...")
    run_sixth_extensions(config)
  }

  invisible(config)
}

policy_override <- Sys.getenv("CROSSWALK_MISSING_WEIGHT_POLICY", unset = "")
if (nzchar(policy_override)) CONFIG$crosswalk_missing_weight_policy <- policy_override

weight_override <- Sys.getenv("CROSSWALK_WEIGHT", unset = "")
if (nzchar(weight_override)) CONFIG$crosswalk_weight <- weight_override

ext_weight_override <- Sys.getenv("EXTENSION_CROSSWALK_WEIGHT", unset = "")
if (nzchar(ext_weight_override)) CONFIG$extension_crosswalk_weight <- ext_weight_override

ext_policy_override <- Sys.getenv("EXTENSION_CROSSWALK_MISSING_WEIGHT_POLICY", unset = "")
if (nzchar(ext_policy_override)) CONFIG$extension_crosswalk_missing_weight_policy <- ext_policy_override

CONFIG$renormalize_crosswalk_weights <- read_bool_env(
  "RENORMALIZE_CROSSWALK_WEIGHTS", CONFIG$renormalize_crosswalk_weights
)
CONFIG$require_balanced_panel <- read_bool_env(
  "REQUIRE_BALANCED_PANEL", CONFIG$require_balanced_panel
)
CONFIG$require_no_vcov_repair_for_main <- read_bool_env(
  "REQUIRE_NO_VCOV_REPAIR_FOR_MAIN", CONFIG$require_no_vcov_repair_for_main
)
CONFIG$allow_vcov_repair <- read_bool_env("ALLOW_VCOV_REPAIR", CONFIG$allow_vcov_repair)
CONFIG$allow_failed_diagnostics <- read_bool_env(
  "ALLOW_FAILED_DIAGNOSTICS", CONFIG$allow_failed_diagnostics
)
CONFIG$standardize_interacted_controls <- read_bool_env(
  "STANDARDIZE_INTERACTED_CONTROLS", CONFIG$standardize_interacted_controls
)
CONFIG$export_crosswalk_maps <- read_bool_env(
  "EXPORT_CROSSWALK_MAPS", CONFIG$export_crosswalk_maps
)
CONFIG$run_crosswalk_sensitivity <- read_bool_env(
  "RUN_CROSSWALK_SENSITIVITY", CONFIG$run_crosswalk_sensitivity
)
CONFIG$run_sixth_extensions <- read_bool_env(
  "RUN_SIXTH_EXTENSIONS", CONFIG$run_sixth_extensions
)
CONFIG$save_single_rds <- read_bool_env(
  "SAVE_SINGLE_RDS", CONFIG$save_single_rds
)
```

```
88
89  if (isTRUE(CONFIG$run_crosswalk_sensitivity)) {
90    message("Running the four-specification crosswalk sensitivity pipeline...")
91    run_crosswalk_sensitivity_pipeline(CONFIG)
92  } else {
93    run_single_pipeline(CONFIG)
94  }
95
96  message("Pipeline complete.")
```

9 county_bridge_exceptions.csv

```
year,county_fips,county_name_pattern,override_source_decade,override_county_fips_source,issue_class,reason,apply
1984,04012,La Paz,1990,04012,county_boundary_change_needs_bridge,"La Paz County, AZ was created after 1980 but exists in 1990 target geography; use 1990 identity mapping for 1984/1988 presidential returns.",TRUE
1988,04012,La Paz,1990,04012,county_boundary_change_needs_bridge,"La Paz County, AZ was created after 1980 but exists in 1990 target geography; use 1990 identity mapping for 1984/1988 presidential returns.",TRUE
1984,35006,Cibola,1990,35006,county_boundary_change_needs_bridge,"Cibola County, NM was created after 1980 but exists in 1990 target geography; use 1990 identity mapping for 1984/1988 presidential returns.",TRUE
1988,35006,Cibola,1990,35006,county_boundary_change_needs_bridge,"Cibola County, NM was created after 1980 but exists in 1990 target geography; use 1990 identity mapping for 1984/1988 presidential returns.",TRUE
2016,46102,Oglala Lakota,2010,46113,county_boundary_change_needs_bridge,"Oglala Lakota County, SD is the renamed Shannon County; map 2016 returns to the 2010/1990 Shannon FIPS where needed.",TRUE
2020,46102,Oglala Lakota,2010,46113,county_boundary_change_needs_bridge,"Oglala Lakota County, SD is the renamed Shannon County; map 2020 returns to the 2010/1990 Shannon FIPS where needed.",FALSE
2004,08014,Broomfield,2010,08014,county_boundary_change_needs_bridge,"Broomfield County, CO was created after the 2000 Census; use the 2010 source crosswalk for 2004 returns so its vote mass is allocated to 1990 counties.",TRUE
2008,08014,Broomfield,2010,08014,county_boundary_change_needs_bridge,"Broomfield County, CO was created after the 2000 Census; use the 2010 source crosswalk for 2008 returns so its vote mass is allocated to 1990 counties.",TRUE
1976,51683,Manassas,1980,51683,county_boundary_change_needs_bridge,"Manassas city appears in 1976 returns but not the 1970 source crosswalk used by the decade rule; use the 1980 source crosswalk.",TRUE
1976,51685,Manassas Park,1980,51685,county_boundary_change_needs_bridge,"Manassas Park city appears in 1976 returns but not the 1970 source crosswalk used by the decade rule; use the 1980 source crosswalk.",TRUE
1976,51735,Poquoson,1980,51735,county_boundary_change_needs_bridge,"Poquoson city appears in 1976 returns but not the 1970 source crosswalk used by the decade rule; use the 1980 source crosswalk.",TRUE
```